GEM21menos65. A Phase III trial for NDMM patients who are candidates for ASCT comparing Extended VRD plus Early Rescue Intervention vs Isatuximab-VRD vs Isatuximab-V-Iberdomide-D.

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The study will be conducted according to the protocol and in compliance with Good Clinical Practice (GCP), with the Declaration of Helsinki, and with other applicable regulatory requirements including but not limited to Institutional Review Board/Ethics Committee (IRB/EC) approval.

Confidentiality Statement

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This study protocol was subjected to critical review and has been approved by the Sponsor. The information it contains is consistent with the current risk/benefit evaluation of the investigational product as well as with the moral, ethical, and scientific principles governing clinical research as set out in the Declaration of Helsinki and the guidelines on Good Clinical Practice.

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I have read this study protocol, including all appendices. By signing this protocol, I agree to supervise and conduct the clinical study, following approval by an Independent Ethics Committee (IEC)/Institutional Review Board (IRB), in accordance with the study protocol, the current International Conference on Harmonization (ICH-E6-R2) Guideline for Good Clinical Practice (GCP), and applicable regulatory requirements. I will ensure that all personnel involved in the study under my direction will be informed about the contents of this study protocol and will receive all necessary instructions for performing the study according to the study protocol.

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I have read and understand this study protocol, including all appendices. By signing this protocol, I agree to conduct the clinical study, following approval by an Independent Ethics Committee (IEC)/Institutional Review Board (IRB), in accordance with the study protocol, the current International Conference on Harmonization (ICH-E6-R2) Guideline for Good Clinical Practice (GCP), and applicable regulatory requirements. I will ensure that all personnel involved in the study under my direction will be informed about the contents of this study protocol and will receive all necessary instructions for performing the study according to the study protocol.

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1 ABBREVIATIONS

Table 1 Abbreviations

Abbreviation	Full terminology
AE	Adverse Event
AESI	Adverse Event of Special Interest
ALT	Alanine Transaminase
ANC	Absolute Neutrophil Count
ASCT	Autologous Stem Cell Transplant
AST	Aspartate Transaminase
BM	Bone marrow
BSA	Body surface area
CONSORT	Consolidated Standards of Reporting Trials
CR	Complete Response
CRA	Contract Research associate
CRF	Case report form
CRO	Contract Research Organization
CT	Computed Tomography
CTCAE	Common terminology criteria for classification of adverse events
DL (-1, 1, 2, 3)	Dose level (-1, 1, 2, 3)
DNA	Deoxyribonucleic acid
ECOG	Eastern Cooperative Oncology Group
eCRF	Electronic case report form
EDTA	Ethylenediamine tetra acetic acid
ELISA	Enzyme-linked immunosorbent assay
EMA	European Medicines Agency
EOT	End of treatment
EPO	Erythropoietin
ERI	Early Rescue Intervention
FISH	Fluorescence in situ hybridization
FLC	Free light chains
GCP	Good Clinical Practice
G-CSF	Granulocyte colony-stimulating factor
HIV	Human immunodeficiency virus

Abbreviation	Full terminology
HCP	Health Care Professionals
HR	Hazard ratio
ICF	Informed consent form
ICH	International Conference on Harmonization
IEC	Independent ethic committee
IF	Immunofixation
Ig	Immunoglobulin
IMWG	International Myeloma Working Group
IP	Investigational product
IRR	Infusion-related reaction
IRRs	Infusion-related reactions
ISS	International Staging System
ITT	Intention-to-treat
IV	Intravenous
LDH	Lactate dehydrogenase
LVEF	Left ventricle ejection fraction
MC	Monoclonal component
MDS	Myelodysplastic Syndromes
MFC	Multiparametric flow cytometry
MM	Multiple Myeloma
MoAbs	Monoclonal antibodies
MP	Melphalan and prednisone
MRD	Minimal Residual Disease
NDMM	Newly-diagnosed multiple myeloma
NGS	Next-generation sequencing
OBDS	On Body Delivery System
OS	Overall Survival
PET	Positron Emission Tomography
PFS	Progression-free survival
PI	Proteasome inhibitors
PO	Per os (oral)
PR	Partial response
RBC	Red Blood Cell

Abbreviation	Full terminology
Rd	Lenalidomide and low dose of dexamethasone
RD	Lenalidomide and standard dose of dexamethasone
R-ISS	Revised International Staging System
RNA	Ribonucleic acid
RNA-Seq	RNA sequencing
RP2D	Recommended phase 2 dose
RRMM	Relapsed refractory multiple myeloma
SAE	Serious Adverse Event
SC	Subcutaneous
sCR	Stringent complete response
ASCT	Autologous Stem Cell Transplantation
sFLC	Serum free light chains
SoA	Schedule of Activities
SmPC	Summary of Product Characteristics
SPEP	Serum Protein Electrophoresis
SPM	Second Primary Malignancies
TTP	Time to progression
ULN	Upper Limit of Normal
UPEP	Urine protein electrophoresis
VGPR	Very good partial response
VID	Iberdomide-bortezomib and dexamethasone
VMP	Bortezomib-melphalan and prednisone
VRD	Lenalidomide-bortezomib and dexamethasone
VTP	Bortezomib-thalidomide, and prednisone
WOCBP	Woman of Childbearing Potential

2 STUDY RATIONALE

We believe that, at least in part, the alleged superiority of the second generation of new drugs over combinations based on lenalidomide and / or bortezomib in the first line of treatment for MM, is a consequence of their administration in continuous mode until progression rather than the effect of an intrinsic superiority face to face.

We also consider that delaying the treatment of an incipient subclinical progression until the tumor reaches the critical mass capable of producing clinical damage is an incorrect strategy. In all likelihood, the possibility of controlling with an alternative treatment an incipient recurrence associated with a low tumor mass until recovering (or reaching de novo) an MRD-ve status is much higher than obtaining the same result when the MM progresses enough to meet the IMWG criteria of Clinical Progression.

But a solid and sustained development of the concepts previously exposed requires:

1. A control that represents the therapeutic standard and that, very predictably, given the times of our proposal, will be made up of the combination of the VRD scheme plus an anti-CD38 monoclonal antibody.

2. The contribution of the new immunomodulators to the first line constitutes in itself a line of clinical research of extraordinary interest. For this reason, the incorporation of another experimental arm in which lenalidomide is replaced by iberdomide ensures the strength of the structure of our clinical trial proposal.

2.0 Justifying the Selection of Isatuximab – VRD

VRD is highly active and well tolerated with a response rate in NDMM patients of 80–100% and CR are seen in 40–50% of patients. In the SWOG S0777 phase III clinical trial comparing VRD with RD in patients of any age, progression free survival (PFS) and overall survival (OS) were significantly longer with VRD compared with Rd(1). In the IFM 2009 randomized study(2) comparing 5 cycles of VRD (14 days of lenalidomide per cycle, 3 pre-ASCT cycles and 2 consolidation cycles) plus transplant vs. 8 cycles of VRD, the CR rate in the 350 transplanted patients reached 58%, with PFS and OS at 3 years, respectively, 61% and 88%. Similarly, VRD efficacy data in the GEM12menos65 clinical trial(3), in which 8 cycles of VRD (lenalidomide 25 mg/d x 21 days) were administered to 486 patients, with an ASCT intercalated between the 6th and 7th cycles,-a design very

similar to the trial PETHEMA/GEM05menos65-, respect to historical results with VTD in this mentioned trial(4), showed better CR rates (VRD: 52% vs. VTD: 35%), a clear advantage over VTD in terms of the frequency and depth of the MRD-ve indices (62% vs. 47%), and lower rates of peripheral neuropathy (grade 3/4, 5.5% vs 15.4%; grade ≥ 2, 20.7% vs 44.6%)(5).

Isatuximab is a monoclonal antibody that binds to a specific epitope on the human cell surface antigen CD38, which is widely and uniformly expressed on myeloma cells(6,7). Isatuximab has antitumor activity via multiple biological mechanisms, including antibody dependent cellular-mediated cytotoxicity, complement-dependent cytotoxicity, antibody-dependent cellular phagocytosis, and direct induction of apoptosis without crosslinking. Isatuximab in combination with pomalidomide and dexamethasone significantly improves the results of pomalidomide-dexamethasone, PFS 11.5 months vs. 6.5 months, as evidenced by the ICARIA MM phase III trial(8), especially in patients already refractory to lenalidomide and a proteasome inhibitor. The most frequent treatment-emergent adverse events (Isatuximab/pomalidomide/dexa vs pomalidomide/dexa) were infusion reactions: 38% vs 0%, upper respiratory tract infections: 28% vs 17%, and diarrhea: 26% vs 20%. Deaths due to treatment-related adverse events were reported for one patient (<1%) in the isatuximab–pomalidomide–dexamethasone group (sepsis) and two (1%) in the pomalidomide–dexamethasone group (pneumonia and urinary tract infection).

Supporting the value of VRD as induction treatment in patients with NDMM, several important new clinical trials designed to evaluate the possibilities of new drugs in NDMM transplant eligible patients use VRD as a control arm, eg: VRD +/- Daratumumab (Perseus Phase III trial, NCT03710603); also, VRD +/- Isatuximab (GMMG HD7, NCT03617731) which are pending preliminary results. Other combination that includes Isatuximab, Lenalidomide and Dexamethasone but with a different Proteasome Inhibitor -Carfilzomib- is currently recruiting patients (IsKia TRIAL, NCT04483739). Similarly, a trial in high risk MM patients defined by Del (17p) or t(4;14) or t(14;16) or > 3 copies 1q21 and ISS stage 2 or 3 (GMMG-CONCEPT, NCT03104842), evaluating this same combination (Isa-KRd) has recently reported excellent result in terms of MRD negativity. Fifty patients (46 TE and 4 NTE) that completed induction achieved 100% ORR, 90% ≥ VGPR, 46% CR/sCR. Importantly, there were 20 MRD negative TE patients among 33 with MRD-assessment during induction treatment(9).

2.0.1 Isatuximab and subcutaneous application method with a wearable on-body delivery system

The local tolerance of isatuximab after a single IV injection, as well as after single intra-arterial, intramuscular, SC, or paravenous injection (unintended routes that might occur accidentally), was assessed in female New Zealand White rabbits (a pharmacologically non-reactive species). Isatuximab produced no compound-related clinical signs or body weight changes and there were no compound-related macroscopic or microscopic findings at the injection sites for any route of administration at concentrations up to 5 mg/mL. In addition, isatuximab did not produce compound-related macroscopic or microscopic changes at the injection sites following once weekly IV infusion for 3 weeks in cynomolgus monkeys up to 20 mg/mL (maximum concentration tested; see IB).
The local tolerance of isatuximab in its clinical SC formulation has been tested in female minipigs (a pharmacologically non-reactive species). The single SC infusion of 12.9 mL isatuximab (140 mg/mL, SC formulation) to minipigs equivalent to a total dose of approximately 1800 mg/minipig at flow rates of 0.5, 1, and 2 mL/min was overall clinically well tolerated. Erythema and hemorrhage at injection sites were noted on the day of infusion, only, and were similar for the three SC flow rates as well as for isatuximab and saline. Swelling at injection site was more evident for isatuximab infused at 1 and 2 mL/min compared with 0.5 mL/min and was considered related to a physical phenomenon (large volume injected in a short period of time). It declined at 4 hours and had disappeared at 24 hours following administration. There were no compound-related microscopic findings at the examination of skin biopsies of SC injection sites sampled 7 and 28 days following each injection at all administration rates tested (see IB). Finally, plasma compatibility and the hemolytic potential of isatuximab in human whole blood were tested *in vitro* at concentrations up to 2.5 mg/mL. No hemolytic effect or plasma incompatibility was observed (see IB).

2.0.1.1 Clinical development Isatuximab SC administration

In the ongoing multi-center, open-label, Phase 1b study TCD15484 (NCT04045795), safety, PK and efficacy of SC and IV isatuximab in combination with pomalidomide/dexamethasone are assessed25. At a cutoff date of 12 March 2021, a total of 34 patients have been treated in the TCD15484 study: 12 participants in the IV isatuximab cohort, 12 participants in the 1000 mg SC isatuximab cohort using a syringe pump and 10 participants in the 1400 mg SC isatuximab cohort using a syringe pump. The median relative dose intensity (RDI) of isatuximab was >90% in all cohorts. Overall, the safety data for SC isatuximab at doses of 1000 mg and 1400 mg administered in combination with pomalidomide/dexamethasone were both consistent with the IV isatuximab arm. One dose limiting toxicity (DLT) was reported in each SC isatuximab cohort (Grade 4 neutropenia in the SC isatuximab 1000 mg cohort and Grade 3 pulmonary infection in the SC isatuximab 1400 mg cohort). DLTs reported with SC

isatuximab were consistent with the DLTs reported in the dose-escalation study of IV isatuximab in combination with pomalidomide/dexamethasone (TCD14079, NCT02283775) in which one DLT was reported at each dose level (5 mg/kg: Grade 4 neutropenia; 10 mg/kg: Grade 4 neutropenic infection; and 20 mg/kg: Grade 3 confusional state)26. IRs for SC isatuximab were infrequent. There has been a single episode of IR (all Grade 2) in each SC/IV isatuximab cohort, which occurred at the first administration and lasted one (IV isatuximab and SC isatuximab 1400 mg) to two days (SC isatuximab 1000 mg). In addition, the local tolerability profile of SC isatuximab has been very good. Only one participant experienced an ISR, Grade 2 erythema and pain (SC isatuximab 1400 mg). There were more participants presenting injection site erythema in cohort SC isatuximab 1400 mg (50.0%) versus SC isatuximab 1000 mg (16.7%). All episodes of injection site erythema (not associated with ISR) were Grade 1 and occurred first at cycle 1.

Despite a shorter duration of follow-up at the SC isatuximab 1400 mg dose level due to sequential nature of the enrollment of the trial, the ORR was 80.0% in the SC isatuximab 1400 mg dose level and 67% in both cohorts, SC isatuximab 1000 mg dose and 10 mg/kg IV isatuximab. Rates of very good partial response (VGPR) or better were similar across the three cohorts (33.3 - 41.7%), as well as the rates of complete responses (16.5% - 25.0%). Overall, preliminary ORR and at least VGPR rates were consistent with the phase III ICARIA trial (NCT02990338)14,15.

Therefore, and based on PK modeling25 (see IB), actual SC isatuximab bioavailability is likely to be close to that observed for the 1400 mg SC isatuximab cohort and SC isatuximab at a dose of 1400 mg will be used in subsequent trials to investigate SC isatuximab in combination with other agents.

Additional Reference:

Study TCD15484: Subcutaneous (SC) isatuximab administration by an on-body delivery system (OBDS) in combination with pomalidomide-dexamethasone (Pd) in patients with relapsed/refractory multiple myeloma (RRMM). (NCT04045795 - J Clin Oncol 40, 2022 (suppl 16; abstr 8025); DOI 10.1200/JCO.2022.40.16_suppl.8025). Sponsor: Sanofi-Aventis Recherche & Development.

Study IgPro20_1001: Comparison of 2 Infusion Devices with Respect to Pharmacokinetics, Safety, and Tolerability of Hizentra®: Investigational Wearable Infusor and the Crono S-PID-50 Infusion Pump (EudraCT Number: 2016-003798-16 – Immunotherapy. 2022 Sep 21. doi: 10.2217/imt-2022-0097). Sponsor: CSL Behring.

Hang Quach, Gurdeep Parmar, Enrique M. Ocio, H. Miles Prince, Albert Oriol, Nobuhiro Tsukada, Kazutaka Sunami, Pierre Bories, Sumit Madan, Dorothee Semiond, Disa Yu, Sandrine Macé, Florence Suzan, Philippe Moreau; Subcutaneous Isatuximab Administration By an on-Body Delivery System (OBDS) in Combination with Pomalidomide and Dexamethasone in Patients with Relapsed/Refractory Multiple Myeloma: Phase 1b Expansion Study Results. Blood 2022; 140 (Supplement 1): 4412–4414. doi: https://doi.org/10.1182/blood-2022-166840

Of the background publications, the publication on TCD15484 study by Quach, et al. can be considered most relevant as it deals with the administration of Isatuximab subcutaneous. That interim analysis found

SC Isatuximab administered by OBDS shows a safety profile consistent with IV administration with no infusion reactions and excellent local tolerability

Isatuximab SC administration by OBDS is well tolerated

The IgPro20_1001 study by Wasserman, et al also found no new safety events associated with the OBDS (referenced as IWI, Investigational Wearable Injector, in the study) and a comparable safety profile to subcutaneous immunoglobulin administered by standard of care (syringe pump).

In combination, the two publications provide background evidence that the OBDS can successfully deliver doses of drug product into the subcutaneous biospace (as evidenced by the pK data cited in each study) without introducing any significant safety signals that could adversely impact the risk/benefit analysis for the underlying therapy.

2.0.1.2 Clinical investigation on SC administration of isatuximab with OBDS

In the current trial, SC injection of isatuximab will be performed using an on-body delivery system (OBDS) also known as the isatuximab wearable injector system. The OBDS has been developed in accordance with the Medical Device Regulation, (EU) 2017/745, and

compliance with the general safety and performance requirements (GSPR) defined in Annex I of Medical Device Regulation, (EU) 2017/745. As the OBDS is not marketed, precautions have been taken to protect the health and safety of patients. The OBDS for delivery of isatuximab is currently being studied in a Phase 1b clinical trial, TCD15484 (NCT04045795), a first in human study to assess the safety, PK and efficacy of SC isatuximab and IV isatuximab in combination with pomalidomide / dexamethasone, in patients with rrMM. 22 patients were recruited into the 1400 mg SC with the OBDS. As of January 2022, data analysis cut off, no patient discontinued treatment due to AE. Overall, data in the OBDS cohort are still maturing and no unexpected signal was observed in the 1400 mg SC isatuximab OBDS cohort. SAE occurrence in SC isatuximab cohorts was consistent with the IV isatuximab arm, with a lower occurrence in the SC isatuximab OBDS cohort that may be due to the shorter follow-up. Two patients, both in the 1400 mg SC isatuximab OBDS cohort, died during the on-treatment period. Causes of death were disease progression and AE (cardio-respiratory arrest, sudden death not considered to be related to the treatment). No injections were interrupted in the 1400 mg SC isatuximab OBDS cohort. There was no occurrence of infusion-related reactions in the SC isatuximab OBDS cohort. Five (22.7%) patients from the 1400 mg SC isatuximab OBDS cohort had at least one TEAE in the “injection site reaction” (ISR) as per customized medical dictionary for regulatory activities (MedDRA) query grouping with a total of 7 episodes (all CTCAE grade 1) out of 305 injections (2.3%): 5 injection site erythemas, 1 injection site hemorrhage, and 1 injection site induration.

As of 03 May 2022, 6 product complaints were reported out of ~400 SC isatuximab OBDS administrations in TCD15484: 1) one case of needle not retracting at end of injection and one possible delayed needle retraction. Additional controls are being implemented to improve needle retraction. 2) One use error leading to lockout of the device prior to use. Updates to instructions for use (see IB/IFU) are being applied. 3) Three prolonged SC injection times (>20 minutes). Investigation of these events are ongoing. None of the product complaints were associated with an AE/SAE. All isatuximab doses were fully delivered[27].

In addition to the clinical use of OBDS in TCD15484, a version of the on-body injector device has been used in clinical trials with other medicinal product(s). While the clinical trial studies specific information is not available to the sponsor of this clinical trial, the following summary is provided: the on-body injector has been used in two prior clinical trials. The total experience with the on-body injector technology in prior clinical trials

includes one study of 22 patient participants using approximately 270 on-body injectors to administer medicinal product and a second study of 8 healthy volunteers using approximately 54 on-body injectors to administer medicinal product. All prior clinical studies were conducted outside of Europe, with one clinical study being performed in the United States and one in Australia. Hence, the manufacturer Enable Injections, Inc (Cincinnati, OH, United States) is not permitted to disclose clinical trial results due to confidentiality and legal obligations with their other pharmaceutical partner(s). Enable Injections has confirmed that there were no device deficiencies that led to reportable events from the use of the OBDS in those clinical trials.

2.0.1.3 Rationale

SC administration of isatuximab via OBDS within approximately 10 minutes allows for a shorter duration of administration compared to the currently approved IV route, optimizing the convenience of administration and minimizing the time the participant spent in healthcare facility, which is particularly relevant in the context of the current COVID-19 pandemic and in line with recommendations of hematology/myeloma societies[28-30]. As a result, SC administration of drugs as compared to IV administration has the potential to significantly increase patient satisfaction as measured by patient reported outcomes (PROs), which is increasingly important for patients, their families and caregivers.

2.0.1.4 Benefit risk-assessment

SC administration can offer a number of benefits to both health care professionals (HCPs) and patients, including shorter injection duration and simplified user interface, and could be used by a qualified HCP in the outpatient setting. Risk management activities are being performed throughout the OBDS development to support the combined use of isatuximab drug product with the OBDS, resulting in risk analysis, risk mitigation and evaluation. With the implemented risk control measures, no unacceptable risks remain. The majority of all risks are acceptable. Some risks (5 in total) remain intermediate. All residual intermediate risks are grouped in the lowest occurrence category, but are associated with highest severity of possible harm. The five specific risks relate to either the administration of an incorrect medicine (instead of isatuximab SC; 2x) or to the potential exposure of a user to a contaminated needle (3rd party needle stick; 3x). However, the risk of administration of an incorrect medicine is a risk that exists for all medication and the implemented risk control measures are state of the art. Further,

the risk of exposure to a contaminated needle exists for all injection devices (including SC and IV) and the implemented risk control measures are state of the art. Investigators and clinical study staff should adhere to all warnings and instructions in accordance with the SC administration of isatuximab using the wearable injector system (OBDS) IB, IFU, pharmacy manual, and study protocol

2.1 Justifying the Selection of the Extended VRD Scheme

The clinical data of new second generation drugs with novel mechanisms of action represented by Daratumumab, Carfilzomib and Ixazomib have generated expectations about their possibilities in the first line of treatment because of their excellent results as rescue treatments in phase III clinical trials(10–14).

The key consideration to highlight is that a significant number of patients included in control arms of the clinical trials for Daratumumab, Carfilzomib and other drugs in RRMM had previously received these same drugs, a factor that clearly limited the rescue treatment´s effectiveness in control arms of these trials.

The conclusions drawn from the evidence in RRMM setting support our initial hypothesis, that is:

i) The efficacy of what are known as the second generation of new drugs in the first line of treatment, in general, is not "*a priori*" higher than the currently recognized standards of care in the treatment of NDMM patients, and in particular in patients who are candidates for HDT/ASCT. In NDMM, with the same duration and intensity of first-line treatment, KRD and VRD combinations seem equivalent.

ii) The apparent superiority of these new drugs compared to the current standards based on lenalidomide and bortezomib could be more related to their use in extended treatment models, rather than to a presumed greater face to face efficacy.

iii) The attempt to improve current results through the application of extended therapeutic models based on bortezomib and lenalidomide, is justified, since if they had a similar efficacy to that of the second generation of new drugs, the economic arguments that hampers access to new therapies become even more relevant, while at the same time it continues to makes sense to preserve therapeutic resources with proven efficacy in the still frequent settings of MM at relapse or progression.

Concerning efficacy, tolerance and toxicity of a first-line transplant program combined with extended VRD, a phase II trial conducted by the Atlanta Group(15) supports this approach. In this trial, VRD was administered following ASCT for up to 3 years as maintenance/consolidation regimen in genetic high risk or plasma cell leukemia. In this patient's population, the ≥CR rates with respect to post-ASCT status increased by 30%. With a median follow up of 26 months, no patient developed grade 3 or 4 neuropathy and no patient discontinued maintenance therapy due to adverse events. Dose modifications were required in 40% of patients.

2.2 Justifying the Early Redirecting of 1st Line to Alternative Therapy if Unfavorable Response Kinetic

In MM, the majority of relapse / progression events are first detected by serologic studies in subclinical phases. The figures are variable, and logically depend on the frequency of follow-up controls, but in up to 45% -85% of patients, recurrences/progressions are initially subclinical and tend to reproduce the behavior that MM exhibited at diagnosis. Rapid progressions are associated with poor prognosis data, such as plasma cell leukemia, extramedullary plasmacytomas or hypercalcemia, high LDH values, and other parameters of similar significance. Rapid proliferation patterns occur in 15-18% of patients. At the other extreme, approximately 10% of patients experience slow and indolent progressions with a MGUS-like profile and excellent prognosis(16–19).

The 1998 and 2006(20) IMWG criteria differentiated asymptomatic or biological relapses from symptomatic progressions with active MM data. Recommending therapeutic abstention in the first case, was probably adequate at that time. Since from the biological relapse, the average time to clinical progression was around 6 months(17), it was somehow considered that this time until the next treatment increased PFS and quality of life. On the other hand, the reproducibility of the concept Progression of Clinical Significance apparently strengthened the reproducibility of this clinical status in clinical trials.

From 1998 and subsequent revisions(20–22), the IMWG criteria consider the category Progressive Disease, associated with an increase of 25% from baseline in serum M-component (the absolute increase must be 0.5 g / dl and in the case of urine the absolute increase of Urine M-component must be 200 mg / 24 h), also more than 10% plasmatic cells in the bone marrow, and / or new lytic lesions or plasmacytomas. But even these

circumstances did not condition the start of treatment in all cases. This category would only be useful for the calculation of TTP, while only Clinical Relapse and “not all cases” of Progressive Disease would be treatment subsidiaries, an indication that requires markers of increasing disease and/or end organ dysfunction (CRAB features). The latest edition of the IMWG Criteria(21) considers two categories of relapse: from CR or from an MRD-ve status. The clinical utility of these categories of early relapse or early progression (both words are synonyms), according to IMWG, however, is limited to the calculation of disease-free survival, without conditioning the initiation of equally early rescue treatments. Considering that in the MM all the prognostic factors that affect the results of the first-line treatment, also the tumor mass, likewise condition the results of the rescue treatments, the position IMWG(21,23,24) previously commented and still in force, could, from our point of view, be improved if therapy is started earlier in the course of the relapse.

2.2.1 Relapse vs. Progressive Disease or Progression of Clinical Significance

To know the real significance of biological relapses before reaching Progressive Disease or Clinical Relapse in MM, we analyzed the clinical course of 333 of the 458 patients included in the PETHEMA sequential trials GEM12menos65 (Induction VRDx6, ASCT, Consolidation VRDx2) and subsequent GEM14MAIN (Lenalidomide-Dexa +/- Ixazomib maintenance) clinical trials, using the following criteria for relapse definition: 1) Increase of component M <0.5 g/dl, 2) In Bence Jones the increase should be less than 200 mg in urine/24h, 3) Increase below 50% in the difference between serum levels of affected and not affected free light chains (FLC), 4) Positive immunofixation after a previously negative immunofixation, 5) Detection of MRD after a previous MRD negative determination. Under the rules of these clinical trials, the emergence of Relapse criteria, without reaching the status of Progressive Disease or Clinical Recurrence, according to the IMWG definition, do not imply the patient leaving the study, allowing to analyze the patient’s subsequent clinical course. This analysis is ongoing, data are preliminary and have not been published.

Overall, 104 (31.2%) patients have relapsed within one of these two clinical trials: 71 cases within the GEM12menos65, and 33 patients during the GEM14MAIN. Of these, in the GEM14MAIN, 29 patients (88%), have progressed, which contrasts strikingly with only 29 patients (28%) of the 71 cases that during the course of the GEM12menos65

trial presented data of early relapse. These differences have an obvious explanation. The permanence in the clinical trial despite developing early biological relapse data has allowed us to observe that in 51 of the 53 patients who were transplanted in biological relapse after the 6th induction cycle (29 VGPR, 19 PR, 3 CR & MRD+), the relapse was aborted while, simultaneously, significant progress was observed in the quality of the responses (Post-ASCT response evaluations: PR 15, VGPR 13, 23 CR&MRD+, 10 CR&MRD-), a response that all of them, except 8 patients who subsequently progressed, have maintained or improved throughout maintenance. Altogether, 75 patients have progressed within both clinical trials. Of these, 56 (75%) patients had previously developed subclinical relapse criteria, with a median TTP of 4 months. In remaining 29 (25%) patients the progression emerged abruptly, with no previous data of biological progression in the evaluations that had been performed in each cycle from the moment of diagnosis.

2.2.2 The meaning of the evolution of MRD in the context of biological recurrence.

The central connotation of reaching a negative MRD response status has been firmly established having shown that the best prognosis of CR is conditioned by this parameter(25), especially when molecular or high sensitivity flow cytometry techniques allow the detection of pathological plasma cells at levels $<10^{-6}$, even $<10^{-7}$. Although the significance of the kinetics of the MRD evolution has not been sufficiently analyzed yet, the available data confirm that the evolution of the MRD is as important or more than the static determinations, and that the kinetics of the MRD must be taken into account in a therapeutic algorithm in which an early increase in MRD values must be considered a failure of the 1st line treatment. Indeed, an Italian study on patients included in the GIMEMA VEL-03-096 clinical trial(26), in which MRD studies were performed by PCR, showed that MRD reappearance heralds' relapse, with a TNT comparable to that of MRD persistence and a median between MRD reappearance and need for salvage treatment of 9 months. Likewise, a Spanish study(27) confirms that the reappearance of MRD was followed by a clinically significant progression in 86% of patients. These data suggest the usefulness of MRD monitoring in MM patients and the need for alternative treatments to revert to a negative MRD status and ensure prolonged responses. This vision is being gradually assumed by a growing number of authors, who consider that it is time to evaluate the potential advantages of incorporating the evolution of MRD into therapeutic decision making(28,29).

2.2.3 Justifying the evaluation of a second experimental arm, in which, with respect to the control lenalidomide is substituted for iberdomide.

Iberdomide (CC-220) is a novel CELMoD cereblon E3 ligase modulator that co-opts cereblon to enable enhanced degradation of target proteins, including Ikaros and Aiolos, with 20 times higher affinity versus LEN and POM(30). Iberdomide enhanced direct antimyeloma and immune stimulatory activity in preclinical models, is active in myeloma cell lines resistant to LEN and POM(31), and synergizes with BORT and DARA, demonstrating enhanced apoptosis and antibody-dependent cellular cytotoxicity(32).

Iberdomide has been evaluated in a phase I / II dose escalation trial, in combination with dexamethasone, the Overall Response rate was 32-33% in all subgroups. IBER + DEX has shown remarkable clinical activity and favorable tolerability in heavily pre-treated patients with relapsed refractory multiple myeloma (RRMM). Iberdomide + DEX had notable clinical activity in a population of heavily pre-treated RRMM patients, including patients with quad-class refractory disease. This regimen had a favorable safety profile and a recommend phase II dose (RP2D) of 1.6 mg for IBER in combination with DEX has been selected. Safety, efficacy, and mechanism of action support the clinical development of IBER as a foundation of combination therapy for MM. In combination with the administration of anti-CD38 monoclonal antibodies, the recommended dose was 1.6 mg/d. Recommended Dosage: 1.6 mg/day, in a continuous form in combination with dexamethasone(33).

In addition to the data previously presented, recent preclinical studies show that iberdomide can circumvent resistance to other immunomodulators, exhibits an immunostimulatory effect superior to lenalidomide or pomalidomide in combination with daratumumab or elotuzumab, and contributes to the recovery of NK cells and T cell proliferation after treatment with daratumumab or proteasome Inhibitors(34,35). This preclinical view is confirmed in very recent reported results in patients with RRMM showing that iberdomide in combination with bortezomib, carfilzomib or daratumumab(33,36) maintains relevant clinical efficacy, even in patients previously treated with other IMiDs and regardless of their previous refractory status to bortezomib or daratumumab. For this reason, the investigation of the clinical potential of the drug in more advanced stages is justified, replacing lenalidomide by iberdomide in the VRD

scheme. In our proposal, the experimental arm C introduces this unique change with respect to the control arm A.

2.2.4 Study Assumptions

- In the immediate future, the association of VRD with an anti CD38 monoclonal antibody will constitute the reference standard in the treatment of patients with newly diagnosed MM who are candidates for transplantation. Therefore, this model, including isatuximab, is used as a control arm in this proposed clinical trial.

- The efficacy of the VRD combination can improve its results if applied in extended regimens, as is the case with combinations based on the second generation of new drugs.

- The economic cost of extended therapeutic models based on a bortezomib and lenalidomide combination provides an unequalled access to treatment compared to second-generation drugs-based therapies.

- In NDMM patients with early relapse/unfavorable response kinetic (IMWG criteria), an Early Rescue Intervention (ERI) that redirects from 1st Line to alternative therapy (Iberdomide plus Isatuximab), restores the response and even induces MRD negativity. Therapeutic resources with proven efficacy are preserved in the still frequent setting of MM relapse or progression.

- In combination with an anti-CD38 monoclonal antibody (Isatuximab), Iberdomide can advantageously replace lenalidomide in combination with bortezomib and dexamethasone in NDMM patients.

3 OBJECTIVES AND ENDPOINTS

3.0 Primary Objective

To compare the efficacy of extended VRD[#] + ASCT plus ERI[&] (Arm B) vs. Isatuximab-VRD[#] + ASCT (Arm A) in terms of proportion of patients who are MRD-negative by next-generation flow cytometry (NGF) after 18 cycles + ASCT*

[#]VRD: Lenalidomide-Bortezomib-Dex;

[&]ERI: Early Rescue Intervention with Iberdomide-Isatuximab.

* 18 cycles would be:

- Arms A & C: Induction x4 + ASCT + Consolidation x2 + Continuous x12

- Arm B: Induction x6 + ASCT + Consolidation x2 + Extended x10

3.0.1 Primary Endpoints

1.1 Improvement in the percentage of patients MRD-negative in Arm A vs. Arm B when considering B1 endpoint.

B1 endpoint: VRD extended impact only, without considering ERI impact: any patient that require an ERI per protocol before finishing the 18 cycles + ASCT, will be considered MRD-positive at that time, independently of the ERI impact.

1.2 Non-inferiority in the percentage of patients MRD-negative in Arm A vs. Arm B when considering B2 endpoint.

B2 endpoint: VRD extended plus ERI impact: any patient who achieves MRD negativity with ERI before finishing the 18 cycles + ASCT will be considered MRD-negative at that time.

3.1 Secondary Objectives

Safety secondary objective 2.1:

To assess the safety of Isatuximab-VID* + ASCT (Arm C).

*VID: Iberdomide-Bortezomib-Dex.

 Note: Due to the experimental nature of the Arm C combination Isatuximab-Iberdomide-Bortezomib-Dex, a safety analysis will be performed with the first 5 patients included in each arm, a second one when 25 patients in each arm have completed 3 cycles of therapy, and subsequent ones might be required if considered by the Primary-Investigator, Co-Primary-Investigator, Co-coordinators and Co-Investigator (hereinafter "Coordinators") .

Key secondary efficacy objective 2.2:

To explore preliminary efficacy comparison of Arm C (Isatuximab-VID* + ASCT) vs. Arm A (Isatuximab-VRD# + ASCT) in terms of proportion of patients who reach NGF MRD-negative after Induction (4 cycles) + ASCT+ consolidation (two cycles) and 12 continuous treatment cycles.

Key secondary efficacy objective 2.3:

To explore preliminary efficacy comparison of Arm C (Isatuximab- VID*ASCT) vs. Arm B (VRD extended plus ERI) in terms of proportion of patients who reach NGF MRD-negative.

Note: The interim safety and efficacy assessment will be the basis for adapting elements of the trial design and qualifying key secondary efficacy objectives 2.2 and 2.3 as confirmatory.

Other secondary objectives

2.4 Progression-Free Survival (PFS)

2.5 Overall Response Rate (ORR): PR or better (PR, VGPR, CR, sCR).

2.6 Complete Response Rate (CRR): CR or better (CR, sCR).

2.7 Time to Response (TTR), among participants who achieve confirmed PR or better.

2.8 Duration of Response (DoR) among participants who achieved PR or better.

2.9 PFS2

2.10 Overall Survival (OS)

3.1.1 Key Secondary Endpoints

2.2 Improvement in the percentage of patients MRD-negative in Arm C vs. Arm A

2.3 Improvement in the percentage of patients MRD-negative in Arm C vs. Arm B (B2)

For objectives 2.4 to 2.10 the IMWG response criteria or well-established chronological definitions

3.2 Correlative or translational studies

The general objective of the correlative studies is to generate biomarkers and datasets for improving outcomes in MM based on precision medicine and better understanding of tumor and immune cell biology.

Specific objectives of the correlative studies:

3.2.1 Identify genetic risk groups of patients that will benefit the most or that are primary refractory to the treatment schema proposed in the clinical trial.

3.2.2 Evaluate the clinical significance of MRD status using next-generation techniques and define undetectable MRD as the endpoint of the treatment schema proposed.

3.2.3 Understand MRD resistance by using bulk- and single-cell sequencing techniques to compare the genomic signature of matched diagnostic vs. MRD clones isolated by next-generation flow cytometry.

3.3.4 Create an algorithm based on patients' immune profile and clinical data that enables individualized prediction of outcome based on simultaneous MRD and immune monitoring.

3.3.5 Genetic characterization of tumor cells at diagnosis.

3.3.5 Molecular clonal structure of tumor cells at diagnosis and after therapy.

3.3.6 Molecular characterization of MRD cells persisting to different regimens.

3.3.7 MRD monitoring based on next-generation sequencing (NGS).

3.3.8 MRD monitoring based on next-generation flow (NGF) cytometry.

3.3.9 Next-generation immune monitoring.

3.3.10 Response assessment in patients with IgG kappa multiple myeloma can be misinterpreted due to the interference between the patients M protein and Isatuximab, especially when reaching very good partial response, which is expected in the majority of them. To resolve this interference, we plan to use mass spectrometry, a method able to discriminate between the patient monoclonal protein and Isatuximab and therefore to properly assign treatment responses.

3.3.11 Evaluate the incidence of infections and their characteristics in the different arms of treatment (category, severity, the moment in which they occur, etc.) as well as other related data like antibiotic prophylaxis, previous vaccination, etc.

4 TRIAL DESIGN

This is a Phase III open-label, 3-arm, parallel, randomized, controlled trial.

The allocation ratio 1:1:1 and outcome assessment are blind to group allocation.

The trial tests two co-primary hypothesis (a superiority hypothesis related to primary end-point B1; and a non-inferiority hypothesis related to primary end-point B2); and two key secondary superiority hypotheses.

The trial is adaptive to the extent that some design elements may be requalified following the early or interim safety and efficacy analyses.

Patients will receive treatment until unacceptable toxicity, disease progression, patient withdrawal, loss to follow-up, end of study or death.

Participants will be evaluated for study eligibility per protocol as defined in the Inclusion/Exclusion criteria (Section 7). The screening period takes place 21 days prior to the baseline visit.

4.0 Study Structure

- Arm A: (Isatuximab-VRD + ASCT)
- Arm B: (VRD extended + ASCT plus ERI)
- Arm C: (Isatuximab-VID + ASCT)

Dose interruptions or reductions may be required to address potential drug-associated toxicities. Guidelines for dose modification and dose delays for the three study drugs will be provided in Section 10.

Safety and disease assessments will be performed regularly according to Schedule of Activities (SoA, Section 11).

After the treatment period is complete, an end-of-treatment visit will be conducted at least 30 days after receiving the last dose of the study drug. Patients will be followed after this time point in the following circumstances: until the resolution of AEs; follow up for progression in case of

discontinuing treatment due to a reason different to disease progression; for other events of special interest such as SPM development; or for survival status.

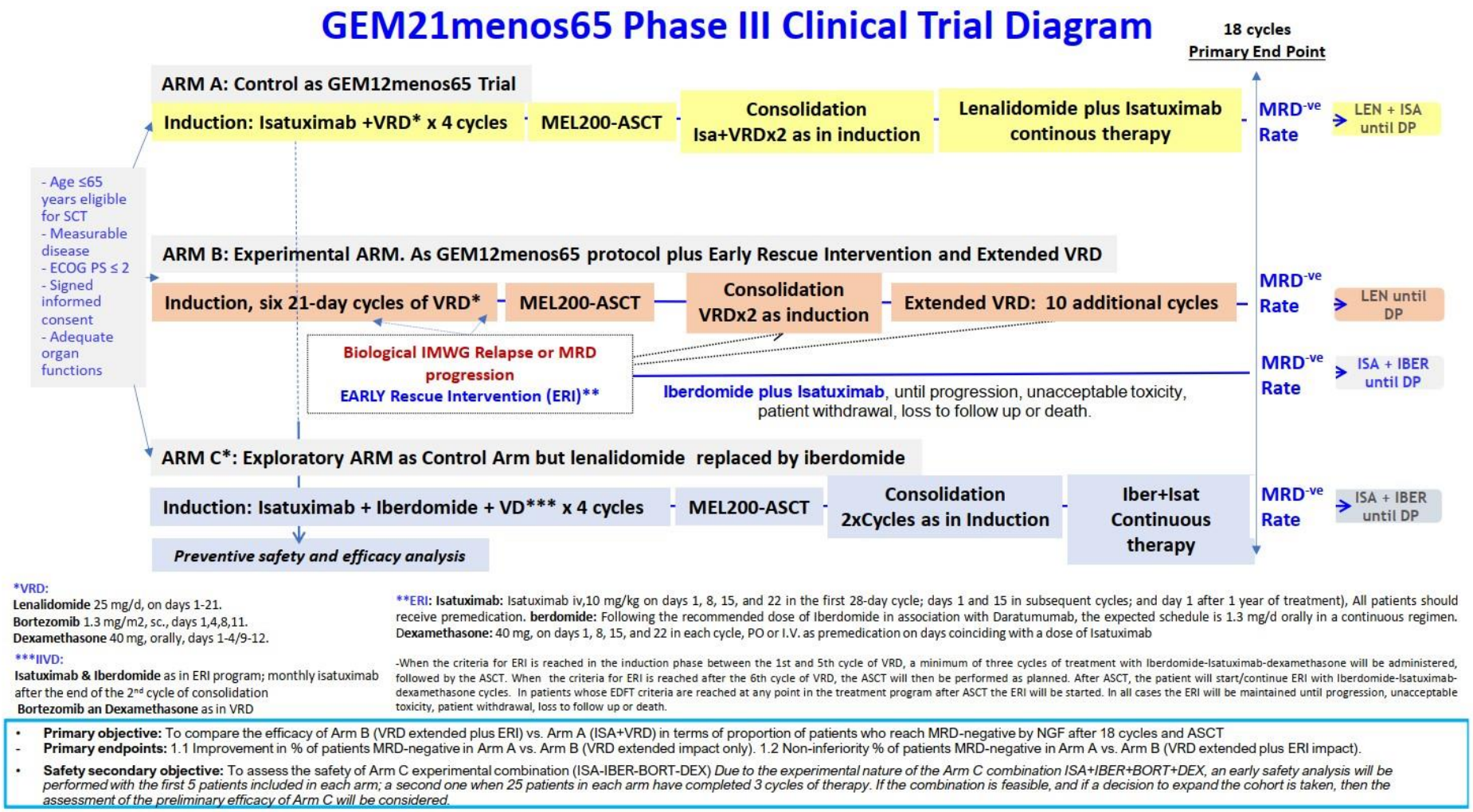


Figure 1 Study Schema

5 STATISTICAL CONSIDERATIONS

5.0 Introduction

The section describes the statistical methods to be used to analyze the safety and efficacy as described above in the objectives. These methods may be revised and updated due to reasons such as regulatory requirements or need for further clarifications. The analysis plan will be documented in a formal statistical analysis plan (SAP) that will be written prior to the formal interim analysis and must be finalized before the database lock. The SAP will include how the variables will be derived, how missing data will be handled, how censoring procedures will be applied, how multiplicity adjustments and sensitivity analyses will be performed as well as the details on the statistical methods to be used for safety and efficacy analysis. The final clinical study report will discuss deviations from the SAP, if any.

5.1 Study Populations

For efficacy analyses, two populations will be used, Intent to treat (IIT) and per protocol (PP) populations. The IIT set will include all study participants to whom study treatment scheme has been assigned by randomization. Patients will be analyzed according to the treatment they are randomized to receive. The PP population is a subset of the ITT population. The PP set will include all study participants with completed baseline evaluation of disease status, who received at least one complete cycle of treatment and one follow-up evaluation of disease status and that did not have major protocol violations. All decisions to exclude patients from the PP population will be made before starting the final analyses.

Study variables will be analyzed/summarized based on the IIT population as well as PP population as appropriate. The primary analysis will be performed on the ITT population. The analysis will be repeated for the PP population as a supportive analysis.

Patients will be evaluable for safety if they have received at least one total or partial infusion of any of the study drugs. Subjects who have been randomized and did not take at least 1 dose of study medication will not be included in the safety set. For the safety analysis, subjects will be analyzed according to the study treatment they actually received.

5.2 Statistical Methods

In general, summary tabulations will be presented by treatment arm and will display descriptive statistics such as the number of observations, mean, standard deviation, median, minimum, and maximum for continuous variables, and the number and percent per category for categorical data. The Kaplan-Meier survival curves and 25th, 50th (median), and 75th percentiles will be provided along with their 2-sided 95% CIs for time-to-event data.

For the primary and key secondary efficacy objectives, we will estimate the Minimal Residual Disease negative risk ratio between study arms and build 95% confidence intervals.

Progression free survival, time to progression and overall survival curves will be estimated by Kaplan-Meier method. Percentile 25, median and 75th percentile PFS, TTP, OS and corresponding 95% confidence interval will be reported by treatment arm. Log-rank test will be used by to compare PFS, TTP and OS curves across two treatment arms.

Further details of the main outcomes and other secondary analyses will be provided in the statistical analysis plan (SAP), including hierarchical hypotheses testing, pre-specified subgroup and sensitivity analyses.

5.3 Sample Size Calculation, Justification and Study Power

Number of participants:

This is a Phase III open-label 3-arm parallel randomized controlled trial. With the predetermined sample size, this clinical trial will address the primary objective (with two different end-points in Arm B, B1 and B2) and the key secondary efficacy objectives 2.2 and 2.3 which respectively correspond to study hypotheses H1, H2, H3, H4.

A sample size of 160 participants per arm, overall sample size 480 participants, is required for testing primary hypotheses H1 and H2. The study sample size is large enough to test key secondary efficacy hypotheses H3 and H4. The first 25 out of the 160 participants included in arm C will be part of the initial safety analysis set and they could be replaced in case it was required to implement a significant change in the therapeutic schema after that safety analysis. Justification of the sample size and power for each study hypothesis is detailed below.

Justification:

The numerical assumptions used for the sample size calculations are the following:

- In Arm A, MRD-negative rate after 18 cycles + ASCT is expected to be 70%;

- in Arm B, MRD-negative rate after 18 cycles + ASCT is expected to be 55% for endpoint B1 (based on a 15% relative increase from the 48% obtained in the previous GEM trial); and 68.75% for outcome B2 (based on a 25% relative increase compared to B1);

- in Arm C, MRD-negative rate after 18 cycles + ASCT is expected to be 84% (based on a 20% relative increase compared to arm A);

Primary hypotheses H1 and H2 and key secondary efficacy hypotheses H3 and H4 will be tested after 18 treatment cycles excluding autologous transplantation. In treatment arms A and C, the 18 cycles include 4 induction cycles, 2 consolidation cycles, and 12 cycles of continuous treatment. Treatment arm B includes 18 treatment cycles with VRD.

Hypothesis testing and study power:

- H1, Primary superiority hypothesis for end-point B1 vs. A: MRD negative rate in Arm A is superior to Arm B considering B1 end-point.

B1 endpoint: VRD extended impact only, without considering ERI impact: any clinical situation that require early rescue intervention per protocol before finishing the 18 cycles + ASCT will always be considered MRD-positive at that time, independently of the Early Rescue Intervention impact.

Sample size is large enough to estimating a MRD-negative risk ratio of 1.27 with a 95% confidence level and 80% power;

- H2, Primary non-inferiority hypothesis for end-point B2 vs. A: MRD negative rate in Arm B is not inferior to Arm A considering B2 end-point.

B2 endpoint: VRD extended plus ERI impact: any patient who achieves MRD negativity with ERI before finishing the 18 cycles + ASCT will be considered MRD-negative at that time.

Sample size is large enough to be 80% sure that the 95% confidence interval will exclude a difference of more than 15%; allowing for 10% lost to follow-up.

- H3, Secondary superiority hypothesis C vs. A: MRD negative rate in Arm C is superior to Arm A;

Sample size is large enough to estimate a MRD-negative risk ratio of 1.20 with a 95% confidence level and 80% power; allowing for 10% lost to follow-up.

- H4, Secondary superiority hypothesis C vs. B2: MRD negative rate in Arm C is superior to Arm B, considering B2 end-point.

Sample size is large enough to estimate a MRD-negative risk ratio of 1.22 with a 95% confidence level and 80% power; allowing for 10% Lost To Follow-Up.

Multiple testing adjustment:

This is a parallel multi-arm phase III study with multiple primary endpoints designed to maximize efficiency while minimizing the number of participants that need to be randomized.

The study objectives, design, planned analysis and sample size calculation are clearly presented in the protocol. The comparisons will be conducted in parallel, with no prioritized sequence of tests, inference will made in each comparison. The interpretation of the results of one comparison will have no direct bearing on the interpretation of the others. Investigators will transparently report all comparisons undertaken, planned and unplanned, and provide appropriate interpretations of the results. In such instance, many authors consider no adjustment for multiple comparisons is deemed appropriate because such adjustments would not be necessary if the interventions were compared in separate trials.

Nonetheless, and in agreement with standard recommendations for multi-arm clinical trials, we will consider tightly controlling the family-wise error rate and minimize the chance of a false positive conclusion. The SAP will detail the standard procedures to be implemented to control the Type I error rate at two-sided alpha of 0.05.

5.4 Level of Significance

We will use a two-sided binomial test at significance level of 0.05 to estimate superiority between two arms. For the non-inferiority hypothesis, we will use a one-sided significance level of 0.025.

5.5 Procedure for Accounting for Missing Data

Details on the approach to handling missing data in the efficacy and safety analyses will be provided in the SAP. In general, missing data will be treated as missing and no data substitutions will be applied, unless otherwise specified. All available efficacy and safety data will be included in data listings and tabulations. Data that are potentially spurious or erroneous will be examined according to standard data management operating procedures.

Subjects with missing data on MRD status will be considered positive for efficacy assessment. For time to event (PFS) analyses, a censoring rule will be developed and sensitivity analyses will be performed. For time to event (OS) analyses, follow-up time will be censored at last date known alive. Patients with no efficacy evaluations will be censored at time 0. Subjects with missing data on Response status will be considered non-responders for efficacy assessment. In safety analyses, missing dates will be imputed per the rules outlined in the SAP. Missing severities of AEs will not be imputed and will be considered missing in any tabulations of AE severity. AEs with completely missing dates will be assumed to be treatment-emergent.

Other missing data imputation methods, such as multiple imputation methods, may be explored as sensitivity analyses.

5.6 Interim Analysis

One interim efficacy analysis is planned in this Phase 3 study. The SAP will specify the O'Brien-Fleming boundaries and stopping rules with approximately half of MRD outcome information. The interim efficacy analysis will be performed when 50% of randomized patients as planned in the sample size calculations will reach post-consolidation. In operational terms, recruitment of 225 patients will take 18 months approx., induction 6M, transplant 6M, consolidation 2M which adds up to 32M from start of recruitment to time for interim efficacy analysis. The interim efficacy analysis

will address the two hypothesis of the primary objective of the trial, a non-inferiority hypothesis and a superiority hypothesis.

Non-inferiority. The non-inferiority comparison in this trial is motivated by the assumption that the experimental therapy (arm B, endpoint B2) has benefits compared to the standard therapy (arm A). The B2 vs A comparison in this trial aims to provide conclusive evidence that substituting the experimental therapy for the standard therapy will not decrease efficacy by an unacceptable amount. It is recommended to plan for interim analyses for harm or futility, so as to minimize patient exposure to inferior therapies. Nonetheless, it is usually recommended (DOI: 10.1093/annonc/mdx788) that unless early evidence strongly suggests that the experimental arm is superior to the standard of care, early stopping with the conclusion of non-inferiority may be counterproductive. We will consider the implementation of the Earliest Information Time (EIT) to improve the timing of the interim analysis (when one-third of the required number of events occur in the experimental arm, i.e. 225/3=75).

The first results on MRD-negative rates for each trial arm will be assessed, including the first ever efficacy estimates with the Isatuximab-Iberdomide-bortezomib-dexamethasone quadruplet (arm C). In addition, the interim analysis will provide response recovery rates with and without rescue intervention impact in arm B, and preliminary data on response course profiles (MRD-ve) for the three therapy programs, and further exploratory secondary objectives.

At this point, and together with the interim efficacy analysis, an interim safety analysis will be performed. The overall interim safety and efficacy assessment will be the basis for adapting elements of the trial design and qualifying key secondary efficacy objectives 2.2 and 2.3 as confirmatory.

6 PATIENT SAFETY AND EFFICACY ASSESSMENTS

To be evaluable for efficacy, patients must have completed baseline evaluation of disease status, received at least one complete cycle of treatment and one follow-up evaluation of disease status.

Assessment of efficacy will be done using the IMWG Criteria. For this purpose, serum monoclonal protein value and urine light chain excretion (and FLC when required) will be determined in the screening visit, prior to the initiation of each cycle and in the end of treatment visit. For the evaluation of the complete response, serum and urine immunofixation, plasma cell infiltration in bone marrow and serum FLC ratio will also be determined.

Patients will be evaluable for safety if they have received at least one total or partial infusion of any of the study drugs.

The safety variables described in the safety analyses will include Adverse Events (to be graded in severity according to the guidelines outlined in the National Cancer Institute [NCI] Common Terminology Criteria for Adverse Events [CTCAE] version 5), physical examination, safety clinical laboratory variables (including hematology, serum chemistry, urinalysis, renal and hepatic function), Eastern Cooperative Oncology Group Performance Status (ECOG /PS) and any other safety variables that may be required. Subjects will be monitored closely by appropriately trained personnel.

Safety will be evaluated using clinical examinations, which will comprise vital signs analysis, clinical assessment of AEs, changes in laboratory parameters (hematological and biochemical, including liver function tests) and any other analyses that may be considered necessary. It will be based on the evaluation of the incidence of AEs and its relationship with the Investigational Product (IP) and severity as graded by the NCI-Common Terminology Criteria for Adverse Events (CTCAE) v. 5.0.

Participants will be followed until all AEs related to the treatment have recovered to grade ≤ 1 or have reached baseline or have been considered irreversible by the investigator. The number and percentage of patients that experience one or more AEs will be summarized by treatment group, their relationship to the study and their severity.

6.0 Initial Safety analysis

The experimental nature of the combination Isatuximab-VID requires an initial safety analysis prior to confirming patient inclusion for the efficacy comparisons involving arm C. In addition, an interim safety analysis will be performed in parallel with the interim efficacy analysis.

The safety analyses will be performed on the cohort of patients assigned in parallel, synchronously, to the three randomization arms and will focus, particularly, on the description of Arm C with Arms A and B as a reference. Two initial safety analyses are planned

The purpose of the 1st safety analysis is to address the secondary objective 2.1 related to the combination Isatuximab-VID safety. The output of this safety analysis will provide supporting evidence for the Coordinators decision on the continuation of recruitment in Arm C. This initial safety analysis will be scheduled after completion of the first induction cycle of the first 5 patients of each arm. Recruitment will be temporarily interrupted when five patients in each arm have been included until all these patients have completed the 1st cycle of treatment and safety of this initial cohort has been evaluated. A second initial safety analysis will be scheduled after the first 25 patients of each arm have completed at least 3 cycles of treatment (9 months from first inclusion, approx.). At this point, recruitment must not necessarily be stopped, unless it is decided by the Coordinators.

The Coordinators will assess the frequency and severity of the safety findings in these initial cohort of patients (5 per arm) and would decide either to continue recruitment without changes, to implement any modification in the therapeutic schema, or to stop recruitment in Arm C. Moreover, the Coordinators could also recommend extending the safety analysis by another 5-10 patients in each study arm in the 1st safety cohort.

Similar situation would apply for the 2nd initial safety analysis once 25 patients have completed 3 cycles in each arm. However, as indicated, no recruitment interruption would be required at this time. In case the Coordinators consider it necessary, the release of the safety report could wait until having long term safety results after the 6th induction cycle (18 months from first inclusion, approx.).

If the Coordinators report is positive and it does not suggest a modification of the Arm C combination (scenario 1), the cohort of patients described in the safety analysis will be part of the efficacy cohort. If the Coordinators report is positive but it does suggest a modification of the Arm C combination, the cohort of patients described in the safety analysis might not be part of the efficacy cohort if the

Coordinators considers that efficacy results could be influenced by the treatment modification. Consequently, in addition to the 480 patients calculated for the efficacy assessment, up to 25 more patients may be required for completing Arm C for the safety cohort leading to a total sample size of 480 or up to 505 patients.

An unfavorable safety report regarding Arm C could motivate the discontinuation of recruitment in arm C and its continuation in Arm A and B of the clinical trial to address the primary objective.

Following a positive safety report that include suggestions for modifying the Arm C combination, the allocation ratio will be adjusted from 1:1:1 to 1:1:2 to compensate for differential inclusion in view of efficacy analysis. Once balance between the three arms will be achieved, the allocation ratio will be readjusted from 1:1:2 to 1:1:1. If the above mentioned report of the Coordinators was to be based on 20 patients, the balance between arms would be reached when 40 patients would be recruited in each arm. If the positive report of the Coordinators was to be based on 50 patients, the balance between arms would be reached when 100 patients would be recruited in each arm.

7 PARTICIPANT SELECTION

7.0 Inclusion Criteria

a. Patient is, in the investigator's opinion, willing and able to comply with the protocol requirements.
b. Patient must be able to understand the study procedures.
c. Patient has given voluntary written informed consent before performance of any study-related procedure non part of normal medical care, with the understanding that consent may be withdrawn by the patient at any time without prejudice to their future medical care.
d. Newly diagnosed multiple myeloma patient who requires start active treatment according to the 2014 IMWG criteria, namely clonal bone marrow plasma cells ≥10% or biopsy-proven bony or extramedullary plasmacytoma and any one or more of the following myeloma defining events: evidence of end organ damage that can be attributed to the underlying plasma cell proliferative disorder, specifically: Hypercalcaemia, Anaemia, Renal Insufficiency, or Bone lesions (one or more osteolytic lesions on skeletal radiography, CT, or PET-CT), and any one or more of the following biomarkers: clonal BMPC% ≥60%, i/u free light ratio ≥100 or > 1 focal lesions on MRI or PET/CT) [Lancet Oncol. 2014;15(12): e538-e548]
e. Patient must have a measurable secretory disease defined as either serum monoclonal protein of ≥ 0,5 g/dl or urine monoclonal (light chain) protein ≥ 200 mg/24 h. For patients whose disease is only measurable by serum FLC, the involved FLC should be ≥ 10mg/dL (100 mg/L), with an abnormal serum FLC ratio.
f. Patient must have an Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 2.
g. Patient must be ≤ 65 years of age.
h. Patient must have adequate organ function, defined as follows (**Table 2**):

Table 2 Laboratory values that define adequate organ function for inclusion in study.

System	Laboratory Values
Hematologic	
Absolute neutrophil count (ANC) [a]	≥1.0 X 10^9/L without G-CSF use in the prior 7 days
Hemoglobin	≥8.0 g/dL (prior red blood cell (RBC) transfusion or recombinant human erythropoietin use is permitted)
Platelets	≥ 75 x 10^9/L in participants in whom <50% of bone marrow nucleated cells are plasma cells and ≥ 50×10^9/L in participants in whom ≥50% of bone marrow nucleated cells are plasma cells (without transfusion support or thrombopoietin receptor agonist within 7 days before the laboratory test)
Metabolic	
Calcium	Corrected serum calcium ≤13.5 mg/dL (≤3.4 mmol/L); or free ionized calcium ≤6.5 mg/dL (≤1.6 mmol/L) (See APPENDIX 8).
Hepatic	
Total bilirubin	≤2 X ULN
ALT	≤2.5 X ULN
AST	≤2.5 X ULN
Renal	
eGFR[a]	≥40 mL/min/ 1.73 m^2
Cardiac	
LVEF (echo)	≥ 50%

[a]As calculated by Modified Diet in Renal Disease (MDRD) formula

ALT=alanine aminotransferase; AST=aspartate aminotransferase; eGFR=estimated glomerular filtration rate; RBC=red blood cell; ULN=upper limit of normal

i. Female patient: contraceptive use should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies (Please refer to **APPENDIX 9** and **APPENDIX 11**).

A female patient is eligible to participate if she is not pregnant or breastfeeding, and at least one of the following conditions applies:

- Is not a woman of childbearing potential (WOCBP), i.e., fertile, following menarche and until becoming post-menopausal unless permanently sterile. Permanent sterilization methods include hysterectomy, bilateral salpingectomy and bilateral oophorectomy

  OR

- Is a WOCBP and

  - ✓ She understands the potential teratogenic risk to the unborn child
  - ✓ She understands the need for effective contraception as stated in the protocol, without interruption, 28 days before starting study treatment, throughout the entire duration of study treatment, during dose interruptions and for at least 28 days after the last dose of study treatment.
  - ✓ She understands and agrees to inform the Investigator if a change or stop of method of contraception is needed.
  - ✓ She must be capable of complying with effective contraceptive measures.
  - ✓ She is informed and understands the potential consequences of pregnancy and the need to notify her study doctor immediately if there is a risk of pregnancy.
  - ✓ She understands the need to commence study treatment as soon as it is dispensed following a negative pregnancy test.
  - ✓ She understands and accepts the need to undergo pregnancy testing based on the frequency outlined in this plan (**APPENDIX 9** and **APPENDIX 11**) and in the Informed Consent (provided separately) .
  - ✓ She acknowledges she understands the hazards iberdomide or lenalidomide can cause to an unborn fetus and the necessary precautions associated with the use of study drugs.

The Investigator must ensure that a WOCBP:

Complies with the conditions of the pregnancy prevention plan, including confirmation that she has an adequate level of understanding

Acknowledges the aforementioned requirements.

A WOCBP must have a negative highly sensitive serum pregnancy test (as required by local regulations) within 72 hours before the first dose of study drug.

Nonchildbearing potential is defined as follows (by other than medical reasons):

- Has not achieved menarche at some point.
- Has undergone a hysterectomy or bilateral oophorectomy.
- Has been naturally postmenopausal (amenorrhea following cancer therapy does not rule out childbearing potential) for at least 24 consecutive months (ie, has had menses at any time in the preceding 24 consecutive months).

j. Male patient: contraceptive use should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies.

Male patient is eligible to participate if he agrees to the following from the time of first dose of study until 6 months after the last dose of iberdomide or lenalidomide to allow for clearance of any altered sperm (Please refer to **APPENDIX 9**):

- Understand the potential teratogenic risk if engaged in sexual activity with a pregnant female or a WOCBP
- Understand the need for the use of a condom even if he has had a vasectomy, if engaged in sexual activity with a pregnant female or a FCBP
- Understand the potential teratogenic risk, so the subject should not donate semen or sperm.. Understand that the effects on fertility are currently unknown, therefore all family planning options and/or alternatives should be thoroughly discussed with the study doctor prior to receiving iberdomide.

k. All prior treatment-related toxicities (defined by National Cancer Institute- Common Toxicity Criteria for Adverse Events (NCI-CTCAE), version 5.0 must be ≤ Grade 1 at the time of enrolment except for alopecia.

7.1 Exclusion Criteria

a. Patient has a diagnosis of primary amyloidosis, monoclonal gammopathy of undetermined significance (MGUS), smoldering multiple myeloma (SMM), plasma cell leukemia or active POEMS syndrome at the time of screening.
b. Patient has had clinical evidence of central nervous system (CNS) or pulmonary leukostasis, disseminated intravascular coagulation, or CNS multiple myeloma.
c. Prior history of malignancies, other than multiple myeloma (except for basal or squamous cell carcinoma of the skin, carcinoma in situ of the cervix or the breast), unless the patient has been free of the disease for ≥ 5 years.
d. Any serious medical condition that places the subject at an unacceptable risk if he or she participates in this study; subjects with conditions requiring chronic steroid or immunosuppressive treatment, such as rheumatoid arthritis, multiple sclerosis and/or lupus, that likely need additional steroid or immunosuppressive treatments in addition to the study treatment.
e. Pregnant or breastfeeding females.
f. Men and women of reproductive potential who are not using effective contraceptive methods (double barrier method, intrauterine device, oral contraception).
g. Patient is simultaneously enrolled in other interventional clinical trial.
h. Patient has used an investigational drug within 28 days or five half-lives, whichever is longer, preceding the first dose of study drug.
i. Patient must not have received prior radiotherapy (except localized palliative radiotherapy for pain, palliation or fracture) within 2 weeks of start of study therapy. Participants must have recovered from all radiation-related toxicities, not require corticosteroids, and not have had radiation pneumonitis.
j. Major surgery (except kyphoplasty) ≤ 4 weeks prior to initiating protocol therapy.
k. Patient has peripheral neuropathy or neuropathic pain grade 1 with pain or ≥2, as defined by the National Cancer Institute Terminology Criteria for Adverse Events (NCI CTCAE) Version 5.0 (APPENDIX 5).
l. Patient evidence of cardiovascular risk including any of the following:
    - Myocardial infarction within 6 months before randomization, or an unstable or uncontrolled disease/condition related to or affecting cardiac function (eg, unstable angina, congestive heart failure, New York Heart Association Class III-IV).

- Uncontrolled cardiac arrhythmia.
- Screening 12-lead ECG showing a baseline interval QTcF> 470 msec (exception: subjects with pacemaker).
- Patients with uncontrolled hypertension.

m. Patients who have current unstable liver or biliary disease defined by the presence of ascites, encephalopathy, coagulopathy, hypoalbuminemia, esophageal or gastric varices, persistent jaundice, or cirrhosis. **Note:** Stable chronic liver disease (including Gilbert's syndrome or asymptomatic gallstones) or hepatobiliary involvement of malignancy is acceptable if otherwise meets entry criteria

n. Presence of active renal condition (infection, requirement for dialysis or any other condition that could affect patient's safety). Participants with isolated proteinuria resulting from MM are eligible, provided they fulfil inclusion criteria.

o. Evidence of active mucosal or internal bleeding.

p. Any serious medical condition or psychiatric illness that would interfere in understanding of the informed consent form.

q. Uncontrolled endocrine diseases (i.e. diabetes mellitus, hypothyroidism or hyperthyroidism) (i.e. requiring relevant changes in medication within the last month, or hospital admission within the last 3 months).

r. Patient with acute diffuse infiltrative pulmonary disease and/or pericardial disease.

s. Patient with severe chronic obstructive pulmonary disease (COPD) or asthma with forced expiratory volume in the first minute (FEV1) less than 50%.

t. History of interstitial lung disease or ongoing interstitial lung disease.

u. Subject has gastrointestinal disease that may significantly alter the absorption of iberdomide and/or other oral study treatment.

v. Patient has an active infection requiring systemic antibiotic, antiviral, or antifungal treatment at the time of starting treatment.

w. Patient has known HIV infection

x. Patient seropositive for hepatitis B defined by a positive test for hepatitis B surface antigen (HBsAg). Subjects with resolved infection (i.e., subjects who are HBsAg negative but positive for antibodies to hepatitis B core antigen [anti-HBc] and/or antibodies to hepatitis B surface antigen [anti-HBs]) must be screened using real-time PCR measurement of hepatitis

B virus (HBV) deoxyribonucleic acid (DNA) levels. Subjects who were positive only for HBsAg and had a valid vaccination certificate for hepatitis B could be included in the stutjdy, without the need to determine an undetectable HBV viral load.

y. Patient has positive hepatitis C antibody test result or positive hepatitis C RNA test result at screening or within 3 months prior to first dose of study treatment.

   Note: Participants with positive Hepatitis C antibody due to prior resolved disease can be enrolled, only if a confirmatory negative Hepatitis C RNA test is obtained.

z. Note: Hepatitis RNA testing is optional and participants with negative Hepatitis C antibody test are not required Patient require concurrent administration of a strong inhibitor or inducer of cytochrome P450 (CYP3A4/5) (including within 14 days of initiating study treatment)

aa. Patient has a known immediate or delayed hypersensitivity reaction or idiosyncratic reactions to iberdomide or drugs chemically related to iberdomide.

bb. Patient has a known immediate or delayed hypersensitivity reaction or idiosyncratic reactions to isatuximab or drugs chemically related to isatuximab, hypersensitivity reactions, or idiosyncratic reactions to other molecular antibodies.

cc. Patient has a known immediate or delayed hypersensitivity reaction or idiosyncratic reactions to lenalidomide or dexamethasone or drugs chemically related to lenalidomide or dexamethasone.

7.2 Screen Failures

Screen failures are defined as participants who consent to participate in the clinical study but do not receive the experimental treatment. A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the Consolidated Standards of Reporting Trials (CONSORT) publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, eligibility criteria, and any serious adverse events (SAEs).

Individuals identified as Screen Failures may be rescreened if the failure was based on elements of eligibility that may change, e.g., laboratory test results. Rescreening of a participant more than once requires discussion with the Clinical Coordinators. Rescreened participants must be assigned a new unique participant number that is different from the initial number.

7.3 Lifestyle Considerations

- Administration of live or live-attenuated vaccines are contraindicated 30 days prior to the first dose of study treatment and while receiving study treatment. Killed or inactivated vaccines may be administered, however, the response to such vaccines cannot be predicted.

No other lifestyle restrictions are required for participants in this study.

8 TREATMENT PLAN

This conceptual trial phase III open-label 3-arm parallel randomized controlled trial will include a sample of 160 patients in each of the three arms.

8.0 Method of Treatment Assignment

Eligible patients shall be enrolled in the study under the supervision of the investigator or designated sub-investigators. If possible, patients will receive treatment in an outpatient basis.

Cycles will be of 4 weeks of duration (28 days).

All patients will receive treatment until unacceptable toxicity, disease progression, patient withdrawal, loss of follow up, end of study or death.

Patients will be randomized to:

8.0.1 CONTROL ARM (A)

Includes the sequence: Induction: VRD+Isatuximab x4- ASCT- Consolidation: VRD+Isatuximab X2 followed by continuous treatment with cycles of Lenalidomide plus Isatuximab During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab.

1. INDUCTION: Isatuximab + VRD, 4 cycles. In this period, following established procedures, and preferably coinciding with the 3rd / 4th cycle, the collection and cryopreservation of hemopoietic precursors is included.

2. AUTOLOGOUS STEM CELLS TRANSPLANTATION (ASCT)

After a maximum period of 60 days after the end of the 4th VRD cycle an autologous stem cell transplant (ASCT) is performed. The conditioning regimen is melphalan 200 mg/m^2.

3. CONSOLIDATION: two additional cycles of 28 days each of Isatuximab + VRD, as in induction, starting approximately 2 months after hospital discharge or 3 months after transplantation.

4. Continuous treatment with Lenalidomide and monthly Isatuximab until progression, unacceptable toxicity, patient withdrawal, loss to follow up or death. During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab.

8.0.2 EXPERIMENTAL ARM (B): Extended VRD and Early Detection of Treatment Failure and Early Rescue Intervention.

Includes two experimental lines: 1) VRD extended to 18 cycles: Induction (VRDx6)-ASCT-consolidation (VRDx2)- Extended VRD: VRDx10, followed by lenalidomide plus dexamethasone maintenance 2) Early detection of treatment failure and Early Rescue Intervention (ERI): Isatuximab-Iberdomide-Dexamethasone in continuous treatment.

1. VRD six initial cycles (in this period, following established procedures, and preferably coinciding with the 3rd / 4th cycle of VRD, the collection and cryopreservation of hemopoietic precursors is included)

2. ASCT

After a maximum period of 60 days after the end of the 6th VRD cycle an ASCT is performed. The conditioning regimen is melphalan 200 mg/m^2.

3. VRD CONSOLIDATION. After 6 induction cycles, and after transplantation, 2 additional two additional cycles (7th and 8th) cycles of VRD are administered starting approximately 2 months after hospital discharge or 3 months after transplantation.

4. VRD EXTENDED. After 6 cycles of induction VRD, ASCT and two consolidation VRDs, treatment continues with 10 additional cycles of VRD.

During the extension cycles, VRD changes the bortezomib and dexamethasone regimen. In these 10 cycles, both bortezomib and dexamethasone will be administered, at the same doses as the previous ones, but on a weekly schedule, on days 1, 8, 15 and 22 of each cycle. The lenalidomide regimen remains unchanged.

5. MAINTENANCE: Maintenance treatment with 15 mg/d lenalidomide x 21 days and 20 mg/d dexamethasone on days 1-4 (28-day cycles).

The maintenance will be continued until progression, unacceptable toxicity, patient withdrawal, loss to follow up or death

8.0.2.1 Early Detection of Treatment Failure and Early Rescue Intervention

Patients assigned to Experimental ARM B with unfavourable response evolution, that is, an incipient subclinical progression to a level equivalent to the definition of the IMWG criteria of Progressive Disease but without measurable disease and without other data of clinical significance, but also including exclusive progression of the minimum residual disease (MRD) values [see criteria of Early Detection of Treatment Failure (EDTF)] during induction, after ASCT, consolidation, extended VRD phase or in maintenance period, will be diverted to an alternative rescue treatment, hereinafter named “Early Rescue Intervention (ERI)” , with the combination Iberdomide (CC-220) plus Isatuximab and Dexamethasone, until progression, unacceptable toxicity, patient withdrawal, loss to follow up or death.

Early Detection of Treatment Failure Criteria previous to ERI indication

Biological progression should be considered when:

1. Progression data, but insufficient to define a Progressive Disease status, according to IMWG criteria[1], that is:

- In any response category, including Complete Remission, Very Good Partial Response, Partial Response or Stable Disease, reappearance or increase of the M-Component in serum and / or urine, but the increase in the serum M band should be less than 0.5 g/dl and / or the urine M-component increase should not exceed 199 mg/24h, in the absence of other clinical involvement data. This must be CONFIRMED 2-4 weeks apart.

- Only in patients without measurable serum and urine M-protein levels: the difference between involved and uninvolved FLC levels (absolute increase must be ≥10 mg/dL). The increase must be CONFIRMED 2-4 weeks apart.

1.Rajkumar SV et al. Consensus recommendations for the uniform reporting of clinical trials: report of the International Myeloma Workshop Consensus Panel 1. Blood 2011;117 (18):4691-4695 (24)

2. Conversion from MRD negative into positive or an increase in the MRD level of at least one log (>10^{-1}). In both cases, the changes in MRD must be CONFIRMED in a new bone marrow sample 2-4 weeks apart.

- In the event that the criteria for ERI is reached in the induction phase between the 1st and 5th cycle of VRD, a minimum of three cycles of the alternative treatment with Iberdomide-Isatuximab-dexamethasone will be administered, followed by the ASCT.
- In the event that the criteria for ERI is reached after starting the 6th cycle of VRD, the ASCT will then be performed as planned.
- After the autologous transplant, the patient will start/continue ERI with Iberdomide-Isatuximab-dexamethasone cycles.
- In patients whose EDFT criteria is reached at any point in the treatment program after ASCT (consolidation, extended VRD or maintenance), the ERI will be started after confirmation of the EDTF criteria, starting the cycles of Iberdomide, Isatuximab, and dexamethasone.

In patients in ERI whose alternative treatment is started both before and after the transplant, the number of cycles prior to the primary end point is established at 12 post-transplant cycles, although in some cases the pre-transplant ERI may have conditioned more than 18 cycles in the induction plus consolidation-extension set.

In all cases the ERI will be maintained until progression, unacceptable toxicity, patient withdrawal, loss to follow up or death.

8.0.3 EXPLORATORY ARM (C)

With a similar design to Arm A, lenalidomide in this arm will be replaced by Iberdomide, that is: Induction: Bortezomib- Iberdomide- -Dexamethasone (VID) -Isatuximab x 4 cycles- ASCT- Consolidation: VID-Isatuximab x2 cycles followed by continuous treatment with cycles of Iberdomide plus Isatuximab. During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab.

1. INDUCTION: Iberdomide plus Isatuximab, bortezomib and dexamethasone (four cycles). In this period, following established procedures, and preferably coinciding with the 3rd / 4th cycle, the collection and cryopreservation of hemopoietic precursors is included.

2. ASCT

After a maximum period of 60 days after the end of the 4th cycle an autologous stem cell transplant (ASCT) is performed. The conditioning regimen is melphalan 200 mg/m^2.

3. CONSOLIDATION: two additional cycles of 28 days each of Isatuximab, Iberdomide, Bortezomib and Dexamethasone, as in induction, starting approximately 2 months after hospital discharge or 3 months after transplantation.

4. Continuous treatment with Iberdomide and monthly Isatuximab until progression, unacceptable toxicity, patient withdrawal, loss to follow up or death. During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab.

8.0.4 Primary Endpoint assessment

The primary ENDPOINT, -the MRD rate-, takes as a reference the evaluation after the last extended VRD cycle, this is: 6 cycles for induction, 6 months for transplantation, 2 cycles for consolidation and 10 cycles until completing the 18 cycles of VRD, (in total about 24 months). For this reason, the primary ENDPOINT in Arms A and C are established after a similar treatment time, which includes 4 cycles of induction, ASCT, 2 cycles of consolidation and 12 cycles of continuous treatment with Iberdomide plus Isatuximab (Dexamethasone to be determined).In patients of Arm B included in ERI, due to the great variability of the possible moments of incorporation in this therapeutic program, only rules are established for the moment and the realization or not of the transplant. The evaluation of the results will be carried out separately in the patients included, but also in conjunction with the rest of the patients in Arm B to know the effect of the global strategy.

8.0.5 Continuous or maintenance treatments

After the evaluation of the primary END-POINT, continuous / maintenance treatment continues in Arms A, B and C, including patients in ARM B assigned to the ERI program. Obtaining conventional CR in either arm will require a BM analysis for MRD. In the case of stable response or improvement without RC, MRD controls have been pre-established. Due to the lack of data on tolerance and adherence to long-term treatment with Isatuximab and Iberdomide, changes in the therapeutic programs specified in the protocol are possible throughout the development of this phase of the clinical trial. Also, for this reason, a complete revision of the therapeutic program has been

predetermined at the moment in which the last patient included in the clinical trial reaches 36 months of treatment. At this point, taking into account the updated knowledge about continuous or maintenance treatments, the strategies for a second clinical trial or an extension of this clinical trial will be defined.

8.0.6 Clinical trial closure

The time of closure of the clinical trial is not specified and will depend on the result of the general evaluation carried out at month 36 of the entry into the trial of the last patient. In any case, the anticipated long follow-up for PFS / OS and other related variables could be maintained in the very long term even after the closure of the clinical trial.

8.0.7 Treatment details: Control Arm (A)

1. Induction cycle 1: Isatuximab + VRD

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Isatuximab (SC) 1400mg	■							■							■							■						
Bortezomib (SC) 1.3 mg/m2	■			■				■			■																	
Lenalidomide (PO) 25 mg	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
Dexamethasone (PO) 40 mg	■	■	■	■					■	■	■	■																

1. Induction cycles 2-4: Isatuximab + VRD

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Isatuximab (SC) 1400mg	■														■													
Bortezomib (SC) 1.3 mg/m2	■			■				■			■																	
Lenalidomide (PO) 25 mg	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
Dexamethasone (PO) 40 mg	■	■	■	■					■	■	■	■																

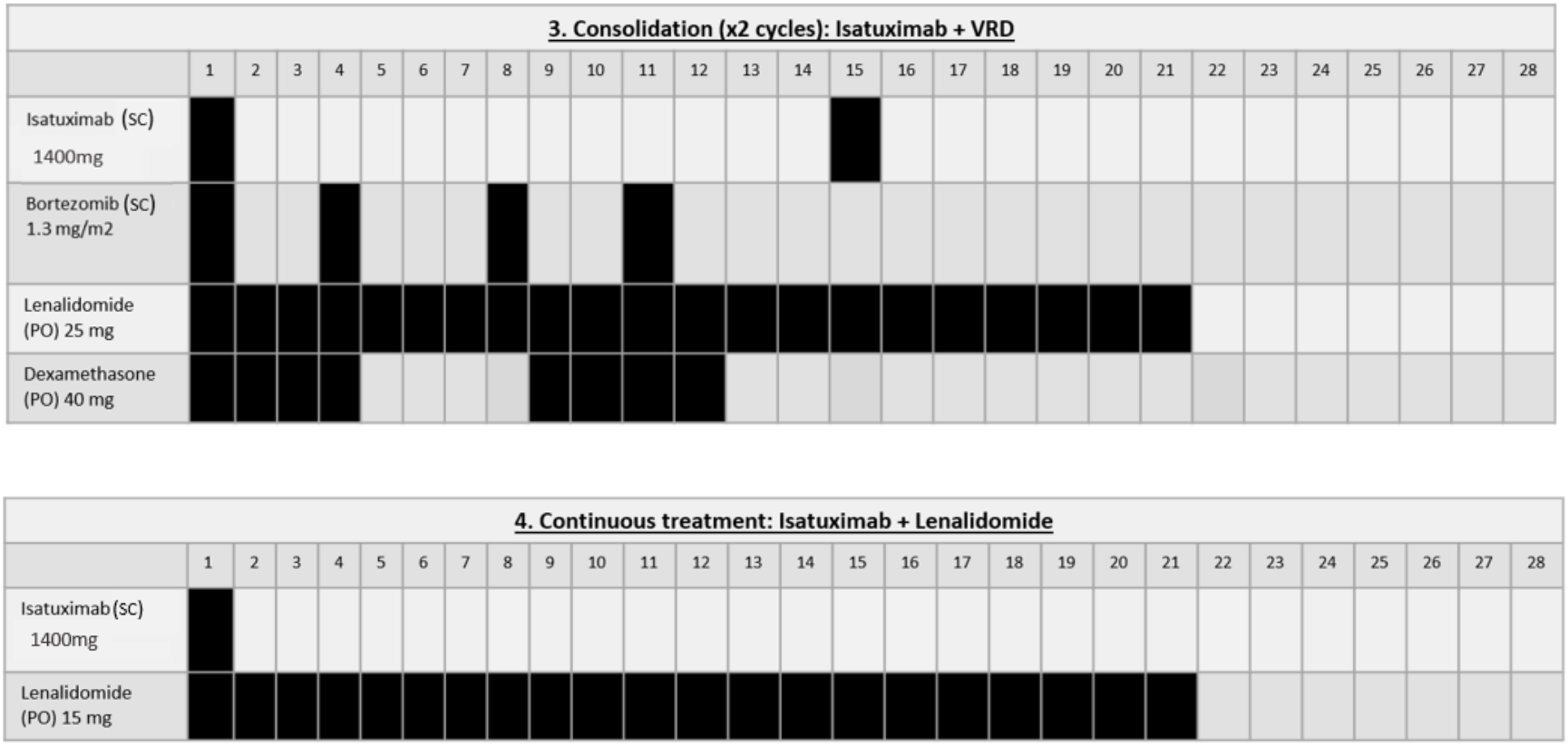


Figure 2 Control Arm (A) treatment: induction (1), ASCT (2), consolidation (3), and continuous treatment (4). Isatuximab will be infused monthly after the end of consolidation.

Table 3 Control arm (A): Induction

Drug	Dose	Days
Isatuximab (SC)	1400 mg	Days 1, 8, 15, and 22 in cycle 1 (Q4W) Days 1 and 15 in cycles 2-4 (Q4W)
Bortezomib (SC)	1.3 mg/m^2	Days 1, 4, 8, and 11 (Q4W)
Lenalidomide (PO)	25 mg	Days 1-21 (Q4W)
Dexamethasone (PO)	40 mg	Days 1 to 4 and 9 to 12 (Q4W)

Table 4 Control arm (A): Consolidation

Drug	Dose	Days
Isatuximab (SC)	1400 mg	Days 1 and 15 (Q4W)
Bortezomib (SC)	1.3 mg/m^2	Days 1, 4, 8, and 11 (Q4W)
Lenalidomide (PO)	25 mg	Days 1-21 (Q4W)
Dexamethasone (PO)	40 mg	Days 1 to 4 and 9 to 12 (Q4W)

Table 5 Control Arm (A): Continuous treatment (During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab).

Drug	Dose	Days
Isatuximab (SC)	1400 mg	Day 1 (Q4W)
Lenalidomide (PO)	15 mg	Days 1-21 (Q4W)

Isatuximab will be administered subcutaneously at 1400 mg on days 1, 8, 15, and 22 in the first 4-week cycle; and days 1 and 15 in subsequent cycles. Isatuximab will be infused monthly since the start of continuous therapy (after the second cycle of consolidation).

Bortezomib will be administered subcutaneously (SC) at 1.3 mg/m^2 on days 1, 4, 8, and 11 of every 4-week cycle.

Lenalidomide 25 mg (PO) will be administered on the days 1 to 21 of every 4-week cycle, in induction and consolidation VRD. Lenalidomide 15 mg (PO) will be administered on the days 1 to 21 of every 4-week cycle during maintenance treatment.

Dexamethasone 40 mg (PO) should be administered on days 1 to 4 and 9 to 12 of every 4-week cycle during induction and consolidation. During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab.

On days where Isatuximab and bortezomib administration coincides, bortezomib administration should follow the Isatuximab infusion and should start after the end of Isatuximab infusion.

On days where dexamethasone administration coincides with administration of Isatuximab, dexamethasone should be administered as part of the premedication.

The actual body weight in kg at baseline (assessed on Cycle 1 Day 1 prior to dosing) will be used for dose calculation of Isatuximab dose in all participants during the treatment period. However, if the change of body weight is greater than 10%, the dose must be recalculated based on the actual body weight at the time of dosing.

Bortezomib dose will be calculated using the patient's actual body surface area (BSA) at baseline. Patients with a BSA >2.2 m^2 will receive a dose based upon a 2.2 m^2 BSA. The calculated dose of bortezomib may be rounded to the nearest tenth of a mg (eg, a calculated dose of 2.34 mg may be rounded to 2.3 mg). If a subject's weight changes by more than 10% from baseline, the dose of bortezomib will be re-calculated.

The amount of lenalidomide and dexamethasone is fixed and it is not to be adjusted to body surface.

See Section 10 for guidance on dose modifications and dose delays.

8.0.7.1 Required concomitant medication for Control Arm (A):

The required concomitant medication is:

- The day of Isatuximab administration, the following order is recommended:

  Dexamethasone 40 mg PO. On days where dexamethasone administration coincides with administration of Isatuximab, dexamethasone should be administered as part of the premedication.

  In the event that isatuximab reactions warrant premedication with dexamethasone, the date of the closest dexamethasone dose should be adapted to that scheduled for isatuximab administration. That is, moving the dexamethasone dose from day 9 to day 8 and from day 12 to day 15, keeping dexamethasone administration unchanged on days 10 and 11. If premedication was necessary on day 22 of the 1st cycle, Isatuximab should be premedicated with a supplemental dose of dexamethasone.

  Acetaminophen (paracetamol) 650 mg to 1000 mg PO

  Ranitidine 50 mg IV (or equivalent). The use of ranitidine or equivalent as IR premedication will be based on Investigators' judgement.

  Diphenhydramine 25 mg to 50 mg IV (or equivalent).

  Montelukast 10 mg PO

  In case of reactions absence after 4 isatuximab infusions, premedication shall be lightened or stopped under medical criteria.
- Oral proton pump inhibitor to prevent gastric duodenal ulcers at least during the days of dexamethasone treatment.
- Acyclovir 400 mg orally, two times a day (or equivalent antiviral drug) to prevent reactivation of herpes zoster during the treatment with bortezomib or isatuximab.
- Lenalidomide: Thromboembolism prophylaxis consisting of low-dose aspirin, low molecular weight heparin, or other equivalent antithrombotic or anticoagulant will be given to all subjects as part of the study.

8.0.8 Experimental Arm B: Extended VRD

1. Induction (x6 cycles): VRD

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Bortezomib (SC) 1.3 mg/m2	■			■				■			■																	
Lenalidomide (PO) 25 mg	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
Dexamethasone (PO) 40 mg	■	■	■	■					■	■	■	■																

3. Consolidation (x2 cycles): VRD

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Bortezomib (SC) 1.3 mg/m2	■			■				■			■																	
Lenalidomide (PO) 25 mg	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
Dexamethasone (PO) 40 mg	■	■	■	■					■	■	■	■																

4. VRD Extended (x10 cycles)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Bortezomib (SC) 1.3 mg/m2	■							■							■							■						
Lenalidomide (PO) 25 mg	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
Dexamethasone (PO) 40 mg	■							■							■							■						

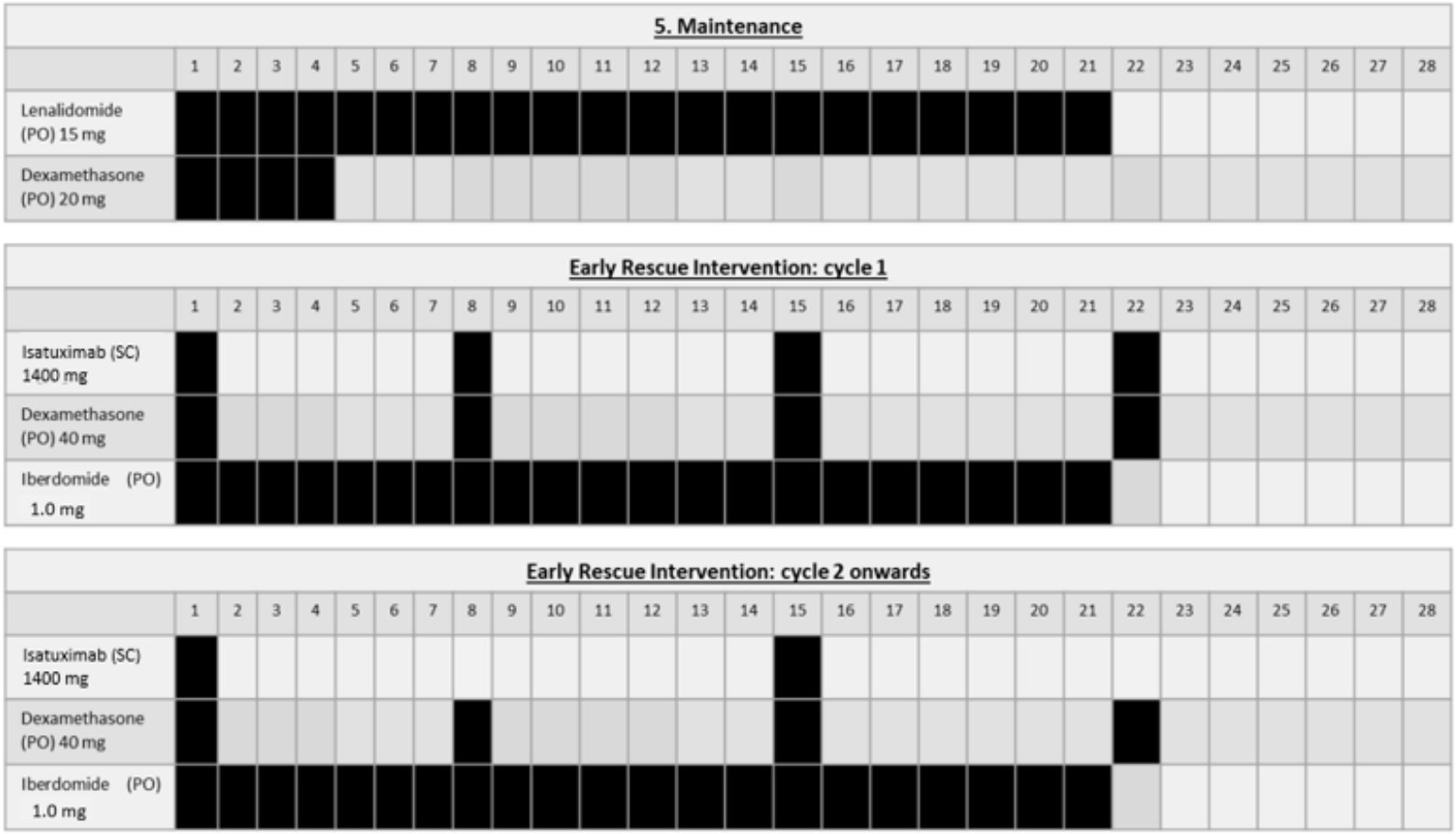


Figure 3 Experimental Arm (B) treatment: VRD (1), ASCT (2), VRD consolidation (3), VRD extended (4), maintenance (5), and ERI. Isatuximab will be infused monthly after 1 year treatment.

Table 6 Experimental Arm B: Induction, Consolidation VRD

Drug	Dose	Days
Bortezomib (SC)	1.3 mg/m^2	Days 1, 4, 8, and 11 (Q4W). In the extension cycles: Days 1, 8, 15, and 22 (Q4W)
Lenalidomide (PO)	25 mg	Days 1-21 (Q4W)
Dexamethasone (PO)	40 mg	Days 1 to 4 and 9 to 12 (Q4W) During the extension cycles: Days 1, 8, 15, and 22 (Q4W)

Table 7 Experimental Arm B: Continuous treatment

Drug	Dose	Days
Lenalidomide (PO)	15 mg	Days 1-21 (Q4W)
Dexamethasone (PO)	20 mg	Days 1 to 4 (Q4W)

Table 8 ERI Continuous Regimen

Drug	Dose	Days
Isatuximab (SC)	1400 mg	Cycle 1: Days 1, 8, 15, and 22 (Q4W) Cycles 2 onwards: Days 1 and 15 (Q4W). Isatuximab will be infused monthly after 1 year treatment (Day 1 Q4W).
Dexamethasone (PO)	40 mg	Days 1, 8, 15, and 22 (Q4W)
Iberdomide (PO)	1.0 mg	Days 1 to 21 (Q4W)

Bortezomib will be administered subcutaneously (SC) at 1.3 mg/m^2 on days 1, 4, 8, and 11 of every 4-week cycle.

Lenalidomide 25 mg (PO) will be administered on the days 1 to 21 of every 4-week cycle. Lenalidomide 15 mg (PO) will be administered on the days 1 to 21 of every 4-week cycle during continuous treatment.

Dexamethasone 40 mg (PO) should be administered on days 1 to 4 and 9 to 12 of every 4-week cycle in the composition of VRD. During continuous treatment, dexamethasone 20 mg should be administered on days 1 to 4 of every 4-week cycle. In the composition of ERI, Dexamethasone 40 mg (PO) should be administered on days 1, 8, 15, and 22 of every 4-week cycle.

Isatuximab will be administered subcutaneously (SC) at 1400 mg on days 1, 8, 15, and 22 in the first 4-week cycle; and days 1 and 15 in subsequent cycles. Isatuximab will be dispensed monthly after 1 year treatment.

Iberdomide will be administered orally (PO) at 1.0 mg on days 1 to 21 in each 4-week cycle.

On days where isatuximab and bortezomib administration coincides, the bortezomib infusion should follow the isatuximab administration starting after the end of isatuximab infusion.

On days where dexamethasone administration coincides with administration of Isatuximab, dexamethasone should be administered as part of the premedication.

The actual body weight in kg at baseline (assessed on Cycle 1 Day 1 prior to dosing) will be used for dose calculation of isatuximab dose in all participants during the treatment period. However, if the change of body weight is greater than 10%, the dose must be recalculated based on the actual body weight at the time of dosing.

Bortezomib dose will be calculated using the patient's actual body surface area (BSA) at baseline. Patients with a BSA >2.2 m^2 will receive a dose based upon a 2.2 m^2 BSA. The calculated dose of bortezomib may be rounded to the nearest tenth of a mg (eg, a calculated dose of 2.34 mg may be rounded to 2.3 mg). If a subject's weight changes by more than 10% from baseline, the dose of bortezomib will be re-calculated.

The amount of lenalidomide, dexamethasone, and iberdomide is fixed and it is not to be adjusted to body surface.

See Section 10 for guidance on dose modifications and dose delays.

8.0.8.1 Required concomitant medication for Experimental Arm (B):

The required concomitant medication during triplet combination treatment is:

- Oral proton pump inhibitor to prevent gastric duodenal ulcers at least during the days of dexamethasone treatment.
- Acyclovir 400 mg orally, two times a day (or equivalent antiviral drug) to prevent reactivation of herpes zoster during the treatment with bortezomib or isatuximab.
- Lenalidomide: Thromboembolism prophylaxis consisting of low-dose aspirin, low molecular weight heparin, or other equivalent antithrombotic or anticoagulant will be given to all subjects as part of the study.

8.0.8.2 Required concomitant medication for ERI in Experimental Arm (B):

- Iberdomide: Thromboembolism prophylaxis consisting of low-dose aspirin, low molecular weight heparin, or other equivalent antithrombotic or anticoagulant will be given to all subjects as part of the study.
- The day of isatuximab administration, the following order is recommended:

  On days where dexamethasone administration coincides with administration of isatuximab, dexamethasone should be administered as part of the premedication.

  In the event that isatuximab reactions warrant premedication with dexamethasone, the date of the closest dexamethasone dose should be adapted to that scheduled for isatuximab administration. That is, moving the dexamethasone dose from day 9 to day 8 and from day 12 to day 15, keeping dexamethasone administration unchanged on days 10 and 11. If premedication was necessary on day 22 of the 1st cycle, Isatuximab should be premedicated with a supplemental dose of dexamethasone.

  Acetaminophen (paracetamol) 650 mg to 1000 mg PO

Ranitidine 50 mg IV (or equivalent). The use of ranitidine or equivalent as IR premedication will be based on Investigators' judgement.

Diphenhydramine 25 mg to 50 mg IV (or equivalent).

Montelukast 10 mg PO

In case of IR absence after 4 isatuximab infusions, premedication shall be lightened or stopped under medical criteria.

- Acyclovir 400 mg orally, two times a day (or equivalent antiviral drug) to prevent reactivation of herpes zoster during the treatment with bortezomib or isatuximab.

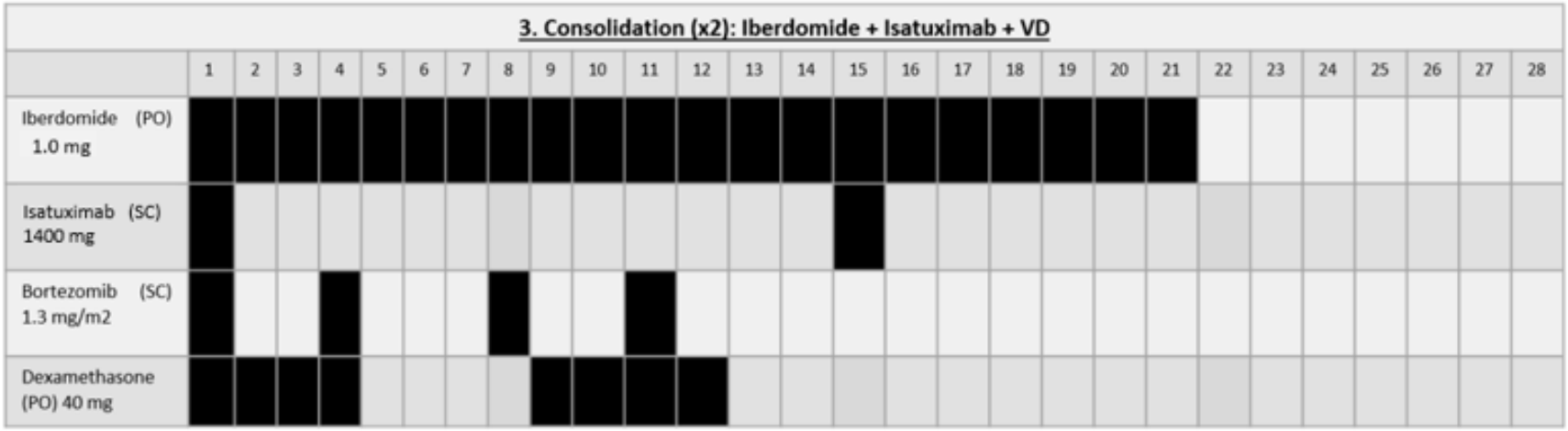
8.0.9 Exploratory Arm (C)

1. Induction cycle 1: Iberdomide + Isatuximab + VD

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Iberdomide (PO) 1.6 mg	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
Isatuximab (SC) 1400 mg	■							■							■							■						
Bortezomib (SC) 1.3 mg/m2	■			■				■			■																	
Dexamethasone (PO) 40 mg	■	■	■	■					■	■	■	■																

1. Induction cycles 2-4: Iberdomide + Isatuximab + VD

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Iberdomide (PO) 1.6 mg	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
Isatuximab (SC) 1400 mg	■														■													
Bortezomib (SC) 1.3 mg/m2	■			■				■			■																	
Dexamethasone (PO) 40 mg	■	■	■	■					■	■	■	■																



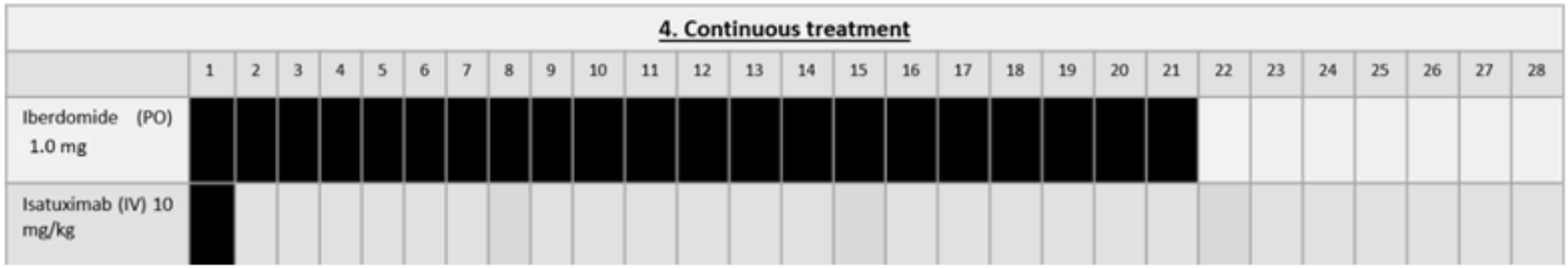


Figure 4 Exploratory Arm (C) treatment: induction (1), ASCT (2), consolidation (3), and continuous treatment (4). Isatuximab will be infused monthly after the end of consolidation.

Table 9 Exploratory arm (C): Induction

Drug	Dose	Days
Isatuximab (SC)	1400 mg	Days 1, 8, 15, and 22 in cycle 1. (Q4W) Days 1 and 15 in cycles 2-4. (Q4W)
Iberdomide (PO)	1.0 mg	Days 1-21 (Q4W)
Bortezomib (SC)	1.3 mg/m^2	Days 1, 4, 8, and 11 (Q4W)
Dexamethasone (PO)	40 mg	Days 1 to 4 and 9 to 12 (Q4W)

Table 10 Experimental arm (C): Consolidation

Drug	Dose	Days
Isatuximab (SC)	1400 mg	Days 1 and 15 (Q4W)
Iberdomide (PO)	1.0 mg	Days 1-21 (Q4W)
Bortezomib (SC)	1.3 mg/m^2	Days 1, 4, 8, and 11 (Q4W)
Dexamethasone (PO)	40 mg	Days 1 to 4 and 9 to 12 (Q4W)

Table 11 Exploratory Arm (C): Continuous treatment. During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab.

Drug	Dose	Days
Isatuximab (SC)	1400 mg	Day 1 (Q4W)
Iberdomide (PO)	1.0 mg	Days 1-21 (Q4W)

Isatuximab will be administered subcutaneously (SC) at 1400 mg on days 1, 8, 15, and 22 in the first 4-week cycle; and days 1 and 15 in subsequent cycles. Isatuximab will be infused monthly since the start of continuous therapy (after the second cycle of consolidation).

Iberdomide will be administered orally (PO) at 1.0 mg on days 1-21 of every 4-week cycle.

Bortezomib will be administered subcutaneously (SC) at 1.3 mg/m^2 on days 1, 4, 8, and 11 of every 4-week cycle.

Dexamethasone 40 mg (PO) should be administered on days 1 to 4 and 9 to 12 of every 4-week cycle. During continuous treatment, dexamethasone 40 mg is used as a standardized premedication for Isatuximab.

On days where Isatuximab and bortezomib administration coincides, the bortezomib infusion should follow the Isatuximab administration starting after the end of isatuximab treatment.

On days where dexamethasone administration coincides with administration of isatuximab, dexamethasone should be administered as part of the premedication.

The actual body weight in kg at baseline (assessed on Cycle 1 Day 1 prior to dosing) will be used for dose calculation of isatuximab in all participants during the treatment period. However, if the change of body weight is greater than 10%, the dose must be recalculated based on the actual body weight at the time of dosing.

Bortezomib dose will be calculated using the patient's actual body surface area (BSA) at baseline. The calculated dose of bortezomib may be rounded to the nearest tenth of a mg (eg, a calculated dose of 2.34 mg may be rounded to 2.3 mg). If a subject's weight changes by more than 10% from baseline, the dose of bortezomib will be re-calculated.

The amount of iberdomide and dexamethasone is fixed and it is not to be adjusted to body surface.

See Section 10 for guidance on dose modifications and dose delays.

8.0.9.1 Required concomitant medication for Exploratory Arm (C):

The required concomitant medication during triplet combination treatment is:

- Oral proton pump inhibitor to prevent gastric duodenal ulcers at least during the days of dexamethasone treatment.
- Acyclovir 400 mg orally, two times a day (or equivalent antiviral drug) to prevent reactivation of herpes zoster during the treatment with bortezomib or isatuximab.
- Iberdomide: Thromboembolism prophylaxis consisting of low-dose aspirin, low molecular weight heparin, or other equivalent antithrombotic or anticoagulant will be given to all subjects as part of the study.
- The day of isatuximab administration, the following order is recommended:

  Dexamethasone 40 mg PO. On days where dexamethasone administration coincides with administration of isatuximab, dexamethasone should be administered as part of the premedication.

  In the event that isatuximab reactions warrant premedication with dexamethasone, the date of the closest dexamethasone dose should be adapted to that scheduled for isatuximab administration. That is, moving the dexamethasone dose from day 9 to day 8 and from day 12 to day 15, keeping dexamethasone administration unchanged on days 10 and 11. If premedication was necessary on day 22 of the 1st cycle, Isatuximab should be premedicated with a supplemental dose of dexamethasone.

  Acetaminophen (paracetamol) 650 mg to 1000 mg PO

  Ranitidine 50 mg IV (or equivalent). The use of ranitidine or equivalent as IR premedication will be based on Investigators' judgement.

  Diphenhydramine 25 mg to 50 mg IV (or equivalent).

  Montelukast 10 mg PO

  In case of complicationsabsence after 4 isatuximab doses, premedication shall be lightened or stopped under medical criteria.

With this quadruplet a high number of neutropenia would be expected. See **Table 21** for instructions if neutropenia develops.

8.1 Mobilization, Harvesting and Autologous Transplantation of Hematopoietic Stem Cells

After the third (ARMS A&C), or fourth (ARM B) cycle of induction, and in the absence of progression or unacceptable toxicity, mobilization of HSCs with G-CSF and subsequent apheresis was to take place.

ASCT in ERI patients: In ARM B, when the criteria for ERI is reached in the induction phase between the 1st and 5th cycle of VRD, a minimum of three cycles of treatment with Iberdomide-Isatuximab-dexamethasone will be administered, followed by the ASCT. When the criteria for ERI is reached after the 6th cycle of VRD, the ASCT will then be performed as planned. In the event that the criterion for ERI occurs before hemopoietic progenitors have been obtained, the moment of the collection is at the discretion of the researcher.

The dose of G-CSF used remains at the discretion of each transplant site. Apheresis was to be initiated 4 to 5 days after stimulation, once the number of CD34+ cells had been quantified.

The minimum number of CD34+ cells needed to carry out the transplant is left at the discretion of each site, although a minimum of 2 x 10^6 CD34+/kg was recommended, as well as cryopreservation, storage, defrosting, and infusion of HSCs. If mobilization failed using the method indicated, the recommended action was to repeat the procedure approximately 2 weeks later and consider using a different mobilization method (preferably cyclophosphamide and G-CSF or G-CSF and plerixafor).

<u>ASCT:</u>

If the patient does not have an active infection, previous data on organ toxicity, or any other contraindication developed after the start of the study to perform the transplant, the start of the procedure takes place after the fourth cycle of induction in ARMS A&C and after the after the sixth cycle of induction in ARM B with a window period of +/- 12 weeks between the end of the last induction cycle and transplantation.

Pre-transplant Conditioning Regimen: Melphalan

Dosage form: Vials of sterile, lyophilized powder containing 50 mg of melphalan, and 10 mL vials of solvent/diluent

Route of administration: IV

Therapeutic group: Cytotoxic agents (alkylating agents)

Supplied by each site (ie, not supplied to the sponsor)

Although melphalan may not have been specifically labeled for the trial, the Pharmacy Service must be advised to record the trade name and batch number of the drug dispensed to each subject in their dispensing registers.

Melphalan 100 mg/m^2 IV on Day -3 and Day -2 relative to infusion of HSCs (Day 0 of the transplant phase), as a slow bolus infusion. Intravenous infusion of melphalan at a dose of 200 mg/m^2 on Day -2 was also permitted, according to each site's standard practice.

Infusion of HSCs was to begin on Day 0 of the transplant phase.

Hydration, antiemetic treatment, prophylaxis for uric acid nephropathy, and treatment of infections, as well as hemotherapy support and treatment with G-CSF post-transplant was to be provided in line with each site's routine clinical practice.

Before including the first patient in the clinical trial:

Those hospitals that do not have the possibility of performing the autologous transplant in their own center, before the inclusion of the first patient in the clinical trial, must agree on the coverage of this procedure with a referral hospital that inevitably has to participate in the trial. The PI of the referral hospital must accept the commitment to carry out the transplant in accordance with the clinical trial rules, including carrying out the transplant within the timeframe established with respect to the date of the end of the induction period and the application of the conditioning regimen. Likewise, the referral hospital must provide the hospital that included the patient in the trial with a clinical report that includes in detail all the parameters necessary to fill in the data referring to transplantation in the CRD of the clinical trial.

8.2 Treatment Administration

8.2.1 Iberdomide Administration

On non-clinic days, iberdomide PO may be self-administered at home.

If necessary, dose modifications for toxicity for each of these planned dose levels are outlined in Section 10.

Table 12 Iberdomide pharmaceutical information

	Iberdomide
Dosage form:	Capsules
Unit dose strengths:	1.0 mg
Route of Administration	Oral
Dosing instructions:	Days 1 to 21 of every 4-week cycle
Manufacturer/ Source of Procurement:	Celgene/BMS

8.2.2 Isatuximab Administration

Patients will receive isatuximab subcutaneously.

No dose reduction for isatuximab will be permitted.

SC isatuximab will be administered via OBDS weekly in the first cycle (Cycle 1) on days 1, 8, 15, 22, and biweekly on the 2 subsequent cycles (Cycles 2+3) at days 1, 15, at the nominal dose of 1400mg. SC isatuximab will be administered by an HCP in the periumbilical region at a single site, using the OBDS (reference to other sections?); injection site will be rotated from one administration to the other (see IB, IFU and pharmacy manual). Preparation and SC administration of isatuximab in the periumbilical region using the SC isatuximab OBDS will be performed by trained study staff / HCP only. The SC isatuximab OBDS includes an adhesive backing to secure it to the injection site during administration. At the end of dosing, the injection needle retracts into the device housing and the device enters a locked-out state to prevent reuse.

The volume to be injected at each administration will be injected at a single site. Further detailed instructions for preparation and administration using the SC isatuximab OBDS are provided in section (reference to other sections), IB, and IFU.

All OBDS incidents, including those resulting from malfunctions of the OBDS, AE (e.g. ISR) and DD must be detected, documented, and reported by the investigators throughout the study (reference to other sections?)

Table 13 Isatuximab pharmaceutical information

	Isatuximab (SAR650984)
Dosage form:	Isatuximab is supplied as one treatment box containing one vial, and as a sterile, nonpyrogenic, injectable, colorless to slightly yellow and essentially free of visible particles. Investigational SC Isatuximab OBDS is supplied in a blister.
Unit dose strengths:	The IMP presented as a 140 mg/mL concentrate in 20mL-sized Type I colorless clear glass vials fitted with elastomeric closures. Each vial contains 140 mg/mL (1400 mg/ 10mL) Isatuximab in 9 mM histidine, 2 %(w/v) sucrose, 110 mM L-Arginine monohydrochloride 0.4% (w/v poloxamer 188), pH 6.2 buffer. The fill volume has been established to ensure removal of 10 mL, corresponding a nominal content of 1400 mg.
Route of Administration	Subcutaneous On-body delivery system (OBDS, for subcutaneous isatuximab). The investigational medical Device OBDS is used for SC administration route. Performance of the Device itself is not an investigational objective of this study.
Dosing instructions:	The preparation of the infusion solution must be done under aseptic conditions. SC isatuximab will be administered by an HCP in the periumbilical region at a single site, using the OBDS.
Manufacturer/ Source of Procurement:	Sanofi-Aventis Groupe

SC injector device for Isatuximab

IMP isatuximab will be injected SC via an on-body delivery system (OBDS) also known as the isatuximab wearable injector system. It is a sterile, single-use, disposable, elastomeric, user-filled medical device designed and manufactured by Enable Injections, Inc (Cincinnati, OH, United States) which will be used to inject isatuximab SC in this clinical trial. The device will be supplied in an inner blister package and an outer box. Both the blister package and the box will be labeled according to the local requirements. The combined use of isatuximab SC with the OBDS (20 mL capacity) includes drug product contained within a vial, a transfer needle, a syringe and the OBDS. The specified transfer devices (20 mL luer lock syringe and 18G stainless steel luer lock needle) will be procured by the clinical trial sites and used to fill the drug product from the vial into the OBDS.

The OBDS includes an on-body delivery device (OBDD) designed for subcutaneous delivery of a defined volume of drug product and an integrated drug product syringe transfer (filling) base (ST Base). The 20 mL OBDS size is designed to accommodate fill volumes between 6 and 20 mL. As the OBDD for the isatuximab SC clinical study will be filled to 10 mL (containing a solution with 1400 mg isatuximab for SC injection), the printed markings on OBDS reflect 10 mL. The SC isatuximab OBDS includes an adhesive backing to secure it to the injection site during administration. At the end of dosing, the injection needle retracts into the device housing and the device enters a locked-out state to prevent reuse. The volume to be injected at each administration will be injected at a single abdominal site; injection site will be rotated from one administration to the next.

HCP must be trained on proper use of the device prior to use. For the purpose of this clinical study, the intended use is limited to in clinic by HCP only. Investigators and clinical study staff should adhere to all warnings and instructions in accordance with the SC administration of isatuximab using the OBDS. All OBDS incidents, including those resulting from malfunctions of the OBDS, AE (e.g. injection site reactions) and ADE must be detected, documented, and reported by the investigators throughout the study. The OBDS has been developed in accordance with the Medical Device Regulation, (EU) 2017/745, and compliance with the general safety and performance requirements (GSPR) defined in Annex I of Medical Device Regulation, (EU) 2017/745

8.2.3 Bortezomib Administration

Patients will receive bortezomib subcutaneously or intravenously, although preferably subcutaneously. Bortezomib dose will be calculated using the patient’s actual body surface area (BSA) at baseline. The calculated dose of bortezomib may be rounded to the nearest tenth of a

mg (eg, a calculated dose of 2.34 mg may be rounded to 2.3 mg). If a subject's weight changes by more than 10% from baseline, the dose of bortezomib will be re-calculated.

In the event that any of the days of bortezomib administration coincides with a public holiday, its administration may be advanced or delayed, taking into account that at least 72 hours must elapse between two doses.

Table 14 Bortezomib pharmaceutical information for subcutaneous administration**.**

	Bortezomib (Velcade® or generic Bortezomib)
Dosage form:	Powder for solution for injection Type 1 glass 10 ml-vial with a grey bromobutyl stopper and an aluminium seal, with a royal blue cap containing 3.5 mg bortezomib. The vial is contained in a transparent blister pack consisting of a tray with a lid. Each pack contains 1 single-use vial.
Unit dose strengths:	Each vial contains 3.5 mg bortezomib (as a mannitol boronic ester). After reconstitution, 1 ml of solution for subcutaneous injection contains 2.5 mg bortezomib.
Route of Administration	Subcutaneous
Dosing instructions:	Instructions for reconstitution: • Each 10 ml vial of VELCADE (or generic bortezomib) must be carefully reconstituted with 1.4 ml of sodium chloride 9 mg/ml (0.9%) solution for injection, by using a syringe of the appropriate size, without removing the vial stopper. Dissolution of the lyophilised powder is completed in less than 2 minutes. • After reconstitution, each ml solution contains 2.5 mg bortezomib. The reconstituted solution is clear and colourless, with a final pH of 4 to 7. The reconstituted solution must be inspected visually for particulate matter and discolouration prior to administration. If any discolouration or particulate matter is observed, the reconstituted solution must be discarded. The reconstituted solution should be used immediately after preparation. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. However, the chemical and physical in-use stability of the reconstituted solution has been demonstrated for 8 hours at 25°C stored in the original vial and/or a syringe. The total storage time for the reconstituted medicinal product should not exceed 8 hours prior to administration.

Manufacturer/ Source of Procurement:	Janssen-Cilag International NV Bortezomib will not be provided by the sponsor

8.2.4 Lenalidomide Administration

Lenalidomide will be orally administered. The amount of lenalidomide is fixed (25 mg or 15 mg) and it is not to be adjusted to body surface.

On non-clinic days, lenalidomide PO may be self-administered at home.

Table 15 Lenalidomide pharmaceutical information

	Lenalidomide (Revlimid®)
Dosage form:	Hard capsules White capsules, size 0, 21.7 mm, marked “REV 25 mg”. Pack size of 21 capsules
Unit dose strengths:	Each capsule contains 25 mg or 15 mg of lenalidomide
Route of Administration	Oral
Dosing instructions:	Lenalidomide 25/15 mg PO should be taken at the same time of the day and may be taken at home when administered PO
Manufacturer/ Source of Procurement:	Celgene Distribution B.V. Lenalidomide will be provided by the sponsor until loss of patent of Revlimid®. From there on, lenalidomide will be provided by each site.

8.2.5 Dexamethasone Administration

Dexamethasone will be orally administered. The amount of dexamethasone is fixed (40 mg weekly) and it is not to be adjusted to body surface.

On non-clinic days, dexamethasone PO may be self-administered at home.

Table 16 Dexamethasone pharmaceutical information

	Dexamethasone
Dosage form:	4 or 8 mg tablets
Unit dose strengths:	40 mg
Route of Administration	Oral
Dosing instructions:	Dexamethasone 40 mg PO should be taken at the same time of the day and may be taken at home when administered PO
Manufacturer/ Source of Procurement:	Dexamethasone may be utilized per a site's standard practice and will not be provided by the sponsor

8.3 Packaging and Labeling

All study drugs will be administered as open-label. No blinding methods will be used in this trial.

Experimental drugs will be supplied to pharmacies packaged and labeled according to the standard requirements for clinical trials.

8.4 Handling and Storage

The investigator or designee must confirm appropriate temperature conditions have been maintained during transit for all study drug(s) received and any discrepancies are reported and resolved before use of the study drug(s).

Only participants enrolled in the study may receive study drug(s) and only authorized site staff may supply or administer study drug(s). All study drugs must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labeled storage conditions with access limited to the investigator and authorized site staff.

8.5 Accountability

The investigator, institution, or the head of the medical institution (where applicable) is responsible for study drug(s) accountability, reconciliation, and record maintenance (i.e., receipt, reconciliation, and final disposition records).

Further guidance and information for the final disposition of unused study drug(s) are provided in the study reference manual.

8.6 Disposal

Precaution will be taken to avoid direct contact with the study drug(s) . A Material Safety Data Sheet (MSDS) describing occupational hazards and recommended handling precautions will be provided to the investigator. In the case of unintentional occupational exposure notify the Coordinators and Sponsor.

Any unused medicinal product should be disposed of at each participating site in accordance with local requirements.

8.7 Study Drug Compliance

When participants are dosed at the site, they will receive study drug(s) directly from the investigator or designee, under medical supervision. The date and time of each dose administered in the clinic will be recorded in the source documents. The dose of study drug(s) and study participant identification will be confirmed at the time of dosing by a member of the study site staff other than the person administering the study drug(s) .

Isatuximab (SC) and bortezomib (SC) will be administered at the site. Administration will be documented in the source documents and reported in the case report form (eCRF). Dexamethasone administration will be documented in the source documents and reported in the case report form (eCRF).

When participants self-administer oral study treatment(s) at home, dosing with iberdomide, lenalidomide, and dexamethasone will be recorded in the patient’s medical history. A record of

the number of doses dispensed to and taken by each participant must be maintained and reconciled with study treatment and compliance records. Treatment start and stop dates, including dates for treatment delays and/or dose reductions will also be recorded in the eCRF.

8.8 Concomitant Therapy/Devices

Concomitant medications that are used in the clinical context of multiple myeloma including those used to treat complications related to this disease, must be recorded along with:

- Reason for use
- Dates of administration including start and end dates
- Dosage information including dose and frequency

Participants will be instructed to inform the investigator prior to starting any of these new medications from the time of first dose of study treatment until the End of Study Treatment Visit.

The Clinical Coordinators should be contacted if there are any questions regarding concomitant therapy.

8.8.1 Permitted Concomitant Therapy

Participants should receive full supportive care during the study, including transfusions of blood products (erythrocytes, platelets), growth factors (G-CSF, erythropoietin or darbepoetin), as well as treatment with antibiotics, anti-emetics, antidiarrheal, and analgesics, as appropriate. Platelet/PRBC transfusions and immunoglobulin infusions are permitted during the study. G-CSF should not be administered to enable eligible laboratory values during the screening period. For example, subjects who fail to meet absolute neutrophil count eligibility criteria during screening should not be retested for the study after being treated with G-CSF.

Participants may receive local irradiation for pain or stability control. Subjects with myeloma-associated bone disease may receive bisphosphonate therapy and/or denosumab therapy prior to study entry. The use of bisphosphonates is permitted throughout the study.

COVID-19 vaccines that have been approved by an appropriate regulatory agency and are considered NOT live (not capable of transmitting infectious virus) are acceptable and should be handled in the same manner as other vaccines approved for immunocompromised cancer patients. If possible, Investigators are encouraged to give the vaccine at least 24 to 72 hours prior to starting study drugs (to watch for and manage vaccine-related acute toxicities).

The effectiveness and safety risk for the currently approved COVID-19 vaccines in patients who are taking the study drugs in this trial, is unknown. Therefore, the Investigator and patient should carefully weigh the benefit: risk ratio prior to making the decision to give an approved COVID-19 vaccine during the treatment phase of this study. Investigators are encouraged to contact the Coordinators to discuss any questions related to the selection or administration of COVID-19 vaccines.

8.8.2 Prohibited Concomitant Therapies

The following medications are considered exclusionary during the study treatment period. The Sponsor or designee must be notified if a subject receives any of these medications during the treatment period.

1. Live vaccines capable of transmitting SARS-CoV-2 or other infectious viruses.

2. Any non-protocol investigational or unapproved therapies for the treatment of MM.

3. Any concurrent chemotherapy, radiotherapy (except palliative radiotherapy for pain

palliation or fracture), immunotherapy, biologic or hormonal therapy for cancer treatment.

4. Immunosuppressive medications including, but not limited to systemic corticosteroids at doses exceeding 10 mg/day of prednisone or equivalent. Use of immunosuppressive medications for the management of AEs or in subjects with contrast allergies is acceptable. In addition, use of

inhaled, topical, intranasal corticosteroids or local steroid injection (eg, intra-articular injection) is permitted. Temporary use of corticosteroids for concurrent illnesses is acceptable upon discussion and agreement with the Coordinators.

5. Concurrent administration of a strong inhibitor or inducer of cytochrome P450 (CYP3A4/5) (including within 14 days of initiating study treatment) for Iberdomide

8.8.3 Specific considerations

Isatuximab Interference with serological testing

Because CD38 protein is expressed on the surface of red blood cells, isatuximab, an anti-CD38 antibody, may interfere with blood bank serologic tests with potential false positive reactions in indirect antiglobulin tests (indirect Coombs tests), antibody detection (screening) tests, antibody identification panels, and antihuman globulin (AHG) crossmatches in patients treated with isatuximab. The interference mitigation methods include treating reagent RBCs with dithiothreitol (DTT) to disrupt isatuximab binding or other locally validated methods. Since the Kell Blood group system is also sensitive to DTT treatment, Kell-negative units should be supplied after ruling out or identifying alloantibodies using DTT-treated RBCs.

Interference with Serum Protein Electrophoresis and Immunofixation Tests

Isatuximab may be detected by serum protein electrophoresis (SPE) and immunofixation (IFE) assays used for the monitoring of M-protein, and could interfere with accurate response classification based on International Myeloma Working Group (IMWG) criteria (see APPENDIX 7)

Bortezomib: Patients should be closely monitored when given bortezomib in combination with potent CYP3A4-inhibitors. Caution should be exercised when bortezomib is combined with

CYP3A4- or CYP2C19 substrates. Normal liver function should be confirmed and caution should be exercised in patients receiving oral hypoglycemics.

8.9 Dose Modification and Delay

Dose delays and reductions are permitted throughout the study as described below. After Cycle 1 Day 1, dose modifications may be made for individual participants based on safety findings. Guidance for dose modifications is provided in section 10.

Dosing delays are also permitted for medical/surgical events or logistical reasons not related to study therapy (e.g., elective surgery, unrelated medical events, participant vacation, and/or holidays) but not for participant's decision to delay treatment.

For potential overlapping toxicities, such as hematologic toxicity, the investigator should use their clinical judgement on which agent should be dose-reduced first, or for severe toxicities, more than one agent can be held/ dose modified. Whenever possible, step-wise dose modification is recommended first in order to minimize the potential to compromise treatment efficacy.

The reason for any dose delay must be documented in the participant's electronic case report form (eCRF) and clinic record.

8.10 Treatment Discontinuation

Participants may continue protocol therapy until one of the following criteria applies:

- Disease progression according to IMWG criteria (**APPENDIX 7**)
- Serious or life-threatening adverse event
- Participant received concurrent treatment for multiple myeloma
- Participant's dose is held for more than 28 days for reasons other than toxicity. In some cases where the delay is caused by complications inherent to the MM or the treatment, for example surgical interventions, severe infectious complications and other equivalent assumptions, the period could be extended after discussion with the trial board.
- Participant experiences unacceptable toxicity

- Severe noncompliance with protocol as judged by the Investigator and/or Sponsor
- Participant decision to withdraw
- Participant becomes pregnant
- Investigator and/or Sponsor becomes aware of conditions or events that suggest a possible risk or hazard to participants if the clinical study continues

If a subject's study treatment must be discontinued, this will not result in automatic withdrawal of the subject from the study. The End-of-Treatment Visit and Follow-up visit assessments should continue as specified in the protocol.

The primary reason for discontinuation of study treatment is to be recorded in the eCRF. For subjects who discontinue study treatment due to disease progression, sites will complete a disease progression form in the eCRF. If the participant voluntarily discontinues from treatment due to toxicity, the AE will be recorded as the primary reason for permanent discontinuation on the eCRF. Once a participant has permanently discontinued from a study treatment, the participant will not be allowed to restart study treatment.

8.11 Duration of Follow Up

Patients must be followed for 28 days following final dose of any of the investigational drugs to monitor for AEs.

8.12 Discontinuation from Study

Participants who permanently discontinue study treatment for reasons other than disease progression will remain in the study and will be followed for PFS until:

- Disease progression according to IMWG criteria (**APPENDIX 7**)
- New anti-cancer therapy is initiated
- Withdrawal of consent
- Loss to follow-up
- Death from any cause
- Termination of the study

Participants with documented disease progression whilst on treatment, or during progression-free survival (PFS) follow-up, will be followed for OS and other events of special interest until death, withdrawal of consent, loss to follow-up or the end of study, whichever occurs first.

The primary reason for discontinuation of each study treatment must be documented independently in the medical record and on the eCRF.

8.13 Lost to Follow Up

A participant will be considered lost to follow-up if he or she repeatedly fails to return for scheduled visits and is unable to be contacted by the study site.

The following actions must be taken if a participant fails to return to the clinic for a required study visit:

- The site must attempt to contact the participant and reschedule the missed visit as soon as possible and counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether or not the participant wishes to and/or should continue in the study.
- Before a participant is deemed lost to follow up, the investigator or designee must make every effort to regain contact with the participant (where possible, via email or telephone call). These contact attempts should be documented in the participant's medical record.
- Should the participant continue to be unreachable, he/she will be considered to have withdrawn from the study.

8.14 Participant Replacement Criteria

Subjects who do not receive at least one dose of study treatment or who are screening failure will be replaced. Subjects who receive any treatment dose will not be replaced.

9 PHARMACEUTICAL INFORMATION

The investigator shall take responsibility for and shall take all steps to maintain appropriate records and ensure appropriate supply, handling, storage, distribution, and usage of these materials in accordance with the protocol and any applicable laws and regulations.

9.0 Isatuximab

9.0.1 Identity

Antineoplastic agents, monoclonal antibody.

Isatuximab (SAR650984) is a naked immunoglobulin G1 kappa (IgG1k) monoclonal antibody (mAb) that selectively binds to the human cell surface antigen molecule classified as cluster of differentiation 38 (CD38). CD38 is highly expressed in MM cells. The drug product is presented as a concentrated solution for administration via SC injection.

9.0.2 Properties

The physical and chemical properties of isatuximab drug substance are described in the Investigator's Brochure (IB). In this trial, the investigational medicinal product (IMP) isatuximab will be administered in combination with Rd or Id, as induction therapy for newly-diagnosed MM

Please refer to the current investigator brochure for the comprehensive and updated list of adverse events.

9.0.3 Packaging, Labeling, and Storage

Isatuximab will be provided by the sponsor. All study treatment supplies must be stored in accordance with the manufacturer's instructions and package labeling. Until dispensed to the participants, the study treatment will be stored in a securely locked area, accessible to authorized personnel only.

Primary packaging description:

Isatuximab SC SAR650984 1400 mg is supplied in a single use vial.

Investigational SC isatuximab OBDS is supplied in a blister.

Secondary packaging description:

Isatuximab SC SAR650984 1400 mg: 1 vial in a box

Investigational SC isatuximab OBDS: 1 single blister in a box

9.0.4 Dispensing

The investigator agrees that study drug(s) will be dispensed by the investigator or sub-investigator(s) named on the investigator agreement or their qualified designees. The investigator, sub-investigators, or qualified designees also agree that the study drug(s) will be dispensed only to study subjects who have provided written informed consent and have met all entry criteria and in accordance with the instructions provided in the storage and handling manual.

9.0.5 Drug Accountability

The local investigator is responsible for ensuring that all IMP Isatuximab SC, received at the site is inventoried and accounted for throughout the study. The dispensing of the IMP to the subject must be documented. Unused study drug will be destroyed at the investigational trial site. Vials should be discarded in a safe manner. Destruction must be documented.

9.0.6 Disposal and Destruction

Any unused medicinal product should be disposed of at each participating site in accordance with local requirements.

Isatuximab should be destroyed at the investigational site if permitted by local regulations.

9.0.0.1 On-Body Delivery System from SC Isatuximab

The investigational medical Device OBDS is used for SC administration route. Performance of the Device itself is not an investigational objective of this study.

isatuximab will be administered SC by an on-body delivery system (OBDS) also known as the isatuximab wearable injector system. It is a sterile, single-use, disposable, elastomeric, user-filled medical device designed and manufactured by Enable Injections, Inc (Cincinnati, OH, United States) which will be used to inject isatuximab SC in this clinical trial. The device will be supplied in an inner blister package and an outer box. Both the blister package and the box will be labeled according to the local requirements. The combined use of isatuximab SC with the OBDS (20 mL capacity) includes drug product contained within a vial, a transfer needle, a syringe and the OBDS. The specified transfer devices (20 mL luer lock syringe and 18G stainless steel luer lock needle) will be procured by the clinical trial sites and used to fill the drug product from the vial into the OBDS.

The OBDS includes an on-body delivery device (OBDD) designed for subcutaneous delivery of a defined volume of drug product and an integrated drug product syringe transfer (filling) base (ST Base). The 20 mL OBDS size is designed to accommodate fill volumes between 6 and 20 mL. As the OBDD for the isatuximab SC clinical study will be filled to 10 mL (containing a solution with 1400 mg isatuximab for SC injection), the printed markings on OBDS reflect 10 mL. The SC isatuximab OBDS includes an adhesive backing to secure it to the injection site during administration. At the end of dosing, the injection needle retracts into the device housing and the device enters a locked-out state to prevent reuse. The volume to be injected at each administration will be injected at a single abdominal site; injection site will be rotated from one administration to the next.

The OBDS has been developed in accordance with the Medical Device Regulation, (EU) 2017/745, and compliance with the general safety and performance requirements (GSPR) defined in Annex I of Medical Device Regulation, (EU) 2017/745. The OBDS is class IIb in accordance with the Medical Device Regulation (EU) 2017/745, as amended, based upon the following:

- Duration: Transient or Short Term
- Medical Device Type: Surgical Invasive Device, Active Therapeutic Device
- Applicable Rules: Annex VIII rules 6 and 12

The OBDS is manufactured at the Enable Injections, Inc (Cincinnati, OH, United States) facility at 2863 East Sharon Rd, Cincinnati OH 45241 under an ISO 13485:2016 certified Quality Management System. For manufacturer's instructions for storage, installation, maintenance, maintaining hygiene standards and for use, refer to the OBDS IB/IFU to be provided as part of study pharmacy manual. Additional information related to storage and handing are included on the device label (see IB/IFU).

Further detailed instructions for preparation and administration using the (investigational) isatuximab injector device are provided in Pharmacy Manual. Study staff must be trained on proper use of the device prior to use based on the Clinical Protocol, the Clinical Investigational Plan, the Pharmacy Manual as well as the Instructions for Use of the injector device; no training required for patients.

Investigators and clinical study staff must adhere to all warnings and instructions in accordance with the SC administration of isatuximab using the wearable injector system (OBDS) IB / IFU / pharmacy manual, and study protocol (reference to other sections?). The SC isatuximab OBDS will only be used for study arm B in this trial and only with isatuximab for SC injection as IMP in this trial. It is not allowed to use the OBDS for any other purposes.

All OBDS incidents, including those resulting from malfunctions of the OBDS, AE (e.g. ISR) and DD must be detected, documented, and reported by the investigators throughout the study (reference to other sections?). Further detailed instructions for preparation and administration using the isatuximab SC injector device are provided in IB/ IFU / pharmacy manual.

9.0.0.0.1 Drug Accountability

The local investigator is responsible for ensuring that all OBDS received at the site is inventoried and accounted for throughout the study. Each OBDS has a batch number. Used OBDS must be disposed in a regular sharps' disposal container for medicinal products. Unused or expired OBDS will be disposed (e.g. in a sharps disposal container for medicinal products). Disposal must be documented. Any OBDS associated with a product technical complaint must be retained by the clinical sites and returned to the manufacturer for investigation per the established process. A product technical complaint form should be completed for any occurrence.

9.0.0.0.2 Disposal and destruction

Used OBDS must be disposed in a regular sharps' disposal container for medicinal products. Unused or expired OBDS will be disposed (e.g. in a sharps disposal container for medicinal products). Disposal must be documented. Any OBDS associated with a product technical complaint must be retained by the clinical sites and returned to the manufacturer for investigation per the established process. A product technical complaint form should be completed for any occurrence.

9.1 Iberdomide

9.1.1 Identity

Antineoplastic agents,

9.1.2 Potential Risks of Iberdomide

Please refer to the current investigator brochure for the comprehensive and updated list of adverse events.

9.1.3 Risks associated with pregnancy

Iberdomide was found to cause birth defects in experimental animals (rat and rabbit). Iberdomide is an analogue of thalidomide. Thalidomide is a known human teratogen that causes severe life-threatening human birth defects. If iberdomide is taken during pregnancy, it may cause birth defects or death to an unborn baby. Therefore, a pregnancy prevention program must be followed.

In a 9-month toxicity study in monkeys given iberdomide (at a dose higher than the highest human equivalent dose [HED] planned for Phase 2 studies), male monkeys had minimal to slight hypospermatogenesis (minimal to slight decrease in the number of spermatozoa and precursor cells). Onset and reversibility of the above-mentioned testicular finding in monkeys are not known. There were no effects observed in male sex organs in monkeys that received a HED higher than the highest dose planned for Phase 2 studies. Female monkeys that received 9 months of

iberdomide dosing (at the same HED as male monkeys) did not exhibit any effects in the sex organs.

In men, the excretion of iberdomide into semen or the effect of CC-220 on spermatogenesis is not known and has not been studied. It is not known if there might be a similar effect in men taking iberdomide, how this might affect their ability to father children and if there is a risk to an unborn baby in females whose male partner is receiving iberdomide.

9.1.4 Packaging, Labeling, and Storage

Iberdomide will be provided by the sponsor. All study treatment supplies must be handled and stored according to the conditions described on the product label. Until dispensed to the participants, the study treatment will be stored in a securely locked area, accessible to authorized personnel only.

9.1.5 Dispensing

The investigator agrees that study drug(s) will be dispensed by the investigator or sub-investigator(s) named on the investigator agreement or their qualified designees. The investigator, sub-investigators, or qualified designees also agree that the study drug(s) will be dispensed only to study subjects who have provided written informed consent and have met all entry criteria and in accordance with the instructions provided in the storage and handling manual.

9.1.6 Drug Accountability

The Investigator or designee is responsible for maintaining accurate dispensing records of the study treatment throughout the clinical study. The study treatment accountability log includes information including a patient identifier, amount and date dispensed, and amount and date returned to the pharmacy, if applicable. Product returned to the pharmacy will be stored under the same conditions as products not yet dispensed but will be marked as 'returned' and kept separate from the products not yet dispensed.

All dispensing and accountability records should be stored in accordance to the Sponsor institution regulations. The pharmacist will dispense study treatment for each participant according to the protocol.

9.1.7 Disposal and Destruction

Any unused medicinal product should be disposed of at each participating site in accordance with local and/or national requirements.

Iberdomide should be destroyed at the investigational site if permitted by local regulations.

9.2 Bortezomib

9.2.1 Identity

Antineoplastics, proteasome inhibitor

9.2.2 Potential Risks of Bortezomib

Please refer to the current investigator brochure for the comprehensive and updated list of adverse events.

9.2.3 Packaging, Labeling and Storage

All study medications should be handled and stored according to the conditions described on the product label. Until dispensed to the participants, the study treatment will be stored in a securely locked area, accessible to authorized personnel only.

9.2.4 Dispensing

The investigator agrees that study drug(s) will be dispensed by the investigator or sub-investigator(s) named on the investigator agreement or their qualified designees. The investigator, sub-investigators, or qualified designees also agree that the study drug(s) will be dispensed only to study subjects who have provided written informed consent and have met all entry criteria and in accordance with the instructions provided in the storage and handling manual.

9.2.5 Drug Accountability

The Investigator or designee is responsible for maintaining accurate dispensing records of the study treatment throughout the clinical study. The study treatment accountability log includes information including a patient identifier, amount and date dispensed, and amount and date returned to the pharmacy, if applicable. Product returned to the pharmacy will be stored under the same conditions as products not yet dispensed but will be marked as 'returned' and kept separate from the products not yet dispensed.

All dispensing and accountability records should be stored in accordance to the Sponsor institution regulations. The pharmacist will dispense study treatment for each participant according to the protocol.

9.2.6 Disposal and Destruction

Any unused medicinal product should be disposed of at each participating site in accordance with local requirements. Bortezomib should be destroyed at the investigational site if permitted by local regulations.

9.3 Lenalidomide

9.3.1 Identity

Antineoplastic agents, immunomodulators.

9.3.2 Potential Risks of Lenalidomide

Please refer to the current investigator brochure for the comprehensive and updated list of adverse events.

9.3.2.1 Pregnancy warning

Lenalidomide is structurally related to thalidomide. Thalidomide is a known human teratogenic active substance that causes severe life-threatening birth defects. Lenalidomide induced in

monkeys' malformations similar to those described with thalidomide. If lenalidomide is taken during pregnancy, a teratogenic effect of lenalidomide in humans is expected.

The conditions of the Pregnancy Prevention Programme must be fulfilled for all patients unless there is reliable evidence that the patient does not have childbearing potential.

9.3.3 Packaging, Labeling and Storage

All study treatment supplies must be stored in accordance with the manufacturer's instructions and package labeling. Until dispensed to the participants, the study treatment will be stored in a securely locked area, accessible to authorized personnel only.

9.3.4 Dispensing

The investigator agrees that study drug(s) will be dispensed by the investigator or sub-investigator(s) named on the investigator agreement or their qualified designees. The investigator, sub-investigators, or qualified designees also agree that the study drug(s) will be dispensed only to study subjects who have provided written informed consent and have met all entry criteria and in accordance with the instructions provided in the storage and handling manual.

9.3.5 Drug Accountability

The Investigator or designee is responsible for maintaining accurate dispensing records of the study treatment throughout the clinical study. The study treatment accountability log includes information including a patient identifier, amount and date dispensed, and amount and date returned to the pharmacy, if applicable. Product returned to the pharmacy will be stored under the same conditions as products not yet dispensed but will be marked as 'returned' and kept separate from the products not yet dispensed.

All dispensing and accountability records should be stored in accordance to the Sponsor institution regulations. The pharmacist will dispense study treatment for each participant according to the protocol.

9.3.6 Disposal and Destruction

Any unused medicinal product should be disposed of at each participating site in accordance with local requirements.

Lenalidomide should be destroyed at the investigational site if permitted by local regulations.

9.4 Dexamethasone

9.4.1 Identity

Glucocorticoids.

9.4.2 Potential Risks of Dexamethasone

Please refer to the current investigator brochure for the comprehensive and updated list of adverse events.

9.4.3 Packaging, Labeling and Storage

Dexamethasone will not be provided by the sponsor and will be used as per routine clinical practice only as an oral drug (8 or 4 mg tablets).

This medicinal product does not require any special storage conditions.

9.4.4 Dispensing

Not applicable. Dexamethasone will not be provided by the sponsor and will not be dispensed by the centers. Patients will acquire dexamethasone in the pharmacy (as per routine clinical practice).

9.4.5 Drug Accountability

Not applicable. Dexamethasone will not be provided by the sponsor as study medication. However, the center will register and document the adequate compliance of dexamethasone administration by the patient in the patient's medical history.

9.4.6 Disposal and Destruction

Not applicable. Dexamethasone will not be provided by the sponsor as study medication.

10 DOSE DELAYS OR MODIFICATIONS

After Cycle 1, dose modifications for individual participants may be required for safety reasons. Dose interruptions and reductions are permitted throughout the study. Toxicities should be attributed, whenever possible, to a specific component of study treatment so that dose interruption, dose skipping, or (if applicable) or dose reduction can be made appropriately.

Dosing delays are also permitted for medical/surgical events or logistical reasons not related to study therapy (e.g., elective surgery, unrelated medical events, participant vacation, and/or holidays) but not simply for participant's choice to delay treatment.

For dose delays or interruptions lasting >28 days counting from the date scheduled for treatment administration, the Coordinators should be contacted to discuss restarting treatment.

The reason for any dose delay must be documented in the participant's eCRF and clinic record. Where safety findings cannot be ascribed solely to one study treatment, follow the guidance provided in **Table 245.**

10.0 Criteria to initiate a new cycle

The following values at least are required in order to start a new treatment cycle:

Haemoglobin ≥7,5 g/dL

Neutrophils ≥ $1x10^9$/L

Platelets ≥ $70x10^9$/L

Support therapy can be used except for C1. If the patient does not show these values, a hemogram will be performed weekly until he or she recovers from the toxicity.

The start of each cycle may occur ±3 days of the scheduled day in order to accommodate the schedule of the site or subject.

Toxicities that do not require a dose reduction:

The following toxicities are exceptions to the above guidelines. It is not necessary to withhold administration of study drugs in these cases:

- Grade 3 nausea, vomiting or diarrhea (unless these events last more than 3 days despite administration of adequate antiemetic or antidiarrheal treatments).
- Grade 3 hyperglycemia related to dexamethasone.
- Grade 3 fatigue (unless it lasts for > 14 days).
- Alopecia.

Dose modifications for individual participants may be required for safety reasons. Dose interruptions and reductions are permitted throughout the study. The following information provides some guidelines for the management of toxicity, but this should be done following the investigators' criteria, and a proactive dose reduction approach is encouraged. However, dose reduction steps for each of the agents should be followed as indicated in **Table 17**, **Table 18**, and **Table 19** .

Adjustment to the dose of lenalidomide is recommended to provide appropriate drug exposure in subjects with moderate or severe renal impairment, because lenalidomide is primarily excreted unchanged by the kidney. Lenalidomide dose adjustment should be instituted for subjects with a creatinine clearance (CrCl) <50 mL/min. The recommended doses for subjects with multiple myeloma and renal impairment are shown in

Table 20. To be enrolled in the study, subjects must have CrCl ≥40 mL/min and Creatinine ≥2mg%. If during treatment a subject's renal status changes, the dose should be adjusted as shown in

Table 20.

Table 17 Bortezomib Dose Reduction steps

Dose Level	**Bortezomib Dose**
Starting Dose	1.3 mg/m^2
Dose Level -1	1.0 mg/m^2
Dose Level -2	0.7 mg/m^2
Dose Level -3	Discontinue Bortezomib

Table 18 Lenalidomide Dose Reduction steps

Starting dose	25 mg during 21 days every 28 days
Dose Level -1 (only applicable for isolated neutropenia)	25 mg during 21 days every 28 days plus G-CSF
Dose Level -2	15 mg daily during 21 days every 28 days
Dose Level -3	10 mg daily during 21 days every 28 days
Dose Level -4	5 mg daily during 21 days every 28 days

Patients who do not tolerate the dose level -4 will be discontinued from the study.

Table 19 Iberdomide Dose Reduction Steps

Dose Level	**Oral Iberdomide Dose** **(Days 1 to 21 of 28-day cycle)**
Starting Dose	1.0 mg
Dose Level -1	0.75 mg
Dose Level -2	0.45 mg
Dose Level -3	0.75 mg

The minimum permitted dose for iberdomide is 0.45 mg. No dose re-escalation is permitted for iberdomide.

10.1 Management of VRD dose reductions

There may be situations where the Investigator will not be able to attribute hematological toxicity to a single agent. In this event,

Table 20 is provided as a suggestion to guide modification of the combination agents for overlapping toxicities.

Table 20 Dose Modification Guidelines for Bortezomib, Lenalidomide, and Dexamethasone

Body System	NCI-CTC Adverse Event and or Symptom and Category	Bortezomib	Lenalidomide	Dexamethasone
Allergic reactions	Allergic reaction or hypersensitivity Grade 2 OR 3	Hold all therapy. If the toxicity resolves to ≤ Grade 1, restart VRD. Reduce by 1 dose-level the suspected medication(s) AND implement appropriate anti-allergic prophylaxis therapy. If the reaction was anaphylactic in nature, do not resume VRD. NOTE: If the reaction was cutaneous in nature, refer to the cutaneous category below.		
	Allergic reaction or hypersensitivity Grade 4	Discontinue VRD		
Cardiovascular	Fluid Retention (i.e., edema) >Grade 3 (limiting function and unresponsive to therapy or anasarca)			Administer diuretics as needed, and decrease dexamethasone dose by 1 dose-level; if edema persists despite above measures, decrease dose another dose-level. Permanently discontinue

				dexamethasone if symptoms persist despite second dose reduction.
Body System	**NCI-CTC Adverse Event and or Symptom and Category**	**Bortezomib**	**Lenalidomide**	**Dexamethasone**
Cutaneous	Non-blistering rash Grade 2	Hold bortezomib therapy. Begin treatment with antihistamines and/or low-dose steroids as per institutional practice. If the toxicity resolves to ≤ Grade 1, reduce dose by 1 level and restart bortezomib. Restart with lower concentration formulation. If recurrent consider IV bortezomib.	Consider holding lenalidomide	
	Non-blistering rash ≥ Grade 3 or 4	Hold bortezomib therapies. Begin treatment with antihistamines and/or low-dose steroids as per institutional practice. If the toxicity resolves to ≤ Grade 1, reduce dose by 1 level and restart bortezomib and lenalidomide and continue antihistamines and/or low-dose steroids as per institutional practice. Restart with lower concentration formulation. If recurrent consider IV bortezomib. For grade 4 toxicity permanently discontinue bortezomib and lenalidomide permanently		
	Desquamating (blistering) rash-any grade or erythema multiform ≥ Grade 3	Discontinue bortezomib and lenalidomide permanently. Hold other therapies.		

		Begin treatment with antihistamines and/or low-dose steroids as per institutional practice. If the toxicity resolves to ≤ Grade 1, restart other medications.		
Body System	**NCI-CTC Adverse Event and or Symptom and Category**	**Bortezomib**	**Lenalidomide**	**Dexamethasone**
Gastrointestinal	Constipation[b] ≥ Grade 3	Hold bortezomib therapy. Upon recovery to ≤ Grade 1, restart bortezomib at 1 dose-reduced level.		
	Diarrhea[c] ≥ Grade 3	Hold bortezomib and consider loperamide therapy. Upon recovery to ≤ Grade 1, restart bortezomib at 1 dose-reduced level.		
	Dyspepsia, gastric or duodenal ulcer, gastritis Grade 1-2 (requiring medical management)			Treat with histamine-2 blockers, sucralfate, or proton pump inhibitor. If symptoms persist despite above measures, decrease dexamethasone dose by 1 dose-level.
	Dyspepsia, gastric or duodenal ulcer, gastritis ≥ Grade 3 (requiring hospitalization or surgery)			Hold dexamethasone and consider treatment with histamine-2 blockers, sucralfate, or proton pump inhibitor. Restart and reduce by 1 dose level if symptoms are adequately controlled. If symptoms persist despite above measures, permanently discontinue.

	Acute Pancreatitis			Permanently discontinue dexamethasone.
Body System	**NCI-CTC Adverse Event and or Symptom and Category**	**Bortezomib**	**Lenalidomide**	**Dexamethasone**
Hematological	Neutropenia Grade 3 (without complications)	No dose reduction required of borte. Consider treatment with G-CSF.	Hold therapy with all drugs until recovery to baseline OR ≤ Grade 2. Consider G-CSF support. Upon recovery if isolated neutropenia, maintain lena at current dose level. If other haematologic toxicities present ore recurrent episode reduce lena by 1 dose level.	
	Grade 3 neutropenia associated with fever (≥38.5ºC) or Grade 4 neutropenia	Hold therapy with all drugs until recovery to baseline OR ≤ Grade 2. Consider G-CSF support. Upon recovery if isolated neutropenia, maintain lena at current dose level. If other haematologic toxicities present reduce lena by 1 dose level. Maintain borte at current dose. If recurrent episode, reduce both lena and borte by 1 dose-level.		
	Thrombocytopenia Grade 3 (without complications)	No dose reduction required for borte.	Reduce lena by 1 dose-level for the remainder of the cycle.	
	Platelet count ≤30 x 10^9 /L or ANC ≤0.75 x 10^9 /L on a borte dosing day	Hold borte dose.		
	Platelet count <25,000/µL (ie, Grade 4) or Grade 3 thrombocytopenia with bleeding	Hold therapy with all drugs until recovery to baseline OR ≤Grade 2. Upon recovery, reduce borte 1 dose level, hold lena for		

		remainder of the cycle and decrease by 1 dose level at start of next cycle.		
Body System	**NCI-CTC Adverse Event and or Symptom and Category**	**Bortezomib**	**Lenalidomide**	**Dexamethasone**
Infection	Herpes Zoster activation or reactivation ANY grade	Hold ALL therapies until lesions are dry. If not already underway, begin antiviral treatment. Once the infection is resolved all medications can be restarted without a dose reduction; however, continued antiviral prophylaxis is required[d].		
Musculoeskeletal	Muscle weakness >Grade 2 (symptomatic and interfering with function +/- interfering with activities of daily living)			Decrease dexamethasone dose by 1 dose-level. If weakness persists despite above measures, decrease dose by 1 further dose-level. If symptoms still persist, permanently discontinue dexamethasone.
Constitutional	Fatigue[a] ≥ Grade 3 (ie, severe fatigue interfering with activities of daily living)	Hold the dose until resolved to Grade ≤2. Consider reduction of lenalidomide/bortezomib by 1 dose-level or change to bortezomib dosing once per week.		
Metabolic	Hyperglycemia ≥ Grade 3			Treatment with insulin or oral hypoglycemics. If uncontrolled despite above measures, decrease dose by 1 dose-level until levels are satisfactory.
Neuro-psychological	Confusion or mood alteration >Grade 2 (interfering with function +/- interfering with activities of daily living)			Hold dexamethasone until symptoms resolve. Restart with 1 dose-level reduction. If symptoms persist despite above measures, permanently

					discontinue dexamethasone.
Body System	**NCI-CTC Adverse Event and or Symptom and Category**		**Bortezomib**	**Lenalidomide**	**Dexamethasone**
Neurological	Peripheral Neuropathy (Sensory or Motor) and/or Neuropathic Pain	Grade 1 (paresthesias and/or loss of reflexes) without pain or loss of function	No action required.		
		Grade 1 with pain or Grade 2 (interfering with function but not with activities of daily living)	Change schedule to once per week; if recurrence, reduce bortezomib by 1 dose-level		
		Grade 2 with pain or Grade 3 (interfering with activities of daily living)	Hold bortezomib until toxicity resolves to <Grade 2. When toxicity resolves, reinitiate with a reduction by 1 dose-level and change bortezomib treatment schedule to once per week		
		Grade 4 (permanent sensory loss that interferes with function) and/or severe autonomic neuropathy	Discontinue bortezomib permanently.		
Thromboembolic	Venous and /or pulmonary thrombo-embolism ≥ Grade 3 [Deep vein thrombosis or cardiac thrombosis intervention indicate; eg:			Stop until toxicity resolves and, if not already given, start anticoagulation therapy. Restart lenalidomide and dexamethasone at full dose after adequate anticoagulation	

	anticoagulation, lysis, filter, invasive procedure.]			
Body System	**NCI-CTC Adverse Event and or Symptom and Category**	**Bortezomib**	**Lenalidomide**	**Dexamethasone**
Renal Impairment	Moderate renal impairment- CrCl[f] 30-49 mL/min		Lenalidomide should be given at a dose of 10 mg daily[g] for Cycles 1-6	
	Severe renal impairment- CrCl[f] <30 mL/min (not requiring dialysis)		Lenalidomide should be given at a dose of 15mg every 48 hrs[g] for Cycles 1-6	
	End-stage renal disease- CrClf <30 mL/min (requiring dialysis)		Lenalidomide should be given at a dose of 5mg[g] daily for Cycles 1-6. On dialysis days, administer dose after dialysis.	
Body System	**NCI-CTC Adverse Event and or Symptom and Category**	**Bortezomib**	**Lenalidomide**	**Dexamethasone**
Other toxicities	Any reported ≥ Grade 3	Determine drug attribution of the toxicity and hold the therapy(ies) as appropriate. If toxicity resolves to ≤ Grade 1, resume therapy with 1 level of dose reduction for suspect drug.		

[a] Determine if fatigue is possibly not medication-related but due to an underlying cause (eg, infection, progression of disease, diarrhea, anemia, depression) and treat these symptoms/causes as appropriate.

[b] Prior to dose reduction of medications, consider/eliminate other possible causes of constipation.

[c] Prior to dose reduction of medications, consider/eliminate other possible causes (ie, bacterial or viral infections) of diarrhea.

[d] In the event that a subject is already receiving antiviral treatment at the time of the Herpes Zoster activation, consider switching to or adding another antiviral agent.

[e] The neurotoxicity-directed questionnaire is a useful tool for determining the presence and intensity of neuropathic pain and/or peripheral neuropathy from the subject's perspective. Neuropathic symptoms are more prominent than abnormalities on the clinical examination. After the subject completes the neurotoxicity-directed questionnaire, the questionnaire should be reviewed to assist with the evaluation of the onset and intensity of peripheral neuropathy and other neurotoxicities that may require intervention or dose modification.

[f] CrCl = creatinine clearance. Estimated by creatinine clearance as calculated by the Cockcroft-Gault formula and adjusted for body weight in subjects with a body mass index >30 kg/m^2. The eGFR (MDRD) or CKD-epi formulas can also be utilized to assess renal function.

[g] Consider escalating dose to 15mg daily after 2 cycles if well tolerated.

10.2 Isatuximab Dose Modifications

No dose reduction is allowed for SC isatuximab. Patients with a grade 4 or reoccurring grade 3 SC isatuximab injection (SC), allergic/hypersensitivity or CRS reaction(s) must have isatuximab SC permanently discontinued.

- A minor delay of the administration of single drug doses for example for logistic reasons is accepted. Minor delay means, that SC isatuximab should be administered in the weekly schedule +/- 1 day, in the biweekly schedule +/- 3 days and in the monthly schedule +/- 1 week.

- No dose reduction is allowed for SC isatuximab. Patients with a grade 4 or reoccurring grade 3 IV/SC isatuximab injection (SC), allergic/hypersensitivity or CRS reaction(s) must have isatuximab SC permanently discontinued. These patients should continue to receive bortezomib, lenalidomide and dexamethasone as per protocol.

Administration adjustments should be made if patients experience infusion reactions.

• In patients necessitating an intervention (Grade 2, moderate infusion reactions), a temporary interruption in the infusion should be considered and additional symptomatic medicinal products can be administered. After symptom improvement to grade ≤1 (mild), SARCLISA infusion may be resumed at half of the initial infusion rate under close monitoring and supportive care, as needed. If symptoms do not recur after 30 minutes, the infusion rate may be increased to the initial rate, and then increased incrementally.

• If symptoms do not resolve rapidly or do not improve to Grade ≤1 after interruption of isatuximab infusion, persist or worsen despite appropriate medicinal products, or require hospitalization or

are life-threatening, treatment with isatuximab should be permanently discontinued and additional supportive therapy should be administered, as needed.

10.2.1 Management of SC isatuximab injection-site reactions (ISR) and infusion reactions (IR)

Patients should routinely receive pre-medications prior to SC isatuximab infusion as described in above in section 8.2.2.1 to reduce the risk and severity of injection site reactions (ISR) and infusion reactions (IR) commonly observed with mAb treatment.

The IR observed in association with SC isatuximab (NCI-CTCAE version 5.0 term ‘allergic/hypersensitivity reactions’ or ‘cytokine release syndrome [CRS]/acute infusion reaction’) are AEs related to SC isatuximab with onset typically within 24 hours from the start of the SC injection.

IR and ISR manifestations reported in patients treated with SC isatuximab are listed in the current edition of the IMP IB. Since most ISR observed with SC isatuximab are Grade 1 no specific topical treatment is needed. However, if supportive local treatment is needed it may be delivered in accordance with the local guidelines for ISR with SC treatment.

In case of an IR while receiving or after SC isatuximab administration, additional medication can be provided for symptom relief as per investigator judgment including diphenhydramine 25 mg IV (or equivalent) and methylprednisolone 100 mg IV, IV fluids, vasopressors, oxygen, bronchodilators, and acetaminophen/paracetamol. These patients must be informed of the potential risk of recurrent IRs.

Patients with a grade 4 or reoccurring grade 3 SC isatuximab IR or ISR must have isatuximab permanently discontinued and appropriate therapy should be administered. Grade ≥3 and higher IRs must be reported as AESIs (see section XY). Recommendations for management of IRs are described in Table XY.

Injection site reaction (ISR) grading	**Recommendation**
Mild / Grade 1 Tenderness with or without associated symptoms (e.g., warmth, erythema, itching)	ISRs: Continuation of isatuximab administration. Report AE and AESI if the reaction lasts >24 hours.
	IRs: Continuation of isatuximab administration is per the judgment of the Investigator following close direct monitoring of the participant's clinical status. Isatuximab administration may be stopped at any time if deemed necessary. If stopped, IR will be classified as Grade 2 as per NCI-CTCAE v5.0 definition. Report AE.

Injection site reaction (ISR) grading	**Recommendation**
Moderate / Grade 2 Pain; lipodystrophy; oedema; phlebitis	ISRs: If during administration: Interrupt isatuximab administration and resume administration with pauses after recovery to G1 or better. Report AE and AESI. If post-administration: Continue isatuximab treatment after recovery to G1 or better. Report AE and AESI. IRs: If during administration: Definitive stop of current isatuximab administration without possibility to restart the injection. Give additional premedication with IV diphenhydramine 25 mg IV (or oral equivalent) and/ or IV methylprednisolone 100 mg (or oral equivalent) and/or other supportive care as needed. Report AE.

Injection site reaction (ISR) grading	Recommendation
Severe or life-threatening / Grade 3/4 Grade 3: Ulceration or necrosis severe tissue damage operative intervention indicated Grade 4: Life-threatening consequences; urgent intervention indicated	ISRs: Definitive discontinuation of SC isatuximab administration. Report AE /AESI. IRs: Definitive discontinuation of isatuximab SC administration. Give additional premedication with diphenhydramine 25 mg IV (or equivalent) and/ or IV methylprednisolone 100 mg (or equivalent) and/or epinephrine as needed. Report AE /AESI.

10.3 Iberdomide Dose Modifications

Guidelines for iberdomide dose modifications and dose reduction steps are provided in **Table 21** and **Table 19**, respectively.

Table 21 Guidelines for Dose Modifications for iberdomide during maintenance

Toxicity	Recommended action
Hematologic Toxicity	
Grade 3 neutropenia (ANC 500 - 1000/µL)	Continue Iberdomide dosing. Initiate G-CSF therapy as per standard practice.

Grade 4 neutropenia (ANC < 500/µL) or febrile neutropenia (fever ≥ 38.5°C and ANC < 1,000/µL)	Interrupt iberdomide. If the subject was not receiving GCSF therapy, initiate GCSF therapy. ANC must return to ≥ 1,000/µL to resume dosing. The dose of iberdomide may be maintained if neutropenia was the only iberdomide-related toxicity requiring a dose modification and GCSF treatments are continued. If a dose reduction is considered appropriate by the investigator, interrupt iberdomide for the remainder of the cycle. Start the reduced dose on Day 1 of the next cycle.
Grade 4 thrombocytopenia (platelet count < 25,000/µL) or Grade 3 thrombocytopenia with bleeding or any requirement for a platelet transfusion	Withhold the dose of iberdomide for remainder of cycle. Decrease by one dose level when restarting treatment. Platelet count must return to ≥ $70x10^9$/L before restarting.
Non-hematologic Toxicity	
Rash = Grade 3	Withhold the dose of iberdomide for remainder of cycle. Rash must be resolved or improved to ≤ Grade 1 before restarting. Decrease by one dose level when restarting treatment. Symptomatic or prophylactic measures might also be applied according to standard practices.
Rash = Grade 4 or blistering	Permanently discontinue IP.
Thrombosis/embolism ≥ Grade 3	Withhold the dose of iberdomide for remainder of cycle. Anticoagulation therapy should be adapted based on the clinical and investigational results.

Iberdomide-related peripheral neuropathy = Grade 3	Withhold the dose of iberdomide for remainder of cycle. Iberdomide may be restarted when recovered to ≤ 2 and with one dose level reduction.
Iberdomide-related peripheral neuropathy = Grade 4	Discontinue the subject from IP.
Other ≥ Grade 3 iberdomide-related adverse events [a, b]	Interrupt iberdomide. If a dose reduction is appropriate, interrupt iberdomide for the remainder of the cycle. Start the reduced dose on Day 1 of the next cycle. Adverse event should be resolved or improved to Grade ≤ 1 or baseline before restarting dose.

AE = adverse events; ANC = absolute neutrophil count; GCSF = granulocyte colony-stimulating factor; IP = investigational product.

[a] For Grade 3/4 AEs that are not considered to be related to iberdomide, the Investigator should consult with the Coordinators for dose interruptions and reductions.

[b] Not applicable to Second Primary Malignancies (SPMs) where dosing modification is at the discretion of the Investigator and in consideration of any SPM-related procedures/therapy.

To initiate a new cycle of iberdomide, the absolute neutrophil count must be ≥ 1,000 µL/mL with or without granulocyte colony-stimulating factor (GCSF), the platelet count must be ≥ 70,000/µL, and non-hematologic AEs must be resolved or improved as outlined in **Table 21**.

If recovery from toxicities is prolonged and iberdomide dose withholding is > 14 days, then the dose of iberdomide may be decreased by one dose level when dosing is resumed in the new cycle.

Dose delays of iberdomide for > 28 days for any reason should be discussed with the Clinical Coordinators.

If the start of a new cycle is delayed > 7 days, an unscheduled visit must be performed for efficacy assessments prior to initiation of the next cycle.

For subsequent cycles following a delayed cycle, efficacy assessments should be performed at the start of each new cycle.

10.4 Lenalidomide Dose Modifications during maintenance

Lenalidomide dose may be decreased as described in **Table 18**; if neutropenia was the only toxicity found requiring dose reduction, then the first dose reduction will be the dose level -1, followed by levels -2, -3 and -4, as described in **Table 22**.

Dose adjustment guidelines for lenalidomide during maintenance treatment are given in the following tables for the purpose of treating any possible toxicity that occurs:

Table 22 Therapeutic guidelines concerning thrombocytopenia and neutropenia during maintenance treatment

Thrombocytopenia	Recommended action
< 30 x 10^9/L platelets	Interrupt lenalidomide treatment Resume lenalidomide at dose level -1 when it recovers to ≥ 50 x 10^9/L
For each subsequent drop below 30 x 10^9/L	Definitely interrupt lenalidomide
Neutropenia	
G4: < 0.5 x 10^9/L (1st occurrence)	Interrupt lenalidomide treatment and start G-CSF. Resume lenalidomide at the same dose when it recovers to ≥ 1 x 10^9/L. Continue G-CSF as per standard practice

G4: < 0.5 x 10^9/L (subsequent occurrences)	Interrupt lenalidomide treatment and start G-CSF. Resume lenalidomide with a decrease in one dose level when it recovers to ≥ 1 x 10^9/L. Continue G-CSF as per standard practice
Any G 3/4 non-hematological toxicity related to lenalidomide	Interrupt Lenalidomide until the toxicity has resolved to ≤ grade 2 or baseline and restart with a dose reduction, bearing in mind that the reduction level -1 applies only for neutropenia

In order to access to the clinical and non-clinical information related to the efficacy and safety of lenalidomide, please refer to the last Investigator Brochure version.

10.5 Bortezomib Dose Modifications

Detailed guidance for bortezomib dose reductions and delays is shown in

Table 20 and **Table 23**. Dose adjustments should be based on the highest grade of toxicity that is attributed to bortezomib.

Table 23 Dose Modification Guidelines for Neuropathic Pain Associated with Bortezomib Treatment

		0	1	2	3	4
		Normal	Asymptomatic: paresthesias, weakness and/or loss of reflexes without pain or loss of function	Sensory alteration or paresthesias interfering with function	Sensory alteration or paresthesias interfering with activities of daily life	Disabled
0	Normal		↓1	↓1	▬ ↓2 ◺	X
1	Mild level without interfering with function	↓1	↓1	↓1	▬ ↓2 ◺	X
2	Moderate pain requiring analgesics and interfering with function but not with activities of daily living	1	2	2	2	X
3	Severe pain requiring analgesics and interfering with activities of daily living	2	▬ ↓2 ◺	▬ ↓2 ◺	X	X
4	Disabled	X	X	X	X	X

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	No action
1	One level dose reduction (from 1.3 to 1.0 mg/m^2)
2	Two levels dose reduction (from 1.3 to 0.7 mg/m^2)
X	Discontinue treatment with bortezomib
▬	Discontinue bortezomib until toxicity has resolved to grade 1 or better
◺	Scheme change: Administer bortezomib once a week; if change already existed, then administer every 15 days

10.6 Dexamethasone Dose Modifications

Dose adjustments should be based on the highest grade of toxicity that is ascribed to dexamethasone as noted in

Table 20.

- Starting dose: 40 mg

- Dose level -1: 20 mg

- Dose level -2: 10 mg

- Dose level -3: discontinue dexamethasone permanently (premedication for isatuximab may continue)

In case dexamethasone should be discontinued, either follow with methylprednisolone for isatuximab premedication or stop definitively steroids if no IR.

10.7 General Dose Modifications and Management Guidelines

See **Table 24** for general dose modification and management guidelines for drug-related adverse events not otherwise specified.

Table 24 General Dose Modification and Management Guidelines for Drug related Adverse Events Not Otherwise Specified[a]

Severity	Management	Follow-up
Grade 1	• Administer symptomatic treatment as appropriate • Continue study drug(s)[a]	Provide close follow-up to evaluate for increased severity, no dose modification necessary
Grade 2	• Administer symptomatic treatment • Investigate etiology • Consider consulting subspecialist, and/or diagnostic procedure	*Symptoms resolved in ≤7 days:* Continue after resolution at the current dose *Symptoms ongoing >7 days or worsening:* ○ Delay study drug [b], or consider dose reduction by 25% If recovery takes >3 weeks, consult with Coordinators • If symptoms continue or worsen to Grade 3-4, see below
Grade 3	• Provide appropriate medical treatment • Consider Consulting subspecialist	Delay treatment till recovery to G1 or less. Consider dose reduction. Exceptions: Participants who develop G3 toxicities which respond to standard treatment and resolve to ≤G1 within 48 hours may continue treatment at scheduled or reduced dose
Grade 4	• Provide appropriate medical treatment • Consider Consulting subspecialist	Interrupt treatment. Further treatment with only allowed on individual basis it is agreed that benefits outweigh the risks for a given participant

a. Treatment-related decisions can be made based on local laboratory results if central results are not available or delayed.

b. In case a dose is delayed, the participant should wait for the next scheduled dose to resume treatment.

10.8 Guidance on Dose Delays

A cycle is defined as a planned dosing period in which either isatuximab + VRD (Arm A), VRD (Arm B) or isatuximab + iberdomide + VD (Arm C) are administered and is 4 weeks in duration. In any dosing delay scenario, Q4W assessments should continue to be carried out as described in the SoA (Section 11). For treatment delays within the treatment window, subsequent treatment dates will not be adjusted.

Patients may have isatuximab/iberdomide/lenalidomide/bortezomib/dexamethasone dose omission within a cycle if toxicity occurs and does not recover or returns to Grade required in the guidelines for dose modifications on the day of planned administration or within the following day for the weekly administration, or within the following 3 days for every 2 weeks and 4 weeks administrations.

Within a cycle, these delays in cases of unresolved toxicity at the time of planned re-administration are permitted, otherwise infusion is omitted. Patients will receive the next infusion after recovering from the toxicity. No dose reductions for isatuximab will be permitted. Bortezomib, dexamethasone, lenalidomide, and iberdomide may be reduced according to toxicity.

11 SCHEDULE OF ASSESSMENTS

A signed, written informed consent form must be obtained from the participant prior to any study-specific procedures or assessments being performed.

Assessments must be performed prior to administration of any study agent. Study assessments and agents should be administered within +/- 3 days of the protocol-specified date, unless otherwise noted.

Table 26 Study schedule

		INDUCTION		PRE-TRANSPLANTATION
EVALUATION	**SCREENING**	**4 cycles (arm A/C) or 6 cycles (arm B) or ERI before ASCT**		**After the completion of the last induction cycle and just before transplantation (*)**
	-21 to -1	**Day 1 of each 4-weekcycle**	**Day 11 (+/- 4 days) of 4-week cycles**	
Informed consent	X			
Medical history	X			
Physical exam (1) and ECOG	X	X		X
Weight, height and BSA	X	X		X
Vital signs	X	X		X
Electrocardiogram, echocardiogram (2)	X			
Blood typing / ABO/Rh	X			
Bone disease evaluation: X-ray, low dose CT or PET-CT (3)	X			
Evaluation of plasmacytomas by PET-CT, CT, MRI (4)	X	(4)		X
Blood sample: hemogram and biochemistry (5)	X	X	X	X
Serology: HBsAg, HBcAb, HCV	X			
Pregnancy test (6)	X	X		X
SPEP/IF and UPEP/IF (7)	X	X		X
sFLC (8)	X	X		X
Serum quantitative Immunoglobulins (immunoglobulin dosage, IgG, IgA, IgM)	X	X		X
B_2-microglobulin, LDH and CRP	X			
Bone marrow aspiration (9)	X			X
Diseasee evaluation based in clinical, paraprotein and biochemical evolution		X		X
Adverse events (10)	X	X	X	X
Survival and Safety Follow-up				

* Applicable after the first four cycles in arms A and C, after six cycles in arm B, and to patients in ERI before transplantation.

	CONSOLIDATION			CONTINUOUS TREATMENT OR MAINTENANCE	EOT	FOLLOW-UP
EVALUATION	**3 months after transplantation (2 months after discharge).**	**2 cycles as induction (plus 10 more cycles for arm B, VRD extended)**		**Continued treatment (Arms A & C and patients in ERI of Arm B) or maintenance (Arm B without ERI)**	**End of study visit, end of treatment or treatment interruption**	
	Day 1 of cycle 1	**Day 1 of cycle 2**	**Day 11 (+/- 4 days) of both consolidation cycles**	**Day 1 of every cycle**		**Every 3 months**
Physical exam (1) and ECOG	X	X		X	X	
Weight, height and BSA	X	X		X	X	
Vital signs	X	X		X	X	
Echocardiogram (a MUGA scan is also valid)						
Bone disease evaluation: X-ray, low dose CT or PET-CT (3)						
Evaluation of plasmacytomas by PET-CT, CT, MRI (4)	(4)			(4)	(4)	
Blood sample: hemogram and biochemistry (5)	X	X	X(12)	X	X	
Pregnancy test (6)	X	X		X		
SPEP/IF and UPEP/IF (7)	X	X		X	X	
sFLC (8)	X	X		X	X	
Serum quantitative Immunoglobulins (immunoglobulin dosage, IgG, IgA, IgM)	X	X		X		
B_2-microglobulin, LDH and CRP (11)	X			X		
Bone marrow aspiration (9)	X	(9)		(9)	X	
Response assessment	X	X		X	X	
Adverse events (10)	X	X		X	X	
Survival and Safety Follow-up						X

(1) A complete physical exam will be carried out during the screening visit and thereafter targeting symptoms in subsequent study visits.

(2) This test will be repeated throughout the study as necessary. A MUGA instead of echocardiogram is also valid.

(3) At screening, a series of bone x-rays or a low-dose CT or PET-CT scan is mandatory. This will be repeated at the investigator's discretion. The bone series will not be performed after screening unless there are clinical data that suggest an increase or appearance of new bone lesions.

(4) Patients suspected of having extramedullary disease will undergo PET/TC, contrasted whole body CT scan or whole-body MRI as per investigation discretion and center availability. The imaging evaluation should be performed every 12 weeks, for disease evaluation or before if it is clinically indicated. The same imaging technique should be use throughout disease course for disease evaluation.

(5) Complete blood count, hemoglobin, white blood cell count with differential count and platelet count. Biochemistry: urea, creatinine, creatinine clearance, uric acid, bilirubin, alkaline phosphatase, LDH, B2-Microglobuline, AST, ALT, albumin, sodium, chloride, calcium, potassium, glucose, magnesium, GGT.

(6) A pregnancy test (serum or urine) will be carried out during the screening visit and repeated every 28 days. The test will not be necessary if menopause is confirmed, defined as the absence of menstruation for more than 2 years, or surgically-induced menopause.

(7) After the end of treatment, UPEP/IF will be evaluated every 3 months according to routine clinical practice, until disease progression, in the case that disease progression has not previously been reported.

(8) Determination of serum free light chains (sFLC) will be carried out at the screening visit and the time of documenting CR by negative IF. For those patients for which this is the detection technique for measurable disease (oligosecretory or non-secretory MM), determination of sFLC will be carried out on day 1 of each new treatment cycle (± 4 days) and at the end of treatment.

(9) BM tests will include morphological analyses, cytogenetic tests (only at screening), molecular biology and flow cytometry. BM aspiration is mandatory during the screening visit, after induction, 3 months after transplantation coinciding with the pre-consolidation evaluation,
and:
Arms A and C
- After being administered the two cycles of consolidation
- After the end of cycle 7 of continuous treatment
- After the end of cycle 12 of continuous treatment, which coincides with the assessment of the primary Endpoint,

And In Arm B, patients without ERI
- After the 2nd consolidation cycle
- After the 5th extended VRD cycle
- After the 10th extended VRD cycle, which coincides with the assessment of the primary Endpoint.

And In Arm B, patients with ERI
- Every 5 cycles of iberdomide / Isatuximab /
dexamethasone.

Beyond the evaluation of the primary endpoint, a BM aspirate will be done yearly, including morphology, flow cytometry, molecular biology (centralized in the reference laboratory).

(10) This applies to all Adverse Events (non-serious related events to the study drugs and related and unrelated serious events), that occur throughout the study and at least 28 days after the last dose of study treatment.

(11) B2-microglobulin, and CRP should be done only at screening.

(12) From cycle 3 of induction forward, it is not compulsory, it depends on the patient's haematological toxicity and the Principal Investigator's (or their designee) criteria.

11.0 Screening/Baseline visit

Each patient must sign and date the informed consent form before the study procedure. However, all procedures that are part of the usual routine in the diagnosis of MM and have been performed before signing the informed consent are valid for selection, as long as they have been performed in a window within the 35 days prior to onset of the treatment. In a candidate patient, pending diagnosis due to lack of bone marrow aspiration, the sample will be sent to the reference laboratories, provided that the informed consent has been previously signed.

During the Selection Visit:

• Presentation of the signed informed consent

• Medical history, including comorbidities. Concomitant medication should not be recorded in the eCRD, except for those that may have anti-myeloma activity such as corticosteroids or clarithromycin

• Complete physical examination, weight, height, vital signs, assessment of functional status (ECOG).

• Electrocardiogram and echocardiogram (a MUGA is also valid)

• Radiological bone series and reference PET / CT, the latter provided that the examination is accessible either in the center or in a subsidized center (for this examination the screening period is extended)

• Any other radiological study (low resolution CT, MRI) that the IP deems necessary to document extramedullary disease. If low resolution CT or MRI are carried out, it will not be necessary to perform radiological bone series.

• Hemogram

• Blood type, Rh, and IAT or phenotyping.

• Serology to rule out HBV, HCV and HIV.

• Dosage of immunoglobulins: IgG, IgA, IgM.

• General biochemistry (including C-reactive protein and beta-2-microglobulin)

• Complete protein study in serum and urine: total proteins, serum electrophoresis, 24-hour proteinuria and electrophoresis in urine, immunoglobulin dosage, immunofixation in serum and urine

• Serum free light chains (sFLC)

• Bone marrow aspirate, including morphology, flow cytometry (with cell cycle analysis), cytogenetics, and molecular biology. In addition to the local evaluation, these studies will be carried out centrally in reference laboratories (see instructions in APPENDIX 3)

• Pregnancy test (see **APPENDIX 9** and **APPENDIX 11**).

11.1 Common procedures to all treatment periods, including induction, ASCT, consolidation, and continued treatment in Arms A/C, maintenance in Arm B, and patients in the ERI program

During the entire Treatment Period: once the disappearance of the monoclonal component has been confirmed, both in electrophoresis and immunofixation, a determination of free light chains in serum and a BM will be necessary. PET/CT will be performed coinciding with MRD evaluations.

The BM must be forwarded to the predetermined reference laboratory. The BM studies will include: morphology, flow cytometry for high resolution minimal residual disease, and other determinations adapted to the lines of biological studies under development (see **APPENDIX 3** and **APPENDIX 4**)

11.2 General procedures for the induction period

Applicable to the first four cycles in arms A and C, to the six cycles in arm B, and to each of the cycles of patients in ERI before transplantation, and for all patients just before ASCT.

On day 1 of each cycle (+/- 4 days), the following evaluations will be carried out:

• Physical exploration

• Assessment of functional status (ECOG)

• Hemogram

• General serum biochemistry

• Serum and urine protein electrophoresis to quantify the monoclonal component. Immunofixation (IF) and FLCs if the electrophoretic spectrum (EEF) is negative.

• Administration of bisphosphonates if clinically necessary.

• Pregnancy test if applicable (see **APPENDIX 9** and **APPENDIX 11**).

• Toxicity / adverse effect data according to common NCI toxicity criteria that occurred during the previous cycle.

• With platelet values in the lower limits of normality prior to the start of the cycle, coinciding with the administration of the 4th dose of bortezomib (cycle day 11), a blood count will be performed following the criterion of the investigator.

• Assessment of the response. Before the beginning of each cycle, an estimate of the response is necessary according to the IMWG criteria.

11.3 Procedures after the induction

Applicable after the first four cycles in arms A and C, after six cycles in arm B, and to patients in ERI before transplantation, and previously ASCT.

At least four weeks after the end of the last induction cycle, the following evaluations will be carried out:

• Physical exploration

• Hemogram.

• General serum biochemistry, including creatinine, calcium, LDH.

• Serum and urine protein electrophoresis to quantify the monoclonal component. Immunofixation (IF) if the electrophoretic spectrum (EEF) is negative.

• Determination of free light chains in serum.

• The bone series will not be performed unless there are clinical data that suggest an increase or appearance of new bone lesions.

Although the radiological bone series is still acceptable in some circumstances, in general, the recommended imaging technique in evaluating response and progression is PET/ CT. See Appendix 10 for the details of the moments in which it is indicated and the description of the procedures to follow to send the images to the centralized reading platform.

• In the case of extramedullary plasmacytomas, the same technique that was used at the time of diagnosis will be performed to evaluate the response. The evolution of the plasmacytoma after the third treatment cycle should also be documented using the same diagnostic test (CT, MRI or

PET/CT). If it is negative at this time, it will not be necessary to confirm it at the end of the treatment. The preference for PET/CT is reiterated

• BM aspirate including morphology, flow cytometry, molecular biology (if applicable). BM aspirate will be centralized in the reference laboratory.

• Toxicity / adverse effects according to the common NCI toxicity criteria presented during the previous cycle.

• If the platelet count is normal on day 1 of each cycle, it is not necessary to perform control blood counts every day of bortezomib administration. In this case, it will only be done on day 11 of the cycle [prior to the 4th dose of bortezomib]

• Assessment of functional status (ECOG).

• Assessment of the response to the disease.

• Pregnancy test (see **APPENDIX 9** and **APPENDIX 11**).

11.4 General procedures during mobilization

The pertinent evaluations will be carried out according to the practice of each Center.

11.5 Procedures during autologous transplantation

• Before transplantation, a complete evaluation of the disease will have been carried out (see section on procedures after induction)

• The rest of the pre-transplant tests and procedures, including reinfusion of hemopoietic progenitors and supportive therapies, to be carried out will be the usual ones according to the practice of each center

During conditioning and transplantation:

• The usual clinical and analytical follow-up will be carried out according to the practice of each center.

• Relevant toxicity will be recorded during the procedure, in particular grade III-IV mucositis, grade of gastrointestinal toxicity, bacteremia, pneumonia, fungal infections, and hepatic veno-occlusive disease.

11.6 Procedures after autologous transplantation

Three months after transplantation (date prior to consolidation), the following evaluations will be performed:

• Hemogram.

• General serum biochemistry, including creatinine, calcium, LDH.

• Protein study (electrophoresis of proteins in serum and urine to quantify the monoclonal component. Immunofixation (IF) if the electrophoretic spectrum (EEF) is negative).

• The bone series will not be performed unless there are clinical data that suggest an increase or appearance of new bone lesions.

Although the radiological bone series is still acceptable in some circumstances, in general, the recommended imaging technique in evaluating response and progression is PET/ CT. See ki0 for the details of the moments in which it is indicated and the description of the procedures to follow to send the images to the centralized reading platform.

• In the case of extramedullary plasmacytomas, the same technique that was used at the time of diagnosis will be carried out to evaluate the response (if in the evaluation after the third cycle or after induction the plasmacytoma had disappeared, the imaging test will not be repeated).

• BM aspirate including morphology, flow cytometry, molecular biology (centralized)

• Toxicity / adverse effects according to the common NCI toxicity criteria presented during the previous cycle.

• Assessment of functional status (ECOG).

• Assessment of the response to the disease.

• Pregnancy test (see **APPENDIX 9** and **APPENDIX 11**).

11.7 Procedures in post-transplant treatments.

Applicable during the 2 cycles of consolidation and during the first 12 cycles of continuous treatment in Arms A and C, as well as during the two cycles of consolidation and the 10 cycles of extended VRD in Arm B (treatments between transplantation and up to evaluation of the primary END POINT) and all patients in ERI, (both those included before or as those included in ERI after transplantation).

After the post-ASCT evaluation, as long as the clinical and analytical conditions of the patient allow, two cycles of consolidation will be administered. Ideally, consolidation will begin three months after ASCT. In the event of a general condition or cytopenia secondary to the transplant procedure, consolidation may be delayed for up to 6 months post-ASCT.

On day 1 of cycle 2 of consolidation and at the beginning of each cycle until reaching the evaluation point of the PRIMARY ENDPOINT:

• Physical examination aimed at the symptoms.

• Hemogram.

• General serum biochemistry, including creatinine, calcium, LDH.

• Serum and urine protein electrophoresis to quantify the monoclonal component. Immunofixation (IF) if the electrophoretic spectrum (EEF) is negative.

• In each cycle and coinciding with the 4th administration of bortezomib [Velcade (day 11)], a complete blood count should be performed.

• Toxicity / adverse effects according to the common terminology toxicity criteria presented during the previous cycle.

• If the platelet count is normal on day 1 of each cycle, it is not necessary to perform control blood counts every day of bortezomib administration. In this case, it will only be done on day 11 of the cycle [prior to the 4th dose of bortezomib]

• Assessment of functional status (ECOG).

• Assessment of the response to the disease.

• Pregnancy test (see **APPENDIX 9** and **APPENDIX 11**).

11.8 Calendar of centralized controls from BM after the transplant and until the assessment of the primary endpoint.

Regardless of the BM carried out for the confirmation of CR, and for the adequate monitoring of the evolution of the MRD, the following centralized BM controls have been foreseen:

After transplantation, approximately 3 months after reinfusion, coinciding with the pre-consolidation evaluation. As noted in Section 11.3, a BM control will be also carried out before ASCT.

• **In Arms A and C**

- After being administered the two cycles of consolidation

- After the end of cycle 7 of continuous treatment

- After the end of cycle 12 of continuous treatment, which coincides with the assessment of the primary Endpoint

• **In Arm B, patients without ERI and patients enrolled in the ERI program before transplantation**

- After the 2nd consolidation cycle (2nd post-ASCT cycle in ERI patients)

- After the 5th extended VRD cycle (7º post-ASCT cycle in ERI patients

- After the 10th extended VRD cycle (12 post ASCT cycle in ERI patients), which coincides with the assessment of the Primary Endpoint

• **In Arm B, patients included in the ERI program after transplantation**

-Every 5 cycles of iberdomide / Isatuximab / dexamethasone

11.9 Procedures during the periods of continued treatment (Arms A & C and patients in ERI of Arm B) or maintenance (Arm B without ERI) beyond the evaluation of the primary endpoint

Applicable to all patients (in continuous treatment from arms A and C or patients in maintenance treatment or in ERI from Arm B) until reconsideration of the duration of continuous / maintenance treatments, expected after 12 months. of the evaluation of the primary endpoint of the last recruited patient.

Day 1 of each continuation or maintenance cycle

• Physical examination aimed at the symptoms

• Hemogram

• General serum biochemistry, including creatinine, calcium, LDH.

• Serum and urine protein electrophoresis to quantify the monoclonal component. Immunofixation (IF) if the electrophoretic spectrum (EEF) is negative.

• Toxicity / adverse effects according to the common NCI toxicity criteria presented during the previous cycle.

• Assessment of functional status (ECOG).

• Assessment of the response to the disease.

• Pregnancy test (see **APPENDIX 9** and **APPENDIX 11**)

• A BM aspirate including morphology, flow cytometry, molecular biology (centralized in the reference laboratory) will be done 6 months after the evaluation of the primary endpoint, time that coincides with the start of continuous treatment (arms A and C) or maintenance (arm B). Subsequently, the evaluations will have an annual periodicity. Coinciding with this BM evaluations, a new PET/TC examination is recommended.

11.10 End of treatment visit (EOT) or treatment interruption visit

Patients who discontinue treatment for any reason should perform an end of treatment visit 2 weeks (+/- 5 days) after the last dose of treatment. A disease workup is required which will include:

- Vital signs (blood pressure, temperature, respiratory rate, heart rate, Oxygen saturation in room air)
- Anamnesis and physical exam.
- ECOG performance status (see **APPENDIX 6**).
- Collection of blood samples for the following laboratory tests: complete blood count and biochemistry.
- SPEP/IF. Collection of blood samples to evaluate monoclonal protein, quantify immunoglobulins and perform immunofixation tests if the monoclonal protein has disappeared during electrophoresis.
- UPEP/IF. Collection of 24-hr urine sample to determine the presence of proteinuria, evaluate the monoclonal protein, and for immunofixation tests.
- BM aspiration to carry out morphological analysis, flow cytometry, and molecular biology tests if reason of discontinuation was disease progression.

- Determination of serum free light chain (sFLC) concentration.
- Documentation of concomitant medications and AEs in accordance with CTCAE criteria v5.0.

A window of +/- 4 days is permitted to complete the required tests.

11.11 Medical history and physical examination

Relevant medical history will include information on the subject's concurrent medical conditions. The medical history will include all the demographic data and a summary of the pathology.

The baseline physical examination will be a complete physical examination. During all visits, a physical examination is mandatory aimed according to the symptoms indicated by the patient.

11.12 Weight, height and body surface

The baseline assessment includes height and weight. Patients must be weighted subsequently every time any change in weight is reported in which a significant change in body surface appears, aimed at a proper medication adjustment. Body surface is calculated according to a nomogram (see **APPENDIX 8**).

11.13 Vital Signs

Vital sign measurements must include systolic and diastolic blood pressure, temperature, pulse rate, respiratory rate and oxygen saturation. Vital signs must be measured after resting for at least 5 minutes. Vital signs must be measured more frequently if warranted by the clinical condition of the participant.

11.14 Performance status

ECOG Performance Status will be determined using the score present in the **APPENDIX 6**.

11.15 Laboratory assessments: hemogram, biochemistry and pregnancy test

The following blood samples for laboratory assessments will be performed during the treatment when required, on the screening visit, and during treatment period, on day 1 of each cycle (± 4 days). These assessments will be repeated at the discretion of the investigator if there is a clinical reason.

The following lab parameters will be assessed:

- Hemogram: Complete blood count, hemoglobin, white blood cell count with differential count and platelet count.

- Biochemistry: urea, creatinine, uric acid, bilirubin, alkaline phosphatase, LDH, AST, ALT, sodium, chloride, calcium, potassium and, glucose, magnesium, GGT.

- Medically supervised pregnancy test, in females with childbearing potential with a minimum sensitivity of 25 IU/ml. A pregnancy test (serum or urine) will be carried out only in women of childbearing potential. The test will not be necessary if menopause is confirmed, defined as the absence of menstruation for more than 2 years, or surgically-induced menopause.

11.16 Electrocardiogram

During the screening visit, an electrocardiogram must be performed, and will be repeated when clinically indicated.

11.17 Echocardiogram

An echocardiogram will be performed when clinically indicated based on patient condition. A MUGA scan is also valid.

11.18 Serum protein electrophoresis (SPEP)/Immunofixation (IF)

Blood samples will be collected for immunoglobulin quantification as well as monoclonal component quantification and identification. Immunofixation will be necessary only if electrophoresis is negative.

11.19 Urine protein electrophoresis (UPEP)/Immunofixation (IF)

24-hour urine samples will be collected for Bence Jones protein identification and quantification during the screening period and during treatment.

11.20 Serum free light chains

Serum free light chain assessments will be performed in patients with oligosecretory/non-secretory MM or pure Bence Jones MM in order to evaluate if their disease is measurable through this technique. In those patients for which this is the detection technique for measurable disease, sFLC assessments will be performed during treatment period. In the rest of patients with measurable disease by conventional techniques, it will only be mandatory at the time of documenting CR by negative IF in order to evaluate if the response qualifies for stringent CR.

11.21 BM Aspirate

A BM aspirate will be performed to all patients during the screening visit, to confirm CR and in the timepoints required in the protocol. BM assessments will include: morphology, flow cytometry, cytogenetic and molecular biology. All assessments will be performed in a centralized manner in reference laboratories (**APPENDIX 4**).

11.21.1 Bone disease evaluation: X-ray, low-dose CT or PET/CT

A complete skeletal survey through radiographic assessments can be performed during the screening visit according to the local guidelines at each center. It is highly recommended to use more sensitive techniques, like low-dose CT or PET-CT for the evaluation of bone disease. PET-CT would be the first recommendation.

Although if available PET / CT is the recommended technique, other evaluation techniques (such as WBMRI or CT scan contrasted WB) may be used for extramedullary disease evaluation per investigator decision. In case of presence of extramedullary disease, the same radiologic method should be repeated every 12 weeks for disease evaluation. In the absence of extramedullary involvement, these tests may be repeated if clinically indicated

Subsequently, it will only be necessary to repeat the skeletal survey if there is clinical data suggesting the appearance of new bone lesions, or at any time, at the discretion of the investigator.

Sometimes patients will present disease progression which manifests through pain symptoms due to bone alterations. Thus, progression of disease may be documented in these cases through bone surveys or other radiographic assessments, depending on the symptoms experienced by the patient. If the progression of disease diagnosis is obvious in the radiographic assessments, it is not necessary to repeat radiographic assessments for confirmation. In those cases, in which changes are more subtle, it is possible that it will be necessary to repeat radiographic assessments during 1-3 weeks.

During the screening visit, it is important to document the myeloma disease locations, especially in extramedullary areas. It is possible that this documentation requires the performance of clinical examinations, computerized tomography or magnetic resonance imaging assessments. In these cases, in order to document the response, it is necessary to repeat the same test performed for its documentation and assessment.

11.22 Beta-2 microglobulin, C reactive Protein, and LDH

Blood samples for beta-2 microglobulin, reactive C protein, and LDH must be performed during the screening visit and at any other time during the study, at the discretion of the investigator.

11.23 Safety Assessments

Safety assessments include the following:

- Complete physical examinations, including height (cm) and weight (kg)
- Vital signs (blood pressure, body temperature, pulse rate, respiratory rate and oxygen saturation)
- ECOG performance status
- Clinical chemistry, hematology, and other laboratory tests
- Serum beta-human chorionic gonadotropin (β -HCG) pregnancy test for female participants of childbearing potential only
- 12-lead ECG at Screening.

Pharmacovigilance during the trial will be carried out pursuant to regulations established in Royal Decree 1090/2015 on Clinical Trials on recording, evaluating and reporting of adverse events. The sponsor, through the principal investigator, will evaluate Serious Adverse Events (SAE) using product safety data sheet. It will report, via the CRO, those SAEs that meet the expedited reporting criteria (serious, unexpected and related to the treatments) to the health authorities, with help from the clinical monitor. The sponsor will also be responsible for notifying the competent authorities [AEMPS (Spanish Agency of Medicines and Health Products)], the reference CREC, hospital CREC, competent authorities of the autonomous communities) and the principal investigators (see **APPENDIX 2**).

Refer to the SoA for the timing and frequency Clinical laboratory tests to be performed. Details for the preparation and shipment of samples for central laboratory assessments will be provided in the SRM/Clinical Lab Manual. The investigator must review the laboratory report, document this review, and record any clinically relevant changes occurring during the study in the AE section of the eCRF. The laboratory reports must be filed with the source documents. Clinically significant abnormal laboratory findings are those which are not associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition. Pregnancy Testing (WOCBP Only)

All females of childbearing potential, including those who commit to total abstinence, must undergo medically-supervised pregnancy tests with a minimum sensitivity of 25 mIU/mL, as detailed below in **APPENDIX 9** and **APPENDIX 11**.

Before beginning treatment with the study drug

Female patients:

All WCPB must have two negative pregnancy tests (with a sensitivity of at least 25 mIU/mL) before starting the study treatment. The first pregnancy test should be performed 10-14 days prior to initiation of treatment, and the second test in the 24 hours prior to initiation of treatment. The patient cannot receive treatment with the study drug until the investigator has verified that the results of these pregnancy tests are negative.

During participation in the study and for 28 days following the end of treatment with the study drug

Female patients:

- All WCPB with regular menstrual cycles or who have no menstrual cycles agree to undergo pregnancy tests during the first 28 days of their participation in the study, every 28 days thereafter throughout the study, after suspension of the study treatment, and at 28 days after treatment has finished. If the patient's menstrual cycles are irregular, pregnancy tests should be carried out once a week during the first 28 days, then every 14 days over the course of study, at the time treatment with the study drug has stopped, and at 14 and 28 days thereafter.

12 ADVERSE EVENT REPORTING

12.0 Definition of Adverse Events

According to GCP, an adverse event (AE) is defined as follows: Any untoward medical occurrence in a subject administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not related to the medicinal (investigational) product.

An AE may be:

-New symptoms/ medical conditions

-New diagnosis

-Changes of laboratory parameters

-Intercurrent diseases and accidents

-Worsening of medical conditions/ diseases existing before clinical trial start

-Recurrence of disease

-Increase of frequency or intensity of occasional diseases.

A pre-existing disease or symptom will not be considered an adverse event unless there will be an untoward change in its intensity, frequency or quality.

12.1 Surgical procedures themselves are not AEs; they are therapeutic measures for conditions that require surgery. The condition for which the surgery is required may be an AE. Planned surgical measures permitted by the clinical trial protocol and the condition(s) leading to these measures are not AEs, if the condition leading to the measure was present prior to inclusion into the trial.Serious Adverse Events (SAEs)

AEs are classified as "non-serious" or "serious".

A serious adverse event (SAE) is one that at any dose:

Results in death

Is life-threatening (Note: the term life-threatening refers to an event in which the subject was at risk of death at the time of event and not to an event which hypothetically might have caused death if it was more severe)

Requires hospitalization or prolongation of existing hospitalization

Results in persistent or significant disability / incapacity or

Is a congenital anomaly/birth defect

Is an important medical event that may not be immediately life threatening or result in death or hospitalization but may jeopardize the patient or may require intervention to prevent one of the other outcomes listed above

Medical and scientific judgement should be exercised in deciding whether expedited reporting is appropriate. Examples of such important medical events are intensive treatment in an emergency room or at home for allergic bronchospasm; blood dyscrasias or convulsions that do not result in hospitalization; or development of drug dependency or drug abuse.

Please note: adverse event of special interest (AESI) also have to be handled like an SAE (see sections 10.1.5 and 10.3).

The following haematological abnormalities need not to be reported as SAE:

Aplasia and thrombopenia during standard HDM treatment regime (leukopenia/neutropenia /thrombocytopenia CTC grade 4)

Any complication from the aplasia and thrombopenia, that itself fulfils the SAE criteria - including (prolongation of) hospitalization or unexpected duration - remains reportable as an SAE.

Hospitalizations as a result of the following causes need not to be reported as SAE: Complications during these hospitalizations remain reportable as SAEs, if these complications itself fulfil the SAE criteria:

Drug application (e.g., chemotherapy, study medication)

Administration of blood or platelet transfusion

Test procedure required in the protocol

Hospitalizations for diagnostic measures only

Technical, practical, or social reasons, in absence of an adverse event

Surgical intervention or other measures and the condition(s) leading to these measures are not AEs, if the condition leading to the measure was present prior to inclusion into the trial

Vertebro- and kyphoplasty

Stay at rehabilitation clinic

PD of MM (incl. complications of MM progression)

In general, an SAE report should be completed for any event where doubt exists regarding its status of seriousness.

12.2 Adverse Event of Special Interest (AESI)

An adverse event of special interest (AESI) is an AE (serious or non-serious) that is of scientific and medical concern specific to the study treatment, for which ongoing monitoring and immediate notification by the investigator to the sponsor is required.

Such AEs may require further investigation in order to be characterized and understood.

Adverse Events of Special Interest (AESI) include the following:

- Pregnancy of a female patient entered in a study as well as pregnancy occurring in a female partner of a male patient entered in a study with lenalidomide and exposure of any pregnant female (e.g., caregiver or pharmacist) to lenalidomide (for the requirements of pregnancy reporting also see section 12.10.1)

- Symptomatic overdose (serious or non-serious) with study treatment (SC isatuximab or lenalidomide or bortezomib or dexamethasone). An overdose (accidental or intentional) with the study treatment is an event suspected by the investigator or spontaneously notified by the patient (not based on systematic pills count) and defined as: Increase of at least 30% of the dose to be administered in the specified duration or if the dose is administered in less than half the recommended duration of administration. Of note, asymptomatic overdose has to be reported as a standard AE.

- Second primary malignancies (SPM) must be considered AESIs regardless of the study treatment arm. This includes any SPM, regardless of causal relationship to study drugs, occurring at any time for the duration of the study, from the time of first randomization up to the end of study participation. Events of SPM are to be reported using the SAE report form and must be considered an “Important Medical Event” even if no other seriousness criteria apply; these events must also be documented in the appropriate pages of the eCRF and patient’s source documents. Documentation on the diagnosis of the SPM must be provided at the time of reporting as a SAE (e.g., any confirmatory histology or cytology results, X-rays, CT scans, etc.). During the follow-up

period (after individual EOS) SPM have to be reported on the appropriate pages of the eCRF and, in addition, any SPM shall also be considered an SAE and expedited reporting by the investigator is required until date of LPO is reached. Thereafter data on SPM during follow up are collected in the GMMG myeloma registry as far as applicable.

Severity of all AESIs will be graded using National Cancer Institute-Common Toxicity Criteria for Adverse Events (CTCAE, v5.0, **APPENDIX 5**). Guidelines for dose modifications and interruptions for management of common toxicities associated with the study treatment(s) are provided in Section 10.

AESIs should be reported on SAE Report Forms (**APPENDIX 14**) whether serious or not.

Embryo-fetal toxicity should be reported as outlined in the Pregnancy reporting section **(APPENDIX 15**).

12.2.1 AESI for Isatuximab

Adverse Events of Special Interest (AESI) for isatuximab include the following:

- Infusion or injection reactions Grade 3 or 4 to SC isatuximab (please refer to the current edition of the IB for infusion / injection related AEs manifestations reported in patients treated with SC isatuximab). An infusion / injection reaction is an AE related to SC isatuximab typically with onset within 24 hours from the start of SC isatuximab application.

12.2.2 Other AESI

12.3 Assessment of Adverse Events

Each AE will be assessed by the investigator for severity and for a causal relationship with the study treatment as outlined below.

12.3.1 Severity Assessment

The grading of AEs in this trial will be carried out on the basis of the 5-grade scale defined in the CTCAE version 5.0:

Grade 1: mild AE

Grade 2: moderate AE

Grade 3: severe AE

Grade 4: life-threatening AE or AE causing disablement

Grade 5: death related to AE

An AE that is assessed as “severe” should not be confused with an SAE. Severe is a category used for rating the intensity of an event; and both AEs and SAEs can be assessed as severe.

12.3.2 An event is defined as 'serious' when it meets at least 1 of the predefined outcomes as described in the definition of an SAE, not when it is rated as severe.Relationship to Study Drug

The investigator will evaluate each AE that occurred after administration of IMP regarding the coherency with the administration of the IMP, investigational device and medical procedure (Table 12).

Table 12: Description of relationship of AEs to IMP, investigational device and medical procedure

Relatedness Description

related: There is a reasonable possibility that the event may have been caused by IMP/investigational device/medical procedure. A certain event has a strong temporal relationship and an alternative cause is unlikely.

probable: An AE that has a reasonable possibility that the event is likely to have been caused by IMP/investigational device/medical procedure. The AE has a timely relationship and follows a known pattern of response, but a potential alternative cause may be present.

possible: An AE that has a reasonable possibility that the event may have been caused by IMP/investigational device/medical procedure. The AE has a timely relationship to the IMP/investigational device/medical procedure; however, the pattern of response is untypical, and an alternative cause seems more likely, or there is significant uncertainty about the cause of the event.

Unlikely (only for IMP): Only a remote connection exists between the IMP/investigational device/medical procedure and the reported adverse event. Other conditions including concurrent illness, progression or expression of the disease state or reaction of the concomitant medication appear to explain the reported adverse event.

not related: An AE that does not follow a reasonable temporal sequence related to IMP/investigational device/medical procedure and is likely to have been produced by the subject's clinical state, other modes of therapy or other known etiology.

not assessable (only for IMP): There is insufficient or incomplete evidence to make a clinical judgement of the causal relationship.

Note: Cases regarding investigational device/medical procedure where relatedness cannot be assessed, or no information has been obtained should also be classified as possible.

In a bi-modal scale which only differentiates "related" and "not related" the following grades of causality are considered as a positive causality with the IMP/investigational device/medical

procedure: ‘related’, ‘probable’, ‘possible’ and ‘not assessable’. In opposite, the grades ‘unlikely’, and ‘not related’ are considered as a negative relationship.

The grades ‘unlikely’ and ‘not assessable’ cannot be chosen for the assessment of the investigational device/medical procedure.

All patients who have reportable AEs must be monitored to determine the outcome. The clinical course of the AE will be followed up until resolution or normalization of changed laboratory parameters or until the investigator attributes the AEs/SAEs to a cause other than the study drug/investigational device/medical procedure or assesses them as chronic or stable or no more clinically relevant. The unresolved events will be followed for a maximum of 6 months. Ongoing

AEs/SAEs of withdrawn patients also have to be monitored to determine the outcome as described above.

The outcome of an AE at the time of the last observation will be classified as:

Recovered/resolved: All signs and symptoms of an AE disappeared without any sequels at the time of the last interrogation,

Recovering/resolving: The intensity of signs and symptoms has been diminishing and/or their clinical pattern has been changing up to the time of the last interrogation in a way typical for its resolution,

Not recovered/not resolved: Signs and symptoms of an AE are mostly unchanged or worsened at the time of the last interrogation,

Recovered/resolved with sequel: Actual signs and symptoms of an AE disappeared but there are sequels related to the AE,

Fatal/Death: Resulting in death. If there are more than one adverse event only the adverse event leading to death (related) will be characterized as ‘fatal‘,

Unknown: The outcome is unknown or implausible and the information cannot be supplemented or verified.

The action taken with IMP will be assigned to one of the following categories for the reported SAEs:

‘Dose not changed’:	no change in the dose of IMP
‘Dose reduced’:	reduction in the dose of IMP
‘Drug temporarily discontinued’:	IMP stopped completely but restarted later
‘Drug withdrawn’:	permanent discontinuation of IMP
‘Unknown’:	the information is unknown or implausible and it cannot be supplemented or verified
‘Not applicable’:	the question is implausible (e.g., the subject is dead)

In addition, action taken due to the investigational device (SC isatuximab OBDS) / medical procedure has to be described. The term ‘countermeasures’ refers to the specific actions taken to treat or alleviate adverse events or to avoid their sequels. Countermeasures have to be

reported for SAEs. The following categories will be used to categorize the countermeasures to adverse events:

None: no action taken

Drug treatment: newly-prescribed medication or change in dose of a medication

12.4 Others: other countermeasures, e.g., an operative procedure

Collection and recording of Adverse Events

AEs may be volunteered spontaneously by the study patient, or discovered by the study staff during physical examinations or by asking an open, nonleading question such as, “How have you been feeling since your last study visit?” The Investigator will document the nature of AE, date of onset of the AE (and time, if known), date of outcome of the AE (and time, if known), severity of the AE, action taken with study drug as a result of the AE, assessment of the seriousness of the AE, and assessment of the causal relationship of the AE to study drug and/or study procedure.

All Adverse Events (non-serious related events and related and unrelated serious events) will be recorded in the CRF over the course of the trial. For AE reporting, the verbatim used in the eCRF by investigators to identify AEs will be coded using the latest version of MedDRA coding dictionary. Description of individual underlying signs and symptoms should be avoided. When a clear diagnosis cannot be identified, each sign or symptom should be recorded as a separate AE.

Concomitant illnesses that existed before entry into the study are to be documented as medical history and will not be considered AEs unless the illness worsens after initiating protocol therapy.

All SAEs will be collected from the administration of the first dose of study drug, and must be throughout the study and for at least 70 days after the last dose of study treatment. SAEs considered by the Investigator to be related to study medication will be reported regardless of the timeframe from last dose of study treatment. Disease progression is an efficacy criterion and is therefore not considered an AE or SAE (even if fatal). Disease progression should be documented but not reported as an SAE. If AEs/SAEs occur in relation to disease progression that are not consistent with the natural progression of the patient’s disease, these AEs/SAEs must be reported per AE/SAE reporting requirements.

Safety will be analysed with respect to adverse events of CTC grade ≥ 2 and serious adverse events of any CTC grade. Infusion-related reactions (IR) and injection site reactions (ISR) will be recorded from CTC grade ≥ 1 on induction therapy. All adverse events (AE) and device deficiencies (DD) of the SC isatuximab OBDS, will be recorded (from grade ≥ 1).

12.5 Expedited reporting of AEs

The event will be reported by the sponsor via the CRO within the timeframes established by current Spanish legislation (i.e., serious adverse and unexpected events shall be reported a maximum of 15 calendar days after the sponsor becomes aware of the event, and a maximum of seven days if the serious adverse or unexpected reaction has caused the patient's death or put the patient's life at risk).

The reference safety information document in which to attribute the expected or unexpected nature of serious adverse reactions, for the purposes of expediting reporting, will be the Investigator's Brochures.

All SAEs will be collected from the start of study intervention until 70 days after discontinuing all study interventions, regardless of initiation of a new cancer therapy or transfer to hospice, at the time points specified in the SoA (Section 11). SAEs considered by the Investigator to be related to study medication will be reported regardless of the timeframe from last dose of protocol therapy.

However, any SAEs assessed as related to study participation (e.g., study intervention, protocol-mandated procedures, invasive tests, or change in existing therapy) or related to an investigational product will be recorded from the time a participant consents to participate in the study up to and including any follow up.

All AEs will be collected from the start of treatment until 70 days following discontinuation of study treatment regardless of initiation of a new cancer therapy or transfer to hospice at the time points specified in the SoA.

- Medical occurrences that begin before the start of study intervention but after obtaining informed consent which are not due to protocol procedures will be recorded on the Medical History/Current Medical Conditions section of the eCRF, not the AE section.

- Medical occurrences that begin before the start of study intervention but after obtaining informed consent which occur due to the protocol (by some procedure such as medication washout period, previous test, etc.) they should be collected as AEs in the CRF.
- All SAEs will be recorded and reported to the sponsor or designee immediately and under no circumstance should this exceed 24 hours, as indicated in Section 10. The investigator will submit any updated SAE data to the sponsor within 24 hours of it being available.
- Investigators are not obligated to actively seek AEs or SAEs after the conclusion of the study participation. However, if the investigator learns of any SAE, including a death, at any time after a participant has been discharged from the study, and he/she considers the event to be reasonably related to the study intervention or study participation, the investigator must promptly notify the sponsor.
- The method of recording, evaluating, and assessing causality of AEs and SAEs and the procedures for completing and transmitting SAE reports are provided in Section 10.

12.6 Follow-Up of Adverse Events

All AEs experienced by a patient, regardless of the suspected causality, will be monitored until the AE or SAE has resolved, until any abnormal laboratory values have returned to baseline or normal levels, until stabilized with a satisfactory explanation for the changes observed or 30 days after the last treatment dose

12.7 Disease-Related Events and/or Disease-Related Outcomes Not Qualifying as SAEs

Disease progression does not need to be reported as a serious adverse event (SAE). However, if the underlying disease (i.e., progression) is greater than that which would normally be expected for the participant, or if the investigator considers that there was a causal relationship between treatment with study treatment(s) or protocol design or procedures and the disease progression, then this must be reported as an SAE. Death due to the disease under study is to be recorded on the Death eCRF.

12.8 Treatment of Overdose

For isatuximab, bortezomib and lenalidomide, overdose is defined as an increase of at least 30% of the dose to be administered in the specified duration or if the dose is administered in less than half the recommended duration of administration. For dexamethasone, overdose is defined as at least 30% of the intended dose to be administered at each cycle

12.8.1 Isatuximab

There has been no experience of overdosage of isatuximab in clinical studies. Doses of subcutaneous isatuximab up to 20 mg/kg have been administered in clinical studies.

There is no known specific antidote for isatuximab overdose. In the event of overdose, monitor the patients for signs or symptoms of adverse reactions and take all appropriate measures immediately.

12.8.2 Iberdomide

BMS/Celgene does not recommend a specific treatment for an overdose of iberdomide.

In the event of an overdose of iberdomide, the investigator must:

- Contact the CRO designed by the Sponsor.
- Monitor the participant closely for AEs/SAEs and laboratory abnormalities until iberdomide can no longer be detected systemically (at least 3 months).
- Document the quantity of the excess dose as well as the duration of the overdosing in the eCRF.
- Obtain an additional plasma sample for PK analysis if requested by the Coordinators (determined on a case-by-case basis).
- Decisions regarding dose interruptions or modifications will be made by the investigator in consultation with the Medical Monitor based on the clinical evaluation of the participant.

12.9 Reporting to the Sponsor

All SAEs and AESIs must be reported to the Sponsor, through the CRO designed, within 24 hours of becoming aware of the initial SAE/AESI or any follow-up information regarding the SAE/AESI using the SAE reporting information below. SAEs/AESIs must be reported after study completion, if the SAE/AESI is assessed as study-drug related.

Table 257 CRO SAE, Pregnancy, AESI Reporting Information

CRO SAE, Pregnancy, AESI Reporting Information
Representative: ADKNOMA HEALTH RESEARCH S.L. **Mailing address of the CRO:** Martí i Julià 6-8 Entlo. 3ª Dcha **Tel.:** 93 206 66 66 **Fax:** 93 206 66 67 **Email:** farmacovigilancia@adknoma.com

12.10 Pregnancy

All pregnancies or suspected pregnancies occurring in either a female subject of childbearing potential or partner of childbearing potential of a male subject are immediately reportable events.

Further describe the requirements, timelines and format to report pregnancy and pregnancy related events for the study.

The exposure of any pregnant female (eg, caregiver, pharmacist, study coordinator or monitor) is also an immediately reportable event.

12.10.1 Females of Childbearing Potential:

Pregnancies and suspected pregnancies (including elevated βhCG or positive pregnancy test in a female subject of childbearing potential regardless of age or disease state) occurring while the subject is on IP, or within at least 28 days of the subject's last dose of IP, are considered immediately reportable events. IP is to be discontinued immediately and the subject instructed to return any unused portion of the IP to the Investigator. The pregnancy/ suspected pregnancy must be reported to Sponsor immediately by email, fax or phone and by sending a completed Pregnancy Report Form or approved equivalent Form to Sponsor.

The exposure of any pregnant female (e.g., caregiver or pharmacist) is also an immediately reportable event.

The female subject should be referred to an obstetrician-gynecologist, preferably one experienced in reproductive toxicity, for further evaluation and counseling.

The Investigator will follow the female subject until completion of the pregnancy, and must notify Sponsor immediately about the outcome of the pregnancy (either normal or abnormal outcome) by email, fax or phone and by sending a completed Pregnancy Report Form or approved equivalent Form to Sponsor.

If the outcome of the pregnancy was abnormal (e.g., spontaneous or therapeutic abortion), the Investigator should report the abnormal outcome as an AE. If the abnormal outcome meets any of the serious criteria, it must be reported as an SAE to CRO within 24 hours of the Investigator's knowledge of the event using the SAE Report Form, or approved equivalent form.

All neonatal deaths that occur within 28 days of birth should be reported, without regard to causality, as SAEs. In addition, any infant death after 28 days that the Investigator suspects is related to the in-utero exposure to the IP should also be reported to CRO as an SAE within 24 hours of the Investigator's knowledge of the event using the SAE Report Form, or approved equivalent form.

12.10.2 Male Subjects

If a female partner of a male subject taking investigational product becomes pregnant, the male subject taking IP should notify the Investigator, and the pregnant female partner should be advised to call their healthcare provider immediately.

Any female participant who becomes pregnant while participating will discontinue study intervention.

12.11 Reporting Product Complaints

Any written, electronic or oral communication that alleges dissatisfaction related to manufactured clinical drug product with regards to its manufacturing, testing, packaging, or shipping, must be reported to the Sponsor or qualified designee within 1 working day of first becoming aware of the possible defect to email address of CRO. The product and packaging components in question, if available, must be stored in a secure area under specified storage conditions until it is determined whether the product is required to be returned for investigation of the defect. If the product complaint is associated with an SAE, the SAE must be reported separately in accordance with the protocol, and the SAE report should mention the product quality complaint.

12.12 Emergency treatment

During and following a patient's participation in the trial, the investigator should ensure that adequate medical care is provided to a patient for any AE including clinically significant laboratory values. The investigator should inform a patient when medical care is needed for intercurrent illness(es) of which the investigator becomes aware. In case of emergency treatment all necessary medications and procedures are permitted and no restrictions will be given (see section 6.10).

13 DATA REPORTING

13.0 Data submission

Sponsor will be responsible for processing and quality control of the data. Data will be handled in accordance with the Data Management Plan, Standard Operating Procedures and applicable regulatory guidelines.

Data management based on GCP refers to activities defined to achieve safe routines for efficient entry of subject information into a database, avoiding errors. The routines include procedures for handling of eCRFs, database set-up and management, data entry and verification, data validation, quality control (QC) of database, and documentation of the performed activities, including information of discrepancies in the process. The database, the data entry screens and the program, will be designed in accordance with the clinical study protocol by sponsor.

All participant data relating to the study will be recorded on printed or electronic CRF unless transmitted to the sponsor or designee electronically (e.g., laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the eCRF.

The eCRFs should always reflect the latest observations on the subjects participating in the study. Therefore, the eCRFs should be completed as soon as possible during or after the subject's visit. Each study site Investigator must verify that all data entries in the eCRFs are accurate and correct.

If some assessments are not done, or if certain information is not available, not applicable or unknown, the Investigator or authorized designee should indicate this in the eCRF.

- The investigator must maintain accurate documentation (source data) that supports the information entered in the eCRF.
- The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source data documents.
- The sponsor or designee is responsible for the data management of this study including quality checking of the data.

- Study monitors will perform ongoing source data verification to confirm that data entered into the eCRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.
- Records and documents, including signed ICF, pertaining to the conduct of this study must be retained by the investigator for 25 years from the issue of the final Clinical Study Report (CSR)/equivalent summary unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor.

13.1 Source Documents

Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator's site.

- Data entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.
- Definition of what constitutes source data can be found in SRM.

The eCRF is essentially considered a data entry form and should not constitute the original (or source) medical records unless otherwise specified. Source documents are all documents used by the Investigator or hospital that relate to the subject's medical history, that verify the existence of the subject, the inclusion and exclusion criteria, and all records covering the subject's participation in the study. They include enrolment and randomization log, investigational product accountability log, laboratory notes, memoranda, material dispensing records, subject files, etc.

Each Investigator is responsible for maintaining source documents. These will be made available for inspection by the study monitor at each monitoring visit. All supportive documentation submitted with the eCRF, such as laboratory data should be clearly identified with the study, visit

and subject number. Any personal information (e.g., subject name, initials) should be removed or rendered illegible to preserve individual confidentiality.

13.2 Web-based eCRF

Data collection for this study will consist of electronic data capture for all eCRF information. Sponsor will supply the eCRF. All study site Investigators agree to maintain accurate eCRFs and source documentation as part of the case histories. All information is to be filled in the subject's eCRF. If an enrolled subject is not randomized into the study (ie, fails screening), only minimum data (such as demographics and consent date) and the reason for failing screening should be reported on the eCRF. In general, no queries for missing data on these subjects will be issued for procedures indicated as 'Not done'.

For randomized subjects, information captured on the source documents will be entered into the subject's eCRF by study site personnel and monitored at the study site. If an item is not available or is not applicable, this fact should be indicated by a missing reason.

Each completed eCRF must be reviewed, signed, and dated by the study site Investigator in a timely manner. The completed eCRF will be reviewed by the study Monitor as soon as practical after completion. A copy of the final, approved and signed eCRF will be provided to the site and should be stored in the appropriate files.

Clinical data (including AEs and concomitant medications) will be entered into an Electronic Data Capture (EDC) application. The data system is password protected and includes internal quality checks, such as automatic range checks, to identify data that appear inconsistent, incomplete or inaccurate. Clinical data will be entered directly from the source documents. Roles and rights of the study site personnel responsible for entering the study data into the eCRF will be determined in advance. Only authorized study site personnel designated by each study site Investigator will complete data collection.

13.3 Data and Safety Monitoring

The clinical monitors are employees of the CRO, or representatives of the sponsor. As such, they have the obligation to follow the trial closely so that all aspects of the trial are carefully monitored for compliance with applicable government regulations and with ICH E6(R2) guidelines.

The clinical monitors will visit the study sites and Investigators at intervals as defined in the monitoring plan, in addition to maintaining necessary contact through telephone, e-mail, and letter. The clinical monitors will maintain current personal knowledge of the trial through observation, review of trial records and source documentation, and discussion of the conduct of the trial with the study site Investigators and staff.

13.4 The Query Process

Each monitor will review the eCRFs and evaluate them for completeness and consistency. Each eCRF will be compared with the respective source documents to ensure that there are no discrepancies between critical data. All entries, corrections and alterations are to be made by the Investigator or his/her authorized designee. The monitor cannot enter data in the eCRFs.

The appropriate investigational staff will answer queries generated in the application. This process is audit trailed meaning that the name of investigational staff, time, and date is logged.

13.5 Audits

The sponsor may choose to schedule and conduct periodic audits of ongoing clinical studies. Audits are independent of, and separate from, routine monitoring or quality control functions and are conducted to assure accuracy and compliance with the protocol, governing regulatory authorities and/or ICH E6(R2) guidelines.

The Investigators and institutions involved in the trial will permit trial-related monitoring, audits, IEC review, and regulatory inspection(s) by providing direct access to all trial records. In the event of an audit, the Investigator agrees to allow the sponsor, representatives of the sponsor, and the governing regulatory agency access to all trial records.

The Investigator should immediately notify the sponsor of any inspections scheduled by any regulatory authorities and promptly forward copies of any inspection reports received to the sponsor.

13.6 Protocol Deviations

Protocol deviations are unintended and/or unanticipated departures from the procedures and/or processes in the protocol as approved by the sponsor, the CA and IEC, and agreed to by the Investigator.

Minor deviations usually have a limited impact to study outcomes and do not involve ICH E6 guidelines, inclusion/exclusion criteria or primary endpoint criteria.

Major deviations are those that may adversely affect the rights, safety or welfare of subjects and / or quality and integrity of data.

Critical deviations are those that adversely affect the rights, safety or welfare of subjects and / or quality and integrity of data.

Each study site Investigator must document and explain in the subject's source documentation any deviations from the approved protocol. Investigators may implement a deviation from the protocol to eliminate an immediate hazard to trial subjects without prior IEC informed consent approval, but the deviation must be reported to the monitor/CRA within 1 working day. Such incidents will be evaluated for potential safety hazards of the ongoing study, and if deemed appropriate, a protocol amendment will be issued.

The monitor/CRA will document protocol deviations throughout the course of monitoring visits. The monitor will notify the Investigator during a visit and a "Protocol Deviation Form" is completed and signed by the investigator and by the monitor.

13.7 Trial Record Retention

Essential documents and eCRF data should be retained during 25 years.

The investigator will maintain all study records in accordance with GCP guidelines as established by ICH and corresponding regulatory requirements.

13.8 Study and Site Closure

Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance to the sponsor and CRO of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to:

• Failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines

• Inadequate recruitment of participants by the investigator

• Discontinuation of further study intervention development

14 ETHICAL, LEGAL, AND ADMINISTRATIVE ASPECTS

14.0 Good Clinical Practice

The current clinical trial will be conducted in accordance with the protocol, the principles established in the current revised version of the Declaration of Helsinki (Annex 1) and the applicable regulatory requirements, particularly the ICH Tripartite Harmonized Guidelines for good clinical practice (1996), the Regulation (EU) 536/2014 relative to clinical trials on medicinal products for human use and the locally applicable regulations (e.g., In Spain, the Royal Decree on Clinical Trials 1090/2015).

14.1 Informed Consent

An approved informed consent form will be provided by the sponsor to investigative site. The written Informed Consent Form (ICF) should be prepared in the local language(s) of the potential subject population. In obtaining and documenting informed consent, the Investigator should comply with the applicable regulatory requirements, and should adhere to GCP and to the ethical principles that have their origin in the Declaration of Helsinki. The consent form and any revision(s) should be approved by the Ethics Committee prior to being provided to potential subjects.

Before a subject's participation in the study, it is the Principal Investigator's (or their designee) responsibility to obtain freely given consent in writing, from the subject after adequate explanation of the aims, methods, anticipated benefits, and potential risks of the study and before any protocol-specific screening procedures or any study drugs are administered. Subjects must have the opportunity to ask questions and receive answers and will have adequate time to decide whether or not to participate in the study. Once the Investigator is assured that the subject understands the implications of participating in the trial, the subject will be asked to give consent to participate in the trial by signing the informed consent. The ICF should be signed and personally dated by the subject and by the physician who conducted the informed consent discussion (Principal Investigator or designee). The subject's written informed consent should also be documented in the subject's medical records.

The Investigator shall provide a copy of the signed informed consent to the subject. A second original form shall be maintained in the Investigator study file at the site.

If the informed consent is revised during the course of the trial, all active participating subjects must sign the revised form approved by the Ethics Committee.

The subject participating in a clinical trial, or his/her legal representative, may withdraw consent at any time without giving any reason and without this involving any penalty or prejudice for the participating subject.

14.2 Protocol Approval and Amendment

ICH guidelines require that approval be obtained from Health Authorities and an Ethics Committee before human subjects can participate in research studies. Prior to the trial onset, the protocol, informed consent, advertisements to be used for subject recruitment (if applicable), and any other written information regarding this trial to be provided to the subject will be approved by the Ethics Committee. The clinical study will only be started when both the Health Authorities and an Ethics Committee have considered that the expected benefits for the trial subject and society justify the risks; in addition, the trial will only be continued if compliance with this criterion is constantly supervised.

The CRO will obtain Ethics Committee approvals on behalf of the sponsor and investigators. All regulatory approvals should be signed by the Ethics Committee Chairman or designee and must identify the Ethics Committee name and address, the clinical protocol by title and/or protocol number and the date approval and/or favorable opinion was granted. Documentation of all Health Authorities and Ethics Committee approvals and of the Ethics Committee compliance with ICH E6(2) will be maintained by the site and will be available for review.

Any change in the approved protocol will require a Protocol amendment. The Investigator must not make any change in the study without favorable opinion from the Ethics Committee and authorization from the Health Authorities, except as necessary to eliminate an impending and obvious risk for the subjects except when necessary to remove an apparent, immediate hazard to subjects. Protocol changes introduced to eliminate an impending and obvious risk may be

implemented immediately, but must subsequently be documented in an amendment, reported to the Ethics Committee and be submitted to the relevant Health Authorities within the required timeframe.

Any substantial amendments to the protocol must be submitted in writing to the Ethics Committee and the Health Authorities for approval before the changes proposed in the amendment are implemented. Depending on the magnitude of the change, the recruitment may be temporally halted.

The sponsor does not have to notify non-substantial amendments to the Health Authorities or the Ethics Committee. However, any non-substantial amendments will be recorded and contained in the documentation when it is subsequently submitted, for example in the subsequent notification of a substantial amendment. Documentation of any non-substantial amendments will be available on request for inspection at the trial site or the sponsor premises as appropriate.

14.3 Obligations of all participants in the trial

14.3.1 Investigator

The responsibilities of the investigator are specified in article 41 of Royal Decree 1090/2015. The investigator must be in agreement with this protocol and possess thorough knowledge of the properties of the products used in the clinical trial.

The investigator should give the patient information sheet to the patient and assist him or her in understanding the explanations provided therein. It is important that the investigator informs the patient that his or her participation in the study is completely voluntary, and that this will not affect the patient/physician relationship. The investigator should also guarantee the patient that all persons involved in the study will maintain the confidentiality of all patient information.

The principal investigator or one of his collaborators will be responsible for collecting, recording and reporting the data correctly, and will guarantee the immediate reporting of serious adverse or unexpected events within 24 hours to Trial Form Support (by fax at: 91- 125 05 51).

The investigator is responsible for preparing and maintaining accurate, complete medical histories, which are designed to document all observations and other relevant study data relating to each participant in the trial. All information recorded in the CRFs for this study must be consistent with each patient's original documentation.

It is the responsibility of the investigator to regularly report to the Clinical Research Ethics Committee on the progress of the study; the investigator will be jointly responsible along with the sponsor for producing the clinical trial report.

14.3.2 The trial monitor

The trial monitor will undertake to directly monitor the conduct of the trial, following Good Clinical Practice guidelines. The responsibilities of the monitor are specified in article 40 of Royal Decree 1090/2015.

All study data recorded in the original documents will be transcribed onto the CRFs. The study data will be entered into a validated and secure data processing system and back-up copies will be stored. Any changes to the study data will be recorded.

During the course of the trial, the monitor will visit the trial sites to verify compliance with the protocol, to compare the CRFs with the patients' individual medical histories, evaluate drug accountability and ensure that the study is being conducted in accordance with the pertinent regulatory requirements. The CRFs will be checked against the original documentation. Medical histories will be verified in such a way that guarantees patient confidentially. Any instances of data omission or of data that cannot be interpreted will be handled by the investigator.

14.3.3 The sponsor

The obligations of the sponsor are specified in article 39 of Royal Decree 1090/2015.

The sponsor of the clinical trial is the physical or legal person who has an interest in the trial being conducted. The sponsor signs authorization requests addressed to the REB and/or the Spanish

Medicine Agency (AEM) and takes responsibility for the trial, including its conduct, initiation, and end. The sponsor is also in charge of ensuring compliance with the pertinent legal regulations.

The PETHEMA Foundation assumes the obligations of the sponsor set out in Article 39 of Royal Decree 1090/2015, and has the means and collaborators necessary to fully guarantee the fulfillment of said obligations.

14.4 Subject Confidentiality and Data Protection

All laboratory specimens, evaluation forms, reports, and other records will be identified in a manner designed to maintain subject confidentiality. Each Investigator will ensure that all site personnel involved will respect the confidentiality of any information about trial subjects. Management of personal data from subjects participating in the trial, particularly as regards consent, will comply with Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) and the Spanish Organic Law 3/2018 of 5th December, on the protection of personal data and digital rights.

At each site, all records will be kept in a secure storage area with limited access. Clinical information will not be released without the written permission of the subject. Subject identity is confidential and may only be known by the Investigator, trial personnel, appointed auditors and monitors, and Health Authorities.

Each Investigator and all employees and coworkers involved with this trial may not disclose or use for any purpose other than performance of the trial, any data, record, or other unpublished, confidential information disclosed to those individuals for the purpose of the trial. Prior written agreement from the sponsor or its designee must be collected for the disclosure of any said confidential information to other parties.

14.5 Data Quality Assurance

All participant data relating to the study will be recorded on printed or electronic CRF unless transmitted to the sponsor or designee electronically (e.g., laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the eCRF.

- The investigator must maintain accurate documentation (source data) that supports the information entered in the eCRF.
- The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source data documents.
- The sponsor or designee is responsible for the data management of this study including quality checking of the data.
- Study monitors will perform ongoing source data verification to confirm that data entered into the eCRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.
- Records and documents, including signed ICF, pertaining to the conduct of this study must be retained by the investigator for 25 years from the issue of the final Clinical Study Report (CSR)/equivalent summary unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor.

14.6 Access to Source Documents

- Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator’s site.
- Data entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may

need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.

- Definition of what constitutes source data can be found in SRM.

14.7 Study Monitoring

The clinical monitors are employees of the CRO or representatives of the Sponsor. As such, they have the obligation to follow the trial closely so that all aspects of the trial are carefully monitored for compliance with applicable government regulations and with ICH E6(R2) guidelines.

The clinical monitors will visit the study sites and Investigators at intervals as defined in the monitoring plan, in addition to maintaining necessary contact through telephone, e-mail, and letter. The clinical monitors will maintain current personal knowledge of the trial through observation, review of trial records and source documentation, and discussion of the conduct of the trial with the study site Investigators and staff.

14.8 Audits and Inspections

The regulatory authorities, the REB and the sponsor or designated representative can request access to all original documents, case report forms and any other study documentation for the purpose of auditing or inspecting the site. The investigator must guarantee direct access to these documents and cooperate at all times in the execution of these activities.

Care will be taken during these procedures to ensure the protection of personal information, in accordance with data protection laws.

If requested, the investigator will grant monitor(s) and auditor(s) from Sponsor or its designees and regulatory authority(ies) access to the patient's original medical records for verification of data gathered on the data capture records and to audit the data collection process. The patient's confidentiality will be maintained and will not be made publicly available to the extent permitted by the applicable laws and regulations.

14.9 Archival

The investigator will maintain all study records in accordance with Good Clinical Practice as set out in the International Conference on Harmonization and corresponding regulatory requirements.

14.10 Publications

The results of this clinical trial, positive or negative, will be presented at scientific conferences and published in scientific journals.

14.10.1 Basic trial guidelines

The principal investigators and/or the participating sites undertake not to use or transfer to third parties, nor distribute or publish the results of this trial without prior written consent from PETHEMA FOUNDATION, sole sponsor of this trial. The following conditions should be observed at all times:

a) The results of this trial shall not be published until the conclusion of the trial, or before if agreed by both parties.
b) The sponsor must not cite the names of the investigators without their permission, except in references to already published work.
c) The sponsor shall permit the publication of data obtained from this study in scientific journals of recognized prestige, and permit the data to be presented at seminars and conferences within the medical field, as long as the conditions set out in paragraph a) of this section are observed, and the sponsor is allowed to review the final draft of the article at least 4 weeks in advance.

It is expected that the results of this trial will be published in a scientific journal and as abstracts/presentations at medical conferences. Current guidelines and publishing recommendations of the ICMJE will be followed.

14.10.2 PETHEMA Group publication conditions

The following guidelines should be observed for any publications based on PETHEMA Group studies:

a) The names of participating sites will appear in all publications, and those individuals who have participated in the design, follow-up, results analysis and writing the publication will be listed as authors.
b) The persons responsible for the protocol will also be responsible for presenting the results at PETHEMA meetings and scientific conferences, as well as for writing papers based on the protocol. The persons responsible for the protocol will have assistance writing these papers if required.
c) The final draft of the text may be submitted for review and approval by another member of PETHEMA with proven expertise, so that his/her suggestions and changes can help to improve the quality of the work. This person can appear as co-author at the discretion of the persons in charge of the protocol. If the reviewer is not listed as a co-author, he/she should be mentioned in the acknowledgements section.

d) Inclusion of the data manager as a co-author is at the discretion of the persons in charge of the protocol. If the data manager is not listed as co-author, she/he will appear in the acknowledgements section.

e) The statistician (if applicable) can be listed as co-author or appear in the acknowledgements section depending on his/her degree of participation, and at the discretion of the persons responsible for the protocol.

f) The statistician and the data manager will not be included in the calculation when determining the maximum number of authors, based on the number of patients included per site.

g) A maximum number of authors will be determined per each participating site who will appear in the article heading. This selection, based on the number of included patients, will be made according to the criteria concerning patients who have contributed to the study (as long as

evaluation and follow-up were carried out up to the final evaluation), and according to the following guidelines:

- Protocols that have included less than 200 patients: each site will have the right to one author per every 5% of patients included out of the total recruited for the study. There will be a limit of 15 authors for any publication. This restriction will be applied to presentations at scientific conferences.

h) If the conference organizers or journal specifies a limit to the number of authors, the number will be reduced starting with the last authors to be cited. The decision of those responsible for the protocol will prevail in the event of discrepancies

i) All institutions and persons that have participated in the protocol will appear in an appendix.

Before submitting the manuscript for publication, a copy will be sent to each author to make any changes they feel are appropriate. If no response is given within the maximum timeline set by the person in charge of the protocol, it will be assumed that the authors agree with the content of the manuscript. Given that there may be a considerable number of authors and that, on occasion, the suggestions may be redundant, contradictory, or minor, the writers of the draft publication will be responsible for producing the final version.

14.11 Insurance policy

The sponsor has contracted an insurance policy to cover the responsibilities of the investigator and other parties participating in the study, according to the applicable legislation.

- Insurance company: HDI Global SE
- Policy Number: 08054309-30128

14.12 Study and site closure

Study sites will be closed upon study completion. Nevertheless, if a site does not include a single patient one year after it was opened, the Sponsor may proceed to replace it with another site with a greater recruiting potential, formalizing the corresponding relevant amendment to that effect.

A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to:

- Failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines
- Inadequate recruitment of participants by the investigator
- Discontinuation of further study intervention development

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Available from: https://doi.org/10.1182/blood-2020-137743

16 APPENDICES

APPENDIX 1 - DECLARATION OF HELSINKI

WMA Declaration of Helsinki - Ethical Principles for Medical Research Involving Human Subjects

Adopted by the

18th WMA General Assembly, Helsinki, Finland, June 1964

and amended by the

29th WMA General Assembly, Tokyo, Japan, October 1975

35th WMA General Assembly, Venice, Italy, October 1983

41st WMA General Assembly, Hong Kong, September 1989

48th General Assembly, Somerset West, Republic of South Africa, October 1996

52nd General Assembly, Edinburgh, Scotland, October 2000

Note of Clarification, added by the WMA General Assembly, Washington 2002

Note of Clarification, added by the WMA General Assembly, Tokyo 2004

59th General Assembly, Seoul, Republic of Korea, October 2008

64th General Assembly, Fortaleza, Brazil, October 2013

Preamble

1. The World Medical Association (WMA) has developed the Declaration of Helsinki as a statement of ethical principles for medical research involving human subjects, including research on identifiable human material and data.

The Declaration is intended to be read as a whole and each of its constituent paragraphs should be applied with consideration of all other relevant paragraphs.

2. Consistent with the mandate of the WMA, the Declaration is addressed primarily to physicians. The WMA encourages others who are involved in medical research involving human subjects to adopt these principles.

General Principles

3. The Declaration of Geneva of the WMA binds the physician with the words, “The health of my patient will be my first consideration,” and the International Code of Medical Ethics declares that, “A physician shall act in the patient's best interest when providing medical care.”

4. It is the duty of the physician to promote and safeguard the health, well-being and rights of patients, including those who are involved in medical research. The physician’s knowledge and conscience are dedicated to the fulfilment of this duty.

5. Medical progress is based on research that ultimately must include studies involving human subjects.

6. The primary purpose of medical research involving human subjects is to understand the causes, development and effects of diseases and improve preventive, diagnostic and therapeutic interventions (methods, procedures and treatments). Even the best proven interventions must be evaluated continually through research for their safety, effectiveness, efficiency, accessibility and quality.

7. Medical research is subject to ethical standards that promote and ensure respect for all human subjects and protect their health and rights.

8. While the primary purpose of medical research is to generate new knowledge, this goal can never take precedence over the rights and interests of individual research subjects.

9. It is the duty of physicians who are involved in medical research to protect the life, health, dignity, integrity, right to self-determination, privacy, and confidentiality of personal information of research subjects. The responsibility for the protection of research subjects must always rest with the physician or other health care professionals and never with the research subjects, even though they have given consent.

10. Physicians must consider the ethical, legal and regulatory norms and standards for research involving human subjects in their own countries as well as applicable international norms and standards. No national or international ethical, legal or regulatory requirement should reduce or eliminate any of the protections for research subjects set forth in this Declaration.

11. Medical research should be conducted in a manner that minimises possible harm to the environment.

12. Medical research involving human subjects must be conducted only by individuals with the appropriate ethics and scientific education, training and qualifications. Research on patients or healthy volunteers requires the supervision of a competent and appropriately qualified physician or other health care professional.

13. Groups that are underrepresented in medical research should be provided appropriate access to participation in research.

14. Physicians who combine medical research with medical care should involve their patients in research only to the extent that this is justified by its potential preventive, diagnostic or therapeutic value and if the physician has good reason to believe that participation in the research study will not adversely affect the health of the patients who serve as research subjects.

15. Appropriate compensation and treatment for subjects who are harmed as a result of participating in research must be ensured.

Risks, Burdens and Benefits

16. In medical practice and in medical research, most interventions involve risks and burdens.

Medical research involving human subjects may only be conducted if the importance of the objective outweighs the risks and burdens to the research subjects.

17. All medical research involving human subjects must be preceded by careful assessment of predictable risks and burdens to the individuals and groups involved in the research in comparison

with foreseeable benefits to them and to other individuals or groups affected by the condition under investigation.

Measures to minimise the risks must be implemented. The risks must be continuously monitored, assessed and documented by the researcher.

18. Physicians may not be involved in a research study involving human subjects unless they are confident that the risks have been adequately assessed and can be satisfactorily managed.

When the risks are found to outweigh the potential benefits or when there is conclusive proof of definitive outcomes, physicians must assess whether to continue, modify or immediately stop the study.

Vulnerable Groups and Individuals

19. Some groups and individuals are particularly vulnerable and may have an increased likelihood of being wronged or of incurring additional harm.

All vulnerable groups and individuals should receive specifically considered protection.

20. Medical research with a vulnerable group is only justified if the research is responsive to the health needs or priorities of this group and the research cannot be carried out in a non-vulnerable group. In addition, this group should stand to benefit from the knowledge, practices or interventions that result from the research.

Scientific Requirements and Research Protocols

21. Medical research involving human subjects must conform to generally accepted scientific principles, be based on a thorough knowledge of the scientific literature, other relevant sources of information, and adequate laboratory and, as appropriate, animal experimentation. The welfare of animals used for research must be respected.

22. The design and performance of each research study involving human subjects must be clearly described and justified in a research protocol.

The protocol should contain a statement of the ethical considerations involved and should indicate how the principles in this Declaration have been addressed. The protocol should include information regarding funding, sponsors, institutional affiliations, potential conflicts of interest, incentives for subjects and information regarding provisions for treating and/or compensating subjects who are harmed as a consequence of participation in the research study.

In clinical trials, the protocol must also describe appropriate arrangements for post-trial provisions.

Research Ethics Committees

23. The research protocol must be submitted for consideration, comment, guidance and approval to the concerned research ethics committee before the study begins. This committee must be transparent in its functioning, must be independent of the researcher, the sponsor and any other undue influence and must be duly qualified. It must take into consideration the laws and regulations of the country or countries in which the research is to be performed as well as applicable international norms and standards but these must not be allowed to reduce or eliminate any of the protections for research subjects set forth in this Declaration.

The committee must have the right to monitor ongoing studies. The researcher must provide monitoring information to the committee, especially information about any serious adverse events. No amendment to the protocol may be made without consideration and approval by the committee. After the end of the study, the researchers must submit a final report to the committee containing a summary of the study's findings and conclusions.

Privacy and Confidentiality

24. Every precaution must be taken to protect the privacy of research subjects and the confidentiality of their personal information.

Informed Consent

25. Participation by individuals capable of giving informed consent as subjects in medical research must be voluntary. Although it may be appropriate to consult family members or community

leaders, no individual capable of giving informed consent may be enrolled in a research study unless he or she freely agrees.

26. In medical research involving human subjects capable of giving informed consent, each potential subject must be adequately informed of the aims, methods, sources of funding, any possible conflicts of interest, institutional affiliations of the researcher, the anticipated benefits and potential risks of the study and the discomfort it may entail, post-study provisions and any other relevant aspects of the study. The potential subject must be informed of the right to refuse to participate in the study or to withdraw consent to participate at any time without reprisal. Special attention should be given to the specific information needs of individual potential subjects as well as to the methods used to deliver the information.

After ensuring that the potential subject has understood the information, the physician or another appropriately qualified individual must then seek the potential subject's freely-given informed consent, preferably in writing. If the consent cannot be expressed in writing, the non-written consent must be formally documented and witnessed.

All medical research subjects should be given the option of being informed about the general outcome and results of the study.

27. When seeking informed consent for participation in a research study the physician must be particularly cautious if the potential subject is in a dependent relationship with the physician or may consent under duress. In such situations the informed consent must be sought by an appropriately qualified individual who is completely independent of this relationship.

28. For a potential research subject who is incapable of giving informed consent, the physician must seek informed consent from the legally authorised representative. These individuals must not be included in a research study that has no likelihood of benefit for them unless it is intended to promote the health of the group represented by the potential subject, the research cannot instead be performed with persons capable of providing informed consent, and the research entails only minimal risk and minimal burden.

29. When a potential research subject who is deemed incapable of giving informed consent is able to give assent to decisions about participation in research, the physician must seek that assent in addition to the consent of the legally authorised representative. The potential subject's dissent should be respected.

30. Research involving subjects who are physically or mentally incapable of giving consent, for example, unconscious patients, may be done only if the physical or mental condition that prevents giving informed consent is a necessary characteristic of the research group. In such circumstances the physician must seek informed consent from the legally authorised representative. If no such representative is available and if the research cannot be delayed, the study may proceed without informed consent provided that the specific reasons for involving subjects with a condition that renders them unable to give informed consent have been stated in the research protocol and the study has been approved by a research ethics committee. Consent to remain in the research must be obtained as soon as possible from the subject or a legally authorised representative.

31. The physician must fully inform the patient which aspects of their care are related to the research. The refusal of a patient to participate in a study or the patient's decision to withdraw from the study must never adversely affect the patient-physician relationship.

32. For medical research using identifiable human material or data, such as research on material or data contained in biobanks or similar repositories, physicians must seek informed consent for its collection, storage and/or reuse. There may be exceptional situations where consent would be impossible or impracticable to obtain for such research. In such situations the research may be done only after consideration and approval of a research ethics committee.

Use of Placebo

33. The benefits, risks, burdens and effectiveness of a new intervention must be tested against those of the best proven intervention(s), except in the following circumstances:

Where no proven intervention exists, the use of placebo, or no intervention, is acceptable; or

Where for compelling and scientifically sound methodological reasons the use of any intervention less effective than the best proven one, the use of placebo, or no intervention is necessary to determine the efficacy or safety of an intervention.

and the patients who receive any intervention less effective than the best proven one, placebo, or no intervention will not be subject to additional risks of serious or irreversible harm as a result of not receiving the best proven intervention.

Extreme care must be taken to avoid abuse of this option.

Post-Trial Provisions

34. In advance of a clinical trial, sponsors, researchers and host country governments should make provisions for post-trial access for all participants who still need an intervention identified as beneficial in the trial. This information must also be disclosed to participants during the informed consent process.

Research Registration and Publication and Dissemination of Results

35. Every research study involving human subjects must be registered in a publicly accessible database before recruitment of the first subject.

36. Researchers, authors, sponsors, editors and publishers all have ethical obligations with regard to the publication and dissemination of the results of research. Researchers have a duty to make publicly available the results of their research on human subjects and are accountable for the completeness and accuracy of their reports. All parties should adhere to accepted guidelines for ethical reporting. Negative and inconclusive as well as positive results must be published or otherwise made publicly available. Sources of funding, institutional affiliations and conflicts of interest must be declared in the publication. Reports of research not in accordance with the principles of this Declaration should not be accepted for publication.

Unproven Interventions in Clinical Practice

37. In the treatment of an individual patient, where proven interventions do not exist or other known interventions have been ineffective, the physician, after seeking expert advice, with informed consent from the patient or a legally authorised representative, may use an unproven intervention if in the physician's judgement it offers hope of saving life, re-establishing health or alleviating suffering. This intervention should subsequently be made the object of research, designed to evaluate its safety and efficacy. In all cases, new information must be recorded and, where appropriate, made publicly available.

APPENDIX 2 - LIST OF SITES/PRINCIPAL INVESTIGATORS

CENTRO DE REALIZACIÓN DEL ESTUDIO	INVESTIGADOR PRINCIPAL
Hospital Universitario 12 de Octubre	Dr. Joaquín Martínez López
Hospital Universitario y Politécnico La Fe	Dr. Mario Arnao Herraiz
Hospital Universitario de Araba	Dr. Xabier Gutierrez López de Ocariz
Hospital Clinic de Barcelona	Dra. Laura Rosiñol Dachs
Hospital Universitario Marqués de Valdecilla	Dr. Enrique Ocio San Miguel
Hospital de la Santa Creu i Sant Pau	Dr. Miguel Granell Rey
ICO Badalona	Dr. Alberto Oriol Rocafiguera
Hospital Universitario Virgen del Rocío	Dra. Estrella Carrillo Cruz
Hospital Universitario Nuestra Señora de Valme	Dra. María del Carmen Couto Caro
Hospital Universitario Virgen de la Arrixaca	Dr. Valentín Cabañas Perianes
Hospital General Universitario Morales Meseguer	Dr. Felipe de Arriba de la Fuente
Hospital General Universitario Santa Lucia	Dra. Marta Romera Martínez
Hospital Virgen de la Salud	Dr. Luis Felipe Casado Montero
Hospital Clínico Universitario de Valencia	Dra. Ana Isabel Teruel Casasús
Hospital Universitario de Salamanca	Dra. María Victoria Mateos Manteca
Hospital Universitario de Gran Canaria Doctor Negrín	Dra. Alexia Suárez Cabrera
Hospital Universitario Son Espases	Dra. Antonia Sampol Mayol
Hospital Universitario Son Llatzer	Dr. Joan Bargay Lleonart
Hospital Universitario Vall d'Hebron	Dra. Mercedes Gironella Mesa
ICO L´Hospitalet	Dra. Anna Sureda Baradi
Complejo Hospitalario de Cáceres	Dr. Ignacio Casas Avilés
Hospital Universitario de Jerez	Dr. Sebastián Garzón López
Hospital Universitario Reina Sofía	Dr. Miguel Ángel Álvarez Rivas
Hospital Clínico Universitario de Santiago	Dra. Marta Sonia González Pérez
ICO Girona (H Dr. Josep Trueta)	Dra. Yolanda González Montes
Hospital Universitario de León	Dr. Fernando Escalante Barrigón

Hospital Universitario Arna de Vilanova de Lleida	Dr. Antonio García Guiñón
Hospital Universitario Lucus Augusti	Dra. María Esperanza Lavilla Rubira
Hospital Universitario La Paz	Dra. Ana López de la Guía
Hospital Universitario Ramón y Cajal	Dra. María Jesús Blanchard Rodríguez
Hospital Universitario Fundación Jiménez Díaz	Dra. Elena Prieto Pareja
Hospital Universitario HM Sanchinarro	Dr. Jaime Pérez de Oteyza
Hospital Universitario Infanta Leonor	Dr. José Ángel Hernández Rivas
Hospital Universitario Virgen de la Victoria	Dr. Ricarda García Sánchez
Hospital Costa del Sol	Dra. María Casanova Espinosa
Clínica Universidad de Navarra	Dra. Paula Rodríguez Otero
Complejo Hospitalario de Navarra	Dr. José María Arguiñano Pérez
Hospital Universitario de Cabueñes	Dra. María Esther González García
Hospital Universitario Central de Asturias	Dra. Ana Pilar González Rodríguez
Hospital Universitario Nuestra Señora de Candelaria	Dr. Pablo Jesús Rios Rull
Hospital Universitario de Canarias	Dr. Sunil Lakhwani
Hospital General De Segovia	Dra. Aranzazu García Mateos
Hospital Universitario de Tarragona Juan XXIII	Dr. Josep Sarrá Escarré
Hospital Universitario Doctor Peset	Dra. Paz Ribas García
Hospital Universitario de Cruces	Dra. María Elena Amutio Díez
Hospital Clínico Universitario Lozano Blesa	Dr. Luis Ramón Palomera Bernal
Hospital Universitario Miguel Servet	Dra. Ana Cristina Godoy Molias
Hospital General Universitario Gregorio Marañón	Dra. Cristina Encinas Rodríguez
Hospital Universitario Mútua Terrasa	Dr. Josep María Martí Tutusaus
Hospital Universitario Fundación de Alcorcón	Dr. Francisco Javier Peñalver Párraga
Hospital Sanitas La Zarzuela	Dr. Daniel García Belmonte
Hospital Arnau de Vilanova	Dr. Aurelio López Martínez
Hospital Regional Universitario de Málaga	Dra. María Magdalena Alcalá Peña
Hospital Universitario Puerta De Hierro	Dr. Rafael Rios Tamayo
Hospital Universitario de Burgos	Dr. Francisco Javier Díaz Gálvez

Hospital Universitario Príncipe de Asturias	Dr. Julio García Suárez
Hospital San Pedro	Dra. María José Nájera Irazu
Complejo Hospitalario Universitario de Ourense	Dr. José Ángel Méndez Sánchez
Hospital Universitario Rey Juan Carlos	Dr. Alberto Velasco Valdazo
Hospital Universitario Infanta Sofía	Dr. Eugenio Giménez Mesa
Hospital Universitario Donostia	Dra. Maialen Sirvent Auzmendi
Hospital Provincial de Pontevedra	Dra. Ana María Dios Loureiro
Hospital Sant Joan de Deu - Althaia	Dra. Mireia Constans
Hospital Universitario de La Princesa	Dr. Adrián Alegre Amor
Hospital Universitario de Fuenlabrada	Dra. Pilar Bravo Barahona
Hospital Universitario de Guadalajara	Dra. Dunia de Miguel Llorente
Hospital HLA Universitario Moncloa	Dra. María Concepción Aláez Usón
Hospital Universitario Quirónsalud Madrid	Dra. Carmen Martínez Chamorro
Complejo Hospitalario Universitario de Albacete	Dr. Irene Gómez Catalán
Hospital Clínico San Carlos	Dra. María Belén Íñigo Rodríguez
Hospital Universitario Virgen de las Nieves	Dra. María Esther Clavero Sánchez

APPENDIX 3 - CORRELATIVE STUDIES

The general objective of the correlative studies is to generate biomarkers and datasets for improving outcomes in MM based on precision medicine and better understanding of tumor and immune cell biology.

Specific objectives of the correlative studies:

1- Identify genetic risk groups of patients that will benefit the most and that are primary refractory to the treatment schema proposed in the clinical trial.

2- Evaluate the clinical significance of MRD status using next-generation techniques and define undetectable MRD as the endpoint of the treatment schema proposed.

3- Understand MRD resistance by using bulk- and single-cell sequencing techniques to compare the genomic signature of matched diagnostic vs. MRD clones isolated by next-generation flow cytometry.

4- Create an algorithm based on patients' immune profile and clinical data that enables individualized prediction of outcome based on simultaneous MRD and immune monitoring.

6- Genetic characterization of tumor cells at diagnosis.

7- Molecular clonal structure of tumor cells at diagnosis and after therapy.

8- Molecular characterization of MRD cells persisting to different regimens.

9- MRD monitoring based on next-generation sequencing (NGS).

10- MRD monitoring based on next-generation flow (NGF) cytometry.

11- Next-generation immune monitoring.

12- Response assessment in patients with IgG kappa multiple myeloma can be misinterpreted due to the interference between the patients M protein and Isatuximab, especially when reaching very good partial response, which is expected in the majority of them. To resolve this interference, we plan to use mass spectrometry, a method able to discriminate between the patient monoclonal protein and Isatuximab and therefore to properly assign treatment responses.

13- Evaluate the incidence of infections and their characteristics in the different arms of treatment (category and severity, the moment in which they occur, etc.) as well as other related data like antibiotic prophylaxis, previous vaccination, etc.

APPENDIX 4 - CENTRALIZATION OF PERIPHERAL BLOOD AND BONE MARROW SAMPLES

Hospital Clínico Universitario de Salamanca (HUSA), Clínica Universidad de Navarra (CUN) and *Hospital Universitario 12 de Octubre* have been designated as the reference laboratories.

Distribution. Participating sites will be distributed between the three reference laboratories. Each site will be informed about the distribution details during the initial visit and by means of the Sample Management documentation that will be provided by the trial coordinators and filed at each site.

Samples. 10cc of bone marrow in EDTA and 10cc in lithium heparin (flow cytometry, cytogenetic and molecular biology), 10cc of serum in gelose (Hevylite and Freelite and storage of serum for correlative studies), and 20cc of peripheral blood in EDTA (flow cytometry and molecular biology). Samples will be shipped at room temperature in less than 24 hours after collection.

Identification of samples. Patient identification number, Protocol code, time of disease (diagnosis, post-induction, 3 months from transplant, first relapse, end-of-study visit and at any time to confirm complete response), shipping hospital and name and telephone number of contact physician.

Time of shipping. Various sets of samples are required: at the time of diagnosis, post- induction, at three months from transplant, first relapse, at the end of study visit, and at any time to confirm complete response.

1. AT DIAGNOSIS

Bone marrow

A bone marrow (BM) sample should be collected at diagnosis (at least the following quantities):

- 10 cc EDTA: this volume will be used for:
  - 1 cc: DNA extraction. Sufficient volume at diagnosis.

  - 9 cc: for flow cytometry tests (MFC) and cell separation using a cell classifier.
  - Extraction of an additional 10ml is also recommended in case of damage to any sample.
- 10 cc heparin: this volume will be used in the separation process with AutoMacs®, before fluorescence in-situ hybridization studies (FISH). The separated sample will be frozen for genetic testing with purified plasma cells.

To ensure the BM samples are valid, they should be extracted from different areas in order to prevent contamination with medullary blood.

Peripheral blood (PB)

Peripheral blood samples will be collected from all patients at the screening visit:

- 10 cc EDTA, (x2):
  - 10 ml for analysis using MFC to identify, quantify, and separate circulating plasma cells. Other possible clonal cells that are identified will also be collected.
  - 10 ml for genetic testing. The plasma will be separated from the cells and frozen for the purpose of extracting nucleic acid and for genetic analyses. The cells will be used to extract DNA from germ cells.
- 10 cc serum (gel or PS tube, no additive):
  - Hevylite® and other protein assays. Serum to be stored at -20ºC at reference laboratories.

2. SAMPLES FOR MONITORING

Bone marrow

Bone marrow will need to be collected at any time the patient achieves complete remission, as long as this does not coincide with the specific times at which a bone marrow collection is already planned. In addition to the screening visit, these times are:

- After induction.

- 3 months after transplantation coinciding with the pre-consolidation evaluation, and:
  - Arms A and C:
    - After being administered the two cycles of consolidation.
    - After the end of cycle 7 of continuous treatment.
    - After the end of cycle 12 of continuous treatment, which coincides with the assessment of the primary Endpoint.
  - And In Arm B, patients without:
    - After being administered the two cycles of consolidation.
    - After the 5th extended VRD cycle.
    - After the 10th extended VRD cycle, which coincides with the assessment of the primary Endpoint.
  - And In Arm B, patients with ERI:
    - Every 5 cycles of iberdomide / Isatuximab / dexamethasone.
- Beyond the evaluation of the primary endpoint, a BM aspirate will be done yearly, including morphology, flow cytometry, molecular biology (centralized in the reference laboratory).
- At the time of suspected CR.

During these specific times a tube of 10ml EDTA will be sent and distributed in the central laboratory as follows:

- 5 ml EDTA, to evaluate the presence of minimal residual disease (MRD) using MFC.
- 5 ml EDTA, to evaluate the presence of MRD in molecular biology tests.

So that MFC and molecular biology tests can be compared, MRD should be analyzed from the same tube, and for this reason the actual volume needed is 10ml. It is recommended that an additional 10ml be collected in case any sample becomes damaged.

Peripheral blood

The following is required:

- 10 ml (x2) EDTA, to evaluate the presence of MRD in the plasma and cells via molecular biology tests.
- 10 ml gel or PS tube with no additives to collect serum samples for protein assays (HevyLite® and QIP-mass spectrometry)

These samples will be collected:

- After induction.
- At the time of suspected CR.
- 3 months after transplantation coinciding with the pre-consolidation evaluation, and:
  - In Arms A and C:
    - After being administered the two cycles of consolidation.
    - After the end of cycle 7 of continuous treatment.
    - After the end of cycle 12 of continuous treatment, which coincides with the assessment of the primary Endpoint.
  - In Arm B, <u>patients without ERI</u>:
    - After being administered the two cycles of consolidation.
    - After the 5th extended VRD cycle.
    - After the 10th extended VRD cycle, which coincides with the assessment of the primary Endpoint.
    - We strongly recommend to obtain **monthly until cycle #18** 10cc of peripheral blood in a gel or PS tube with no additives. Once obtained, the sample will be centrifuged and the collected serum will be frozen and stored at -20ºC (ideally at -80ºC)

- In Arm B, patients with ERI:
  - At the moment when biological progression is confirmed.
  - Every 5 cycles of iberdomide / Isatuximab / dexamethasone.
  - We strongly recommend to obtain **monthly until progression** 10cc of peripheral blood in a gel or PS tube with no additives. Once obtained, the sample will be centrifuged and the collected serum will be frozen and stored at -20ºC. (ideally at -80ºC)

. Beyond the evaluation of the primary endpoint, a BM aspirate will be done yearly, including morphology, flow cytometry, molecular biology (centralized in the reference laboratory).

3. SAMPLES WHEN RELAPSE/PROGRESSION OCCURS

All procedures will be exactly the same as those carried out at the time of diagnosis.

If extramedullary disease is observed, collection of one disease sample (via aspiration or biopsy) which is placed in normal saline with EDTA is recommended.

FLOW CYTOMETRY

Evaluation of minimal residual disease by flow cytometry (MRD-MFC) will consist of a relatively rapid, highly sensitive, reproducible biomarker to detect the presence of phenotypically abnormal clonal plasma cells (MRD). The methodology that will be used to process large volumes of samples (to ensure a high degree of sensitivity) is a bulk lysis-based approach, before performing conventional sorting of activated cells by fluorescence (FACS).

Bone marrow cells will be stained with two combinations of 8 monoclonal antibody colors:

Tube	V450	BV510	FITC	PE	PerCPCy5.5	PECy7	APC	APCC750
1	CD138	CD27	CD38	CD56	CD45	CD19	CD117	CD81
2	CD138	CD27	CD38	CD56	CD45	CD19	cyKAPPA	cyLAMBDA

The strategy based on the use of two tubes will include as a last resort monitoring of myeloma cells using 10-color flow cytometry. This will permit unique, high specificity confirmation of the monoclonality of phenotypically abnormal plasma cells (by light chain restriction). Said cells will have been clearly identified by low antigen expression (CD19, CD27, CD38, CD45, CD81) or overexpression (CD56, CD117, CD138). This strategy is possible thanks to the merge and calculation options available using *Infinicyt* software for analyses of flow data, as described below:

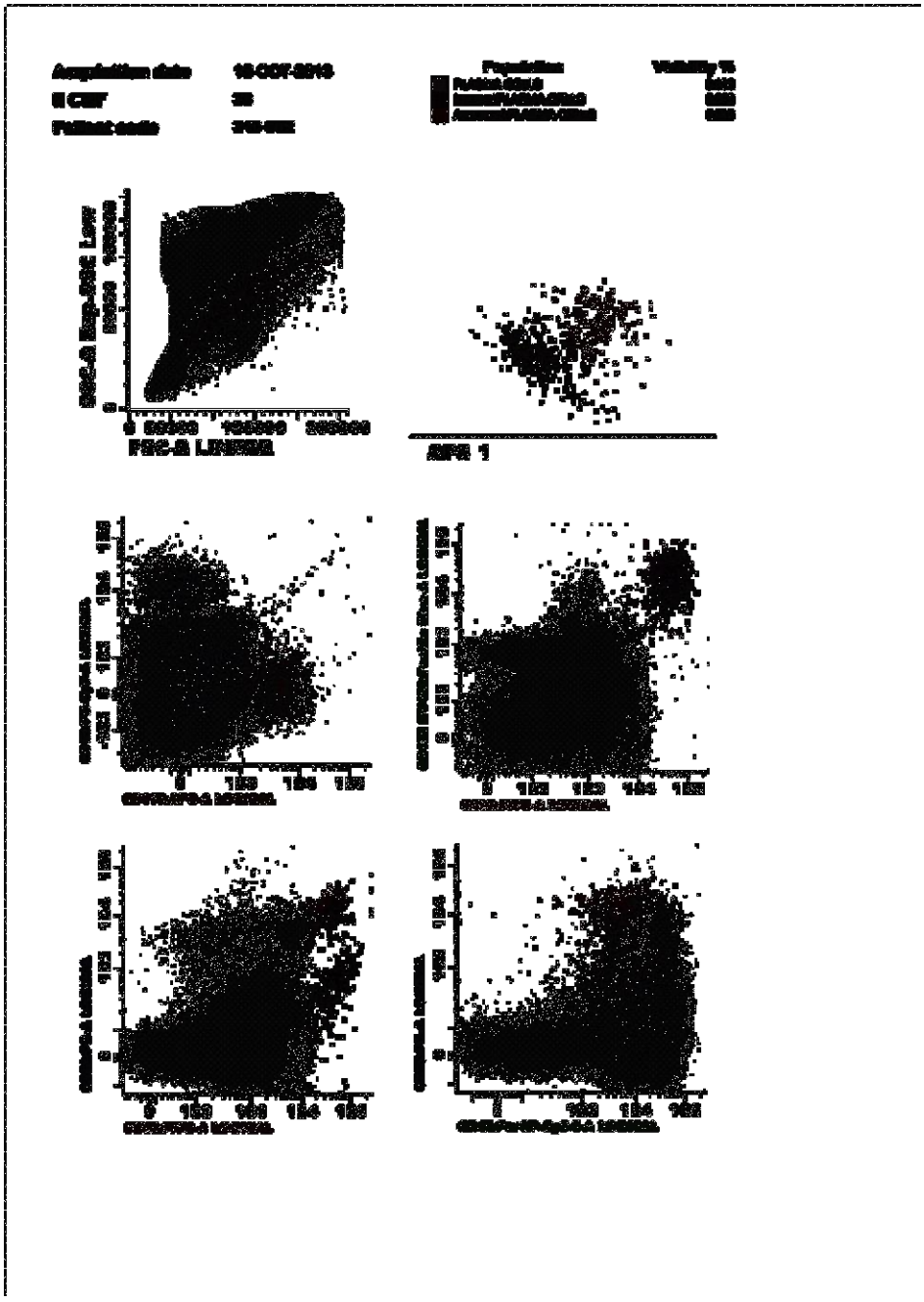
The presence of surface markers (CD138, CD27, CD38, CD56, CD45, CD19) and certain morphological characteristics (FSC and SSC) permit the specific identification of plasma cells (PC). The merge function in Infinicyt™ software will then be used to combine different data files corresponding to the 2 different aliquots of the same sample. This will create a new combined data file (unique file) that will contain all of the information measured in the same sample, including the antigens that have been measured directly and those not evaluated directly (missing values) for an individual cell in a specific aliquot of the sample (for example CD117 and CD81 in tube 2). The “Calculation” function in the Infinicyt™ software will then be used to estimate the “missing values”.

To do this, the software searches for the “nearest” event in each of the other aliquots from the sample, based simply on its unique position in the multidimensional space created by all of the common basic parameters. In other words, for each unique clonal plasma cell in a patient with recently-diagnosed symptomatic MM, 10 parameters will be evaluated which will be integrated into one representative vector for each patient via an automated analysis of the main component, based on a pattern.

This unique strategy will guarantee highly sensitive detection of minimal residual disease (10^{-5}) by flow cytometry, even if the information about the diagnostic sample is not available.

The report for each patient will include dot plots and graphic representations of the analysis of the main component, which will be available to the patient’s referring physician and /or the company.

The project manager will analyze the sample, include all results in the analytical database and generate a graphical report of the sample analysis. In all samples in which MRD has been detected (below 10^{-5} cells) a second project manager will perform a blind analysis of the sample in order to determine the accuracy and reproducibility of the results. The graphical report of the analysis will be printed only if the company requests it, and it will be stored electronically by the laboratory. The graphical report will also include the laboratory serial number and the patient code for the purpose of ensuring the reliability of the chain of custody.



Transportation and contacts. The appropriate containers for the trial will be sent by the Sponsor to the list of participating hospitals. Any errors in distribution or a lack of supply can be resolved by requesting containers from the Sponsor secretary in Madrid, Rocío Aguirre, by email: rocio@fundacionpethema.es , or by leaving a voicemail at this number: 916266232. The following mobile number: 629853789 may be used for urgent requests only.

Business hours. Samples will be accepted during arrival times from Monday to Thursday, and until 10:00 am on Friday (this means that **samples must be taken between Monday and Thursday, never on Friday, Saturday or Sunday)**. Please arrange for your samples so that they will arrive before 10:00 am (at each destination site, check with the courier). You must call to arrange delivery outside of these hours.

The Sponsor will cover the cost of transportation, using a courier company**. Further information is provided in the study Lab sample Manual.**

Samples will be sent to:

Salamanca:

Attn: Dr Noemi Puig / Dr Norma Gutiérrez / Dr Ramón García Sanz.

Unidad de Inmunopatología/Biología Molecular/citogenética.

Servicio de Hematología. Hospital Universitario de Salamanca

Paseo de la Transición Española, Bloque I, Planta 3, 426, 37007 Salamanca.

Tel: 923 291100. Ext: 56616, 56620, 55606. / Fax: 923 294624.

Email: rgarcias@usal.es, normagu@usal.es, noepuig@gmail.com

Pamplona:

Attn: Dr Bruno Paiva / Dr Iria Vazquez

CIMA LAB Diagnostics

Laboratorio de Citometría de Flujo (1.07). Centro de Investigación Médica Aplicada (CIMA)

Av. Pio XII 55, 31008 Pamplona.

Tel: 948 194 700. Ext: 1038. / Fax: 948 194 714.

Email: bpaiva@unav.es / ivurio@unav.es

Madrid:

Attn: Dr. Joaquín Martínez López / Dr. Teresa Cedena

Unidad de Biología Molecular.

Servicio de Hematología. Hospital Universitario 12 de Octubre.

Centro de Actividades Ambulatorias, 6ª planta bloque D.

Avda. de Córdoba s/n, 28041 Madrid

Tel: 91 779 27 87 / 91 779 23 03.

Email: jmarti01@med.ucm.es / mariateresa.cedena@salud.madrid.org

More details about sample management can be found in the study Lab Manual

APPENDIX 5 - NCI-CTC AE-CRITERIA

Common terminology criteria for classification of adverse events version 5.0 (NCI CTC AEv5) are available at the following Internet address:

https://ctep.cancer.gov/protocolDevelopment/electronic_applications/ctc.htm

APPENDIX 6 - ECOG AND KARNOFSKY PERFORMANCE STATUS

Karnofsky and ECOG performance status scales are shown in the table below:

Karnofsky	Description	ECOG
100	Normal, no complaints; no evidence of disease.	0
90	Able to carry on normal activity. Minor signs or symptoms of disease.	0
80	Normal activity with effort; some signs or symptoms of disease	1
70	Cares for self; unable to carry on normal activity or to do active work.	1
60	Requires occasional assistance, but is able to care for most of his/her personal needs	2
50	Requires considerable assistance and frequent medical care.	2
40	Disabled; requires special care and assistance.	3
30	Severely disabled; admission to hospital is indicated although death not imminent.	3
20	Very sick; hospital admission necessary; active supportive therapy required.	4
10	Moribund; fatal processes progressing rapidly.	4
0	Dead	5

*Karnofsky D, Burchenal J, The clinical evaluation of chemotherapeutic agents in cancer. In: MacLeod C, ed. Evaluation of Chemotherapeutic Agents. New York, NY: Columbia University Press; 1949:191–205. **Zubrod C, et al. Appraisal of methods for the study of chemotherapy in man: Comparative therapeutic trial of nitrogen mustard and thiophosphoramide. *Journal of Chronic Diseases*; 1960:11:7-33.

Available at: http://ecog-acrin.org/resources/ecog-performance-status

APPENDIX 7 - RESPONSE CRITERIA

Response will be assessed using IMWG criteria:

Response	Response criteria[1]
CR with sustained negative MRD	CR with MRD negative status in BM (by flow cytometry or molecular testing) as defined below, confirmed with one year of follow-up.
Molecular complete remission (mCR)	CR and absence of clonal plasma cells in BM analyzed using large-scale sequencing techniques with a minimum sensitivity of 10^{-5} (1 cell in 10^5 nucleated cells).
Immunophenotypic complete remission (CR – flow cytometry)	RC and absence of phenotypically abnormal clonal plasma cells in BM, using EuroFlow standard analysis recommendations to detect MRD in MM (minimum sensitivity of 1 cell in 10^5 nucleated cells).
Stringent complete response (sCR)	The following criteria must be met: 1. Disappearance of monoclonal protein in blood and urine, confirmed in immunofixation studies 2. Disappearance of soft tissue plasmacytomas. 3. BMPC < 5%.[2] 4. Normal serum kappa/lambda ratio in sFLC determinations. 5. Absence of clonal cells in BM[3], as shown by immunohistochemistry or immunofluorescence.
Complete Response (CR)	1. Disappearance of monoclonal protein in blood and urine confirmed by immunofixation 2. Disappearance of soft tissue plasmacytomas for at least 6 weeks. 3. < 5% BMPC[2]
Very good partial response (VGPR)	1. Disappearance of serum and urine monoclonal protein on electrophoresis even if detectable by immunofixation or ≥ 90% reduction in serum monoclonal protein and less than 100 mg of monoclonal protein detected in 24-hr urine sample

Partial response (PR)	1. ≥ 50% reduction in serum monoclonal protein and ≥ 90% reduction in urine monoclonal protein, or < 200 mg of monoclonal protein excreted in 24hr urine sample. 2. If the serum and urine M-protein are not measurable, a decrease of ≥ 50% in the difference between involved and uninvolved FLC levels is required in place of the M-protein criteria. 3. If serum and urine M-protein are not measurable, and serum free light assay is also not measurable, ≥ 50% reduction in bone marrow PCs is required in place of M-protein, provided baseline bone marrow plasma cell percentage was ≥ 30%. 4. In addition to the above criteria, if present at baseline, a ≥ 50% reduction in the size of soft tissue plasmacytomas is also required.
Stable disease (SD)	Not meeting criteria for sCR, CR, VGPR or disease progression
Progressive disease (PD)	Any one or more of the following criteria: Increase of 25% from lowest confirmed response value in one or more of the following criteria: · Serum M-protein (absolute increase ≥0.5 g/dl). · Serum M-protein increase ≥1g/dl, if the lowest M component was ≥ 5 g/dl. · Urine M-protein, absolute increase must be ≥ 200mg/24 h. · In patients without measurable serum and urine M-protein levels, the difference between involved and uninvolved FLC levels (absolute increase must be > 10mg/dl). · In patients without measurable serum and urine M-protein levels, and without measurable involved FLC levels, bone marrow plasma-

	cell percentage irrespective of baseline status (absolute increase must be ≥10%). Development of new soft tissue plasmacytomas or bone lesions (osteoporotic fractures do not constitute progression) or definite increase in the size of existing plasmacytomas or bone lesions. A definite increase is defined as a 50% (and ≥1 cm) increase of soft tissue plasmacytomas. Appearance of hypercalcemia (corrected serum calcium > 11.5 mg/dL) that can be attributed exclusively to myeloma.

1 Two consecutive measurements are required for all response categories. All categories require the absence of disease progression and no new bone lesions. These response criteria do not require routine radiological tests.

2 It is not necessary to repeat the bone biopsy to confirm the absence of plasma cells in BM.

3 In all categories related to relapse or progression, two consecutive measurements are required before disease relapse or progression is determined.

APPENDIX 8- ALGORITHMS USED IN THIS PROTOCOL

- **Corrected calcium:**

It will be calculated using the following formula and only for those patients in which during the screening visit albumin was out of range.

If calcium is expressed in mg/dL and albumin is expressed in g/dL:

Corrected calcium (mg/dL) = serum calcium (mg/dL) + 0.8 x (4 – serum albumin [g/dL])

If calcium is expressed in mmol/L and albumin is expressed in g/L:

Corrected calcium (mmol/L) = serum calcium (mmol/L) + 0.02 x (40 – serum albumin [g/L])

Burtis CA, Ashwood ER. Tietz textbook of clinical chemistry. 3rd ed. Philadelphia: WB Sauders; 1998.

- **Body surface area:**

BSA must be calculated using a standard nomogram. An example of a normogram is as follows:

$$\text{BSA } (m^2) = \sqrt{\frac{Height\ (cm) x\ Weight\ (kg)}{3{,}600}}$$

APPENDIX 9- CONTRACEPTION GUIDANCE FOR IBERDOMIDE

Female Subjects of Childbearing Potential

Females of childbearing potential enrolled in this protocol must agree to use two reliable forms of contraception simultaneously or to practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [e.g. calendar, ovulation, symptothermal or post ovulation methods] and withdrawal are not acceptable methods of contraception.) from heterosexual contact during the following time periods related to this study: 1) for at least 28 days before starting iberdomide; 2) while taking iberdomide; 3) during dose interruptions; and 4) for at least 28 days after the last dose of iberdomide.

The two methods of reliable contraception must include one highly effective method and one additional effective (barrier) method. If the below contraception methods are not appropriate for the FCBP, she must be referred to a qualified provider of contraception methods to determine the medically effective contraception method appropriate to the subject. The following are examples of highly effective and additional effective methods of contraception:

Examples of highly effective methods:

- Intrauterine device (IUD)
- Hormonal (birth control pills, injections, implants, levonorgestrel-releasing intrauterine system [IUS], medroxyprogesterone acetate depot injections, ovulation inhibitory progesterone-only pills [e.g., desogestrel])
- Tubal ligation
- Partner’s vasectomy

Examples of additional effective methods:

- Male condom
- Diaphragm
- Cervical Cap

Implants and levonorgestrel-releasing intrauterine systems are associated with an increased risk of infection at the time of insertion and irregular vaginal bleeding. Prophylactic antibiotics should be considered particularly in subjects with neutropenia.

Male Subjects

Male subjects must practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post ovulation methods] and withdrawal are not acceptable methods of contraception.) or agree to use a condom during sexual contact with a pregnant female or a FCBP while taking iberdomide, during dose interruptions and for at least 90 days following the last dose of iberdomide, even if he has undergone a successful vasectomy.

Pregnancy Testing

Medically supervised pregnancy tests with a minimum sensitivity of 25 mIU/mL must be performed for FCBP.

Females of childbearing potential must have two negative pregnancy tests (sensitivity of at least 25 mIU/mL) prior to starting iberdomide. The first pregnancy test must be performed within 10 to 14 days prior to the start of iberdomide and the second pregnancy test must be performed within 24 hours prior to the start of iberdomide. The subject may not receive CC-220 until the study doctor has verified that the results of these pregnancy tests are negative.

Females of childbearing potential with regular or no menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 28 days while taking iberdomide, at study discontinuation, and at Day 28 following the last dose of iberdomide.

Females of childbearing potential with irregular menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 14 days while taking iberdomide, at study discontinuation, and at Days 14 and 28 following the last dose of iberdomide.

Pregnancy Precautions for Iberdomide Use

Before Starting Iberdomide

Female Subjects of Childbearing Potential

Females of childbearing potential must have two negative pregnancy tests (sensitivity of at least 25 mIU/mL) prior to starting iberdomide. The first pregnancy test must be performed within 10 to 14 days prior to the start of iberdomide and the second pregnancy test must be performed within 24 hours prior to the start of iberdomide. The subject may not receive iberdomide until the study doctor has verified that the results of these pregnancy tests are negative.

Females of childbearing potential must use two reliable forms of contraception simultaneously, or practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) from heterosexual contact for at least 28 days before starting iberdomide.

Male Subjects

Male subjects must agree to practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post ovulation methods] and withdrawal are not acceptable methods of contraception.) or agree to use a condom during sexual contact with a pregnant female or a

FCBP while taking iberdomide, during dose interruptions and for at least 90 days following the last dose of iberdomide, even if he has undergone a successful vasectomy.

During and After Study Participation

Female Subjects

- Females of childbearing potential with regular or no menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 28 days while taking iberdomide, at study discontinuation, and at Day 28 following the last dose of iberdomide.

- Females of childbearing potential with irregular menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 14 days while taking iberdomide, at study discontinuation, and at Days 14 and 28 following the last dose of iberdomide.

- At each visit, the Investigator must confirm with the FCBP that she is continuing to use two reliable methods of birth control if not committing to complete abstinence, or confirm commitment to complete abstinence.

- If a FCBP considers the need to change or to stop a method of contraception, the Investigator must be notified immediately.

- Counseling about pregnancy precautions and the potential risks of fetal exposure must be conducted at a minimum of every 28 days.

- If pregnancy or a positive pregnancy test does occur in a subject, iberdomide must be immediately discontinued.

- Pregnancy testing and counseling must be performed if a subject misses her period or if her pregnancy test or her menstrual bleeding is abnormal. Iberdomide must be discontinued during this evaluation.

- Females must agree to abstain from breastfeeding while taking iberdomide and for at least 28 days after the last dose of CC-220.

Male Subjects

- Must practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post ovulation methods] and withdrawal are not acceptable methods of contraception.) or use a condom during sexual contact with a pregnant female or a FCBP while receiving iberdomide, during dose interruptions and for at least 90 days following the last dose of iberdomide, even if he has undergone a successful vasectomy.

- Must not donate semen or sperm while receiving iberdomide, during dose interruptions or for at least 90 days following the last dose of iberdomide.

- Counseling about pregnancy precautions and the potential risks of fetal exposure must be conducted at a minimum of every 28 days.

- If pregnancy or a positive pregnancy test does occur in the partner of a male subject while taking iberdomide, the Investigator must be notified immediately.

Additional Precautions

- Subjects should be instructed to never give iberdomide to another person.

- Subjects should be instructed to return any unused capsules to the study doctor.

- Subjects should not donate blood while receiving iberdomide, during dose interruptions and for at least 28 days following the last dose of iberdomide.

- No more than a 28 day iberdomide supply may be dispensed to a subject.

APPENDIX 10 - EDUCATION AND COUNSELING GUIDANCE DOCUMENTS FOR IBERDOMIDE. IBERDOMIDE EDUCATION AND COUNSELING GUIDANCE DOCUMENT FOR FEMALE SUBJECTS

To be completed prior to each dispensing of Iberdomide.

Protocol Number: _____

Subject Name (Print): _____ DOB: _____/_____/_____ (dd/mmm/yyyy)

Check one risk category:

- ☐ FCBP (Female of childbearing potential): a female who: 1) has achieved menarche (first menstrual cycle) at some point, 2) has not undergone a hysterectomy (the surgical removal of the uterus) or bilateral oophorectomy (the surgical removal of both ovaries) or 3) has not been naturally postmenopausal (amenorrhea following cancer therapy does not rule out childbearing potential) for at least 24 consecutive months (ie, has had menses at any time during the preceding 24 consecutive months)
- ☐ NOT FCBP

Female of Childbearing Potential:

1. I have verified and counseled the subject regarding the following:
   - ☐ Potential risk of fetal exposure to iberdomide: iberdomide was found to cause birth defects in experimental animals (rat and rabbit). A teratogenic potential of iberdomide in humans cannot be ruled out. If iberdomide is taken during pregnancy, it may cause birth defects or death to any unborn baby. Females are advised to avoid pregnancy while taking iberdomide. Females of childbearing potential must agree not to become pregnant while taking iberdomide.
   - ☐ That the required pregnancy tests performed are negative.
   - ☐ The subject confirmed that she is using TWO reliable methods of birth control at the same time, or complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) from heterosexual contact (at least 28 days prior to receiving iberdomide, while

receiving iberdomide, during dose interruptions and for at least 28 days after the last dose of iberdomide).

One highly effective method and one additional method of birth control must be used AT THE SAME TIME. The following are examples of highly effective and additional effective methods of contraception:

- Examples of highly effective methods:
  - Intrauterine device (IUD)
  - Hormonal (birth control pills, injections, implants, levonorgestrel-releasing intrauterine system [IUS], medroxyprogesterone acetate depot injections, ovulation inhibitory progesterone-only pills [e.g. desogestrel])
  - Tubal ligation
  - Partner's vasectomy
- Examples of additional effective methods:
  - Male condom
  - Diaphragm
  - Cervical Cap

☐ The subject confirmed that even if she has amenorrhea she must comply with advice on contraception.

☐ Pregnancy tests before, during administration of iberdomide and at the last dose of iberdomide, even if the subject agrees not to have reproductive heterosexual contact.

☐ Frequency of pregnancy tests to be done:

- Two pregnancy tests will be performed prior to receiving iberdomide, one within 10 to 14 days, and a second within 24 hours of the start of iberdomide.
- <u>Every week</u> during the first 28 days of this study and a pregnancy test <u>every 28 days</u> while the subject is taking iberdomide if menstrual cycles are regular.
- <u>Every week</u> during the first 28 days of this study and a pregnancy test <u>every 14 days</u> while the subject is taking iberdomide if menstrual cycles are irregular.
- If the subject missed a period or has unusual menstrual bleeding.
- When the subject is discontinued from the study and at Day 28 after iberdomide discontinuation if menstrual cycles are regular. If menstrual cycles are irregular, pregnancy tests will be done at discontinuation from the study and at Days 14 and 28 after the last dose of iberdomide.

- ☐ The subject confirmed that she will stop taking iberdomide immediately in the event of becoming pregnant and to call her study doctor as soon as possible.
- ☐ The subject confirmed that she has not and will not breastfeed a baby while taking iberdomide and for at least 28 days after the last dose of iberdomide.
- ☐ The subject has not and will never share iberdomide with anyone else.
- ☐ The subject has not and will not donate blood while taking iberdomide, during dose interruptions and for at least 28 days after the last dose of iberdomide.
- ☐ The subject has not and will not break, chew, or open iberdomide capsules at any point.
- ☐ The subject confirmed that she will return unused iberdomide capsules to the study doctor.

2. I have provided the iberdomide Information Sheet to the subject.

Female Not of Childbearing Potential (Natural Menopause for at Least 24 Consecutive Months, a Hysterectomy, or Bilateral Oophorectomy):

1. I have verified and counseled the subject regarding the following:
   - ☐ Potential risk of fetal exposure to iberdomide: iberdomide was found to cause birth defects in experimental animals (rat and rabbit). A teratogenic potential of iberdomide in humans cannot be ruled out. If iberdomide is taken during pregnancy, it may cause birth defects or death to any unborn baby.
   - ☐ The subject has not and will never share iberdomide with anyone else.
   - ☐ The subject has not and will not donate blood while taking iberdomide, during dose interruptions and for at least 28 days after the last dose of iberdomide.
   - ☐ The subject has not and will not break, chew, or open iberdomide capsules at any point.
   - ☐ The subject confirmed that she will return unused iberdomide capsules to the study doctor.
2. I have provided the iberdomide Information Sheet to the subject.

<u>Do Not Dispense iberdomide if:</u>

- **The subject is pregnant.**
- **No pregnancy tests were conducted for a FCBP.**
- **The subject states she did not use TWO reliable methods of birth control (unless practicing complete abstinence from heterosexual contact) at least 28 days prior**

to receiving iberdomide, while receiving iberdomide and during dose interruptions.

- **The subject stated that she has or does not want to adhere to pregnancy precautions outlined within this PPP.**

Counselor Name (Print): _____

Counselor Signature: _____ Date: _____/_____/_____(dd/mmm/yyyy)

Maintain a copy of the Education and Counseling Guidance Document in the subject's records.

1. IBERDOMIDE EDUCATION AND COUNSELING GUIDANCE DOCUMENT FOR MALE SUBJECTS

To be completed prior to each dispensing of IBERDOMIDE.

Protocol Number: _____

Subject Name (Print): _____ DOB: _____/_____/_____ (dd/mmm/yyyy)

1. I have verified and counseled the subject regarding the following:
   - ☐ Potential risk of fetal exposure to iberdomide: iberdomide was found to cause birth defects in experimental animals (rat and rabbit). A teratogenic potential of iberdomide in humans cannot be ruled out. If iberdomide is taken during pregnancy, it may cause birth defects or death to any unborn baby.
   - ☐ The subject confirmed that he has practiced complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) or used a condom when engaging in sexual contact (including those who have had a vasectomy) with a pregnant female or FCBP, while taking iberdomide, during dose interruptions and for at least 90 days after the last dose of iberdomide.
   - ☐ The subject confirmed that he has not impregnated his female partner while in the study.
   - ☐ The subject confirmed that he will notify his study doctor if his female partner becomes pregnant and the female partner of a male subject taking iberdomide confirmed that she will call her healthcare provider immediately if she becomes pregnant.
   - ☐ The subject has not and will never share iberdomide with anyone else.
   - ☐ The subject confirmed that he has not donated and will not donate semen or sperm while taking iberdomide or during dose interruptions and that he will not donate semen or sperm for at least 90 days after the last dose of iberdomide.
   - ☐ The subject has not and will not donate blood while taking iberdomide, during dose interruptions and for at least 28 days after the last dose of iberdomide.
   - ☐ The subject has not and will not break, chew, or open iberdomide capsules at any point.

- ☐ The subject confirmed that he will return unused iberdomide capsules to the study doctor.

2. I have provided the iberdomide Information Sheet to the subject.

Do Not Dispense iberdomide if:

- **The subject stated that he has or does not want to adhere to pregnancy precautions outlined within this PPP.**

Counselor Name (Print): _____

Counselor Signature: _____ Date: _____/_____/_____ (dd/mmm/yyyy)

Maintain a copy of the Education and Counseling Guidance Document in the subject's records.

2. IBERDOMIDE INFORMATION SHEET

For subjects enrolled in clinical research studies

Please read this iberdomide Information Sheet before you start taking iberdomide and each time you get a new supply. This iberdomide Information Sheet does not take the place of an informed consent to participate in clinical research or talking to your study doctor or healthcare provider about your medical condition or your treatment.

What is the most important information I should know about iberdomide?

1. **iberdomide may cause birth defects (deformed babies) or death of an unborn baby.** iberdomide is similar to the medicine thalidomide. It is known that thalidomide causes life-threatening birth defects. iberdomide was found to cause birth defects in experimental animals (rat and rabbit). iberdomide has not been tested in pregnant women but may cause birth defects.

   If you are a female who is able to become pregnant:

   - **Do not take iberdomide if you are pregnant or plan to become pregnant**
   - **You must practice complete abstinence from sexual contact with a male or use two reliable, separate forms of effective birth control at the same time:**
     - for 28 days before starting iberdomide
     - while taking iberdomide
     - during breaks (dose interruptions) of iberdomide
     - for at least 28 days after the last dose of iberdomide
   - **You must have pregnancy testing done at the following times:**
     - within 10 to 14 days prior to the first dose of iberdomide
     - 24 hours prior to the first dose of iberdomide
     - weekly for the first 28 days
     - if you have regular menstrual periods: every 28 days after the first month
     - if you have irregular menstrual periods: every 14 days after the first month
     - if you miss your period or have unusual menstrual bleeding
     - 28 days after the last dose of iberdomide (14 and 28 days after the last dose if menstrual periods are irregular)
   - **Stop taking iberdomide if you become pregnant while taking iberdomide**

  - If you suspect you are pregnant at any time during the study, you must stop iberdomide immediately and immediately inform your study doctor. Your study doctor will report all cases of pregnancy to Sponsor.
- **Do not breastfeed while taking iberdomide and for at least 28 days after the last dose of iberdomide**
- The study doctor will be able to advise you where to get additional advice on contraception.

If you are a female not able to become pregnant:

In order to ensure that an unborn baby is not exposed to iberdomide, your study doctor will confirm that you are not able to become pregnant.

If you are a male:

In a 9-month study in monkeys given iberdomide (at a dose higher than the highest dose intended in this study), male monkeys had a decrease in the amount of sperm producing cells. There were no observable effects in male sex organs in monkeys that received a dose higher than the highest intended dose in this study. It is not known at this time if this finding can be reversed or at what time after start of iberdomide treatment this change occurred.

It is not known if there might be a similar effect in men taking iberdomide and how this might affect their ability to father children. There were no effects of iberdomide in the sex organs of female monkeys.

- subjects (including those who have had a vasectomy) must practice complete abstinence or must use a condom during sexual contact with a pregnant female or a female that can become pregnant:
  - While you are taking iberdomide
  - During breaks (dose interruptions) of iberdomide
  - For at least 90 days after the last dose of iberdomide
- **Male subjects should not donate sperm or semen** while taking iberdomide, during breaks (dose interruptions) and for at least 90 days after the last dose of iberdomide.

- **If you suspect that your partner is pregnant any time during the study, you must immediately inform your study doctor. The study doctor will report all cases of pregnancy toSponsor. Your partner should call their healthcare provider immediately if they become pregnant.**

2. **All subjects:**

- **Do not share iberdomide with other people. It must be kept out of the reach of children and should never be given to any other person.**
- **Do not donate blood** while you take iberdomide, during breaks (dose interruptions) and for at least 28 days after the last dose of iberdomide.
- **Do not break, chew, or open iberdomide capsules at any point.**
- You will get no more than a 28-day supply of iberdomide at one time.
- Return unused iberdomide capsules to your study doctor.

Additional information is provided in the informed consent form and you can ask your study doctor for more information.

APPENDIX 11 CONTRACEPTION GUIDANCE FOR LENALIDOMIDE

Lenalidomide Risks of Fetal Exposure, Pregnancy Testing Guidelines and Acceptable Birth Control Methods

Risks Associated with Pregnancy

Lenalidomide is structurally related to thalidomide. Thalidomide is a known human teratogenic active substance that causes severe life-threatening birth defects. An embryofetal development study in animals indicates that lenalidomide produced malformations in the offspring of female monkeys who received the drug during pregnancy. A teratogenic effect of lenalidomide in humans cannot be ruled out. Therefore, a pregnancy prevention program must be followed.

Definition of Females of Childbearing Potential

A FCBP is a female who: 1) has achieved menarche at some point, 2) has not undergone a hysterectomy or bilateral oophorectomy or 3) has not been naturally postmenopausal (amenorrhea following cancer therapy does not rule out childbearing potential) for at least 24 consecutive months (ie, has had menses at any time in the preceding 24 consecutive months).

Definition of Females Not of Childbearing Potential

Females who do not meet the above definition of FCBP should be classified as FNCBP.

Counseling

Females of Childbearing Potential

For a FCBP, lenalidomide is contraindicated unless all of the following are met (ie, all FCBP must be counseled concerning the following risks and requirements prior to the start of lenalidomide):

She understands the potential teratogenic risk to the unborn child

She understands the need for effective contraception, without interruption, 28 days before starting lenalidomide, throughout the entire duration of lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide

She understands and agrees to inform the Investigator if a change or stop of method of contraception is needed

She must be capable of complying with effective contraceptive measures

She is informed and understands the potential consequences of pregnancy and the need to notify her study doctor immediately if there is a risk of pregnancy

She understands the need to commence lenalidomide as soon as it is dispensed following a negative pregnancy test

She understands and accepts the need to undergo pregnancy testing based on the frequency outlined in this plan (Section 0) and in the Informed Consent

She acknowledges that she understands the hazards lenalidomide can cause to an unborn fetus and the necessary precautions associated with the use of lenalidomide.

The Investigator must ensure that a FCBP:

Complies with the conditions of the pregnancy prevention plan, including confirmation that she has an adequate level of understanding

Acknowledges the aforementioned requirements.

Females Not of Childbearing Potential

For a FNCBP, lenalidomide is contraindicated unless all of the following are met (ie, all FNCBP must be counseled concerning the following risks and requirements prior to the start of lenalidomide):

She acknowledges she understands the hazards lenalidomide can cause to an unborn fetus and the necessary precautions associated with the use of lenalidomide.

Males

Traces of lenalidomide have been found in semen. Male subjects taking lenalidomide must meet the following conditions (ie, all males must be counseled concerning the following risks and requirements prior to the start of lenalidomide):

Understand the potential teratogenic risk if engaged in sexual activity with a pregnant female or a FCBP

Understand the need for the use of a condom even if he has had a vasectomy, if engaged in sexual activity with a pregnant female or a FCBP

Understand the potential teratogenic risk if the subject donates semen or sperm.

Contraception

Female Subjects of Childbearing Potential

Females of childbearing potential enrolled in this protocol must agree to use two reliable forms of contraception simultaneously or to practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) from heterosexual contact during the following time periods related to this study: 1) for at least 28 days before starting lenalidomide; 2) while taking lenalidomide; 3) during dose interruptions; and 4) for at least 28 days after the last dose of lenalidomide.

The two methods of reliable contraception must include one highly effective method and one additional effective (barrier) method. If the below contraception methods are not appropriate for the FCBP, she must be referred to a qualified provider of contraception methods to determine the medically effective contraception method appropriate to the subject. The following are examples of highly effective and additional effective methods of contraception:

Examples of highly effective methods:

- Intrauterine device (IUD)
- Hormonal (birth control pills, injections, implants, levonorgestrel-releasing intrauterine system [IUS], medroxyprogesterone acetate depot injections, ovulation inhibitory progesterone-only pills [e.g. desogestrel])
- Tubal ligation
- Partner’s vasectomy

Examples of additional effective methods:

- Male condom
- Diaphragm
- Cervical Cap

Because of the increased risk of venous thromboembolism in subjects with multiple myeloma taking lenalidomide and dexamethasone, combined oral contraceptive pills are not recommended. If a subject is currently using combined oral contraception the subject should switch to another one of the highly effective methods listed above. The risk of venous thromboembolism continues for 4 to 6 weeks after discontinuing combined oral contraception. The efficacy of contraceptive steroids may be reduced during co-treatment with dexamethasone.

Implants and levonorgestrel-releasing intrauterine systems are associated with an increased risk of infection at the time of insertion and irregular vaginal bleeding. Prophylactic antibiotics should be considered particularly in subjects with neutropenia.

Male Subjects

Male subjects must practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) or agree to use a condom during sexual contact with a pregnant female or a FCBP while taking lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide, even if he has undergone a successful vasectomy.

Pregnancy Testing

Medically supervised pregnancy tests with a minimum sensitivity of 25 mIU/mL must be performed for FCBP.

Females of childbearing potential must have two negative pregnancy tests (sensitivity of at least 25 mIU/mL) prior to starting lenalidomide. The first pregnancy test must be performed within 10 to 14 days prior to the start of lenalidomide and the second pregnancy test must be performed within 24 hours prior to the start of lenalidomide. The subject may not receive lenalidomide until the study doctor has verified that the results of these pregnancy tests are negative.

Females of childbearing potential with regular or no menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 28 days while taking lenalidomide, at study discontinuation, and at Day 28 following the last dose of lenalidomide.

Females of childbearing potential with irregular menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 14 days while taking lenalidomide, at study discontinuation, and at Days 14 and 28 following the last dose of lenalidomide.

Pregnancy Precautions for Lenalidomide Use

Before Starting Lenalidomide

Female Subjects of Childbearing Potential

Females of childbearing potential must have two negative pregnancy tests (sensitivity of at least 25 mIU/mL) prior to starting lenalidomide. The first pregnancy test must be performed within 10 to 14 days prior to the start of lenalidomide and the second pregnancy test must be performed within 24 hours prior to the start of lenalidomide. The subject may not receive lenalidomide until the study doctor has verified that the results of these pregnancy tests are negative.

Females of childbearing potential must use two reliable forms of contraception simultaneously, or practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) from heterosexual contact for at least 28 days before starting lenalidomide.

Male Subjects

Male subjects must agree to practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) or agree to use a condom during sexual contact with a pregnant female or a FCBP while taking lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide, even if he has undergone a successful vasectomy.

During and After Study Participation

Female Subjects

Females of childbearing potential with regular or no menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 28 days while taking lenalidomide, at study discontinuation, and at Day 28 following the last dose of lenalidomide.

Females of childbearing potential with irregular menstrual cycles must agree to have pregnancy tests weekly for the first 28 days of study participation and then every 14 days while taking lenalidomide, at study discontinuation, and at Days 14 and 28 following the last dose of lenalidomide.

At each visit, the Investigator must confirm with the FCBP that she is continuing to use two reliable methods of birth control if not committing to complete abstinence, or confirm commitment to complete abstinence.

If a FCBP considers the need to change or to stop a method of contraception, the Investigator must be notified immediately.

Counseling about pregnancy precautions and the potential risks of fetal exposure must be conducted at a minimum of every 28 days.

If pregnancy or a positive pregnancy test does occur in a subject, lenalidomide must be immediately discontinued.

Pregnancy testing and counseling must be performed if a subject misses her period or if her pregnancy test or her menstrual bleeding is abnormal. Lenalidomide must be discontinued during this evaluation.

Females must agree to abstain from breastfeeding while taking lenalidomide and for at least 28 days after the last dose of lenalidomide.

Male Subjects

Must practice complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) or use a condom during sexual contact with a pregnant female or a FCBP while taking lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide, even if he has undergone a successful vasectomy.

Must not donate semen or sperm while receiving lenalidomide, during dose interruptions or for at least 28 days after the last dose of lenalidomide.

Counseling about pregnancy precautions and the potential risks of fetal exposure must be conducted at a minimum of every 28 days.

If pregnancy or a positive pregnancy test does occur in the partner of a male subject while taking lenalidomide, the Investigator must be notified immediately.

Additional Precautions

Subjects should be instructed to never give lenalidomide to another person.

Subjects should be instructed to return any unused capsules to the study doctor.

Subjects should not donate blood while receiving lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide.

No more than a 28-day lenalidomide supply may be dispensed with each cycle of lenalidomide.

APPENDIX 12 EDUCATION AND COUNSELLING GUIDANCE DOCUMENTS FOR LENALIDOMIDE

LENALIDOMIDE EDUCATION AND COUNSELING GUIDANCE DOCUMENT FOR FEMALE SUBJECTS

To be completed prior to each dispensing of lenalidomide.

Protocol Number: _____

Subject Name (Print): _____ DOB: _____/_____/_____ (dd/mmm/yyyy)

Check one risk category:

- ☐ FCBP (Female of childbearing potential): a female who: 1) has achieved menarche (first menstrual cycle) at some point, 2) has not undergone a hysterectomy (the surgical removal of the uterus) or bilateral oophorectomy (the surgical removal of both ovaries) or 3) has not been naturally postmenopausal (amenorrhea following cancer therapy does not rule out childbearing potential) for at least 24 consecutive months (ie, has had menses at any time during the preceding 24 consecutive months)

- ☐ NOT FCBP

Female of Childbearing Potential:

1. I have verified and counseled the subject regarding the following:

   - ☐ Potential risk of fetal exposure to lenalidomide: A teratogenic potential of lenalidomide in humans cannot be ruled out. If lenalidomide is taken during pregnancy, it may cause birth defects or death to any unborn baby. Females are advised to avoid pregnancy while taking lenalidomide. Females of childbearing potential must agree not to become pregnant while taking lenalidomide.

   - ☐ That the required pregnancy tests performed are negative.

   - ☐ The subject confirmed that she is using TWO reliable methods of birth control at the same time, or complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) from heterosexual contact (at least 28 days prior to receiving lenalidomide, while receiving lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide).

One highly effective method and one additional method of birth control must be used AT THE SAME TIME. The following are examples of highly effective and additional effective methods of contraception:

Examples of highly effective methods:

- Intrauterine device (IUD)
- Hormonal (birth control pills, injections, implants, levonorgestrel-releasing intrauterine system [IUS], medroxyprogesterone acetate depot injections, ovulation inhibitory progesterone-only pills [e.g. desogestrel])
- Tubal ligation
- Partner's vasectomy

Examples of additional effective methods:

- Male condom
- Diaphragm
- Cervical Cap

☐ The subject confirmed that even if she has amenorrhea she must comply with advice on contraception.

☐ Pregnancy tests before, during administration of lenalidomide and at the last dose of lenalidomide, even if the subject agrees not to have reproductive heterosexual contact.

☐ Frequency of pregnancy tests to be done:

Two pregnancy tests will be performed prior to receiving lenalidomide, one within 10 to 14 days, and a second within 24 hours of the start of lenalidomide.

<u>Every week</u> during the first 28 days of this study and a pregnancy test <u>every 28 days</u> while the subject is taking lenalidomide if menstrual cycles are regular.

<u>Every week</u> during the first 28 days of this study and a pregnancy test <u>every 14 days</u> while the subject is taking lenalidomide if menstrual cycles are irregular.

If the subject missed a period or has unusual menstrual bleeding.

When the subject is discontinued from the study and at Day 28 after the last dose of lenalidomide if menstrual cycles are regular. If menstrual cycles are irregular, pregnancy tests will be done at discontinuation from the study and at Days 14 and 28 after the last dose of lenalidomide.

- ☐ The subject confirmed that she will stop taking lenalidomide immediately in the event of becoming pregnant and to call her study doctor as soon as possible.
- ☐ The subject confirmed that she has not and will not breastfeed a baby while taking lenalidomide and for at least 28 days after the last dose of lenalidomide.
- ☐ The subject has not and will never share lenalidomide with anyone else.
- ☐ The subject has not and will not donate blood while taking lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide.
- ☐ The subject has not and will not break, chew, or open lenalidomide capsules at any point.
- ☐ The subject confirmed that she will return unused lenalidomide capsules to the study doctor.

I have provided the Lenalidomide Information Sheet to the subject.

Female Not of Childbearing Potential (Natural Menopause for at Least 24 Consecutive Months, a Hysterectomy, or Bilateral Oophorectomy):

1. I have verified and counseled the subject regarding the following:

- ☐ Potential risk of fetal exposure to lenalidomide: A teratogenic potential of lenalidomide in humans cannot be ruled out. If lenalidomide is taken during pregnancy, it may cause birth defects or death to any unborn baby.
- ☐ The subject has not and will never share lenalidomide with anyone else.
- ☐ The subject has not and will not donate blood while taking lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide.
- ☐ The subject has not and will not break, chew, or open lenalidomide capsules at any point.
- ☐ The subject confirmed that she will return unused lenalidomide capsules to the study doctor.

I have provided the Lenalidomide Information Sheet to the subject.

<u>Do Not Dispense Lenalidomide if:</u>

The subject is pregnant.

No pregnancy tests were conducted for a FCBP.

The subject states she did not use TWO reliable methods of birth control (unless practicing complete abstinence from heterosexual contact) at least 28 days prior to receiving lenalidomide, while receiving lenalidomide and during dose interruptions.

The subject stated that she has or does not want to adhere to pregnancy precautions outlined within this PPP.

Counselor Name (Print): _____

Counselor Signature: _____ Date: _____/_____/_____(dd/mmm/yyyy)

Maintain a copy of the Education and Counseling Guidance Document in the subject's records.

LENALIDOMIDE EDUCATION AND COUNSELING GUIDANCE DOCUMENT FOR MALE SUBJECTS

To be completed prior to each dispensing of lenalidomide.

Protocol Number: _____

Subject Name (Print): _____ DOB: _____/_____/_____ (dd/mmm/yyyy)

1. I have verified and counseled the subject regarding the following:

   - ☐ Potential risk of fetal exposure to lenalidomide: A teratogenic potential of lenalidomide in humans cannot be ruled out. If lenalidomide is taken during pregnancy, it may cause birth defects or death to any unborn baby.

   - ☐ The subject confirmed that he has practiced complete abstinence (True abstinence is acceptable when this is in line with the preferred and usual lifestyle of the subject. Periodic abstinence [eg calendar, ovulation, symptothermal or post-ovulation methods] and withdrawal are not acceptable methods of contraception.) or used a condom when engaging in sexual contact (including those who have had a vasectomy) with a pregnant female or FCBP, while taking lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide.

   - ☐ The subject confirmed that he has not impregnated his female partner while in the study.

   - ☐ The subject confirmed that he will notify his study doctor if his female partner becomes pregnant and the female partner of a male subject taking lenalidomide confirmed that she will call her healthcare provider immediately if she becomes pregnant.

   - ☐ The subject has not and will never share lenalidomide with anyone else.

   - ☐ The subject confirmed that he has not donated and will not donate semen or sperm while taking lenalidomide or during dose interruptions and that he will not donate semen or sperm for at least 28 days after the last dose of lenalidomide.

   - ☐ The subject has not and will not donate blood while taking lenalidomide, during dose interruptions and for at least 28 days after the last dose of lenalidomide.

   - ☐ The subject has not and will not break, chew, or open lenalidomide capsules at any point.

- ☐ The subject confirmed that he will return unused lenalidomide capsules to the study doctor.

2. I have provided the Lenalidomide Information Sheet to the subject.

Do Not Dispense Lenalidomide if:

The subject stated that he has or does not want to adhere to pregnancy precautions outlined within this PPP.

Counselor Name (Print): _____

Counselor Signature: _____ Date: _____/_____/_____(dd/mmm/yyyy)

Maintain a copy of the Education and Counseling Guidance Document in the subject's records.

LENALIDOMIDE INFORMATION SHEET

For subjects enrolled in clinical research studies

Please read this Lenalidomide Information Sheet before you start taking lenalidomide and each time you get a new supply. This Lenalidomide Information Sheet does not take the place of an informed consent to participate in clinical research or talking to your study doctor or healthcare provider about your medical condition or your treatment.

What is the most important information I should know about lenalidomide?

1. **Lenalidomide may cause birth defects (deformed babies) or death of an unborn baby.** Lenalidomide is similar to the medicine thalidomide. It is known that thalidomide causes life-threatening birth defects.

   If you are a female who is able to become pregnant:

   Do not take lenalidomide if you are pregnant or plan to become pregnant

   You must practice complete abstinence from sexual contact with a male or use two reliable, separate forms of effective birth control at the same time:

   for 28 days before starting lenalidomide

   while taking lenalidomide

   during breaks (dose interruptions) of lenalidomide

   for at least 28 days after the last dose of lenalidomide

   You must have pregnancy testing done at the following times:

   within 10 to 14 days prior to the first dose of lenalidomide

   24 hours prior to the first dose of lenalidomide

   weekly for the first 28 days

   if you have regular menstrual periods: every 28 days after the first month

   if you have irregular menstrual periods: every 14 days after the first month

if you miss your period or have unusual menstrual bleeding

28 days after the last dose of lenalidomide (14 and 28 days after the last dose if menstrual periods are irregular)

Stop taking lenalidomide if you become pregnant while taking lenalidomide

If you suspect you are pregnant at any time during the study, you must stop lenalidomide immediately and immediately inform your study doctor. Your study doctor will report all cases of pregnancy to Celgene Corporation.

Do not breastfeed while taking lenalidomide and for at least 28 days after the last dose of lenalidomide

The study doctor will be able to advise you where to get additional advice on contraception.

If you are a female not able to become pregnant:

In order to ensure that an unborn baby is not exposed to lenalidomide, your study doctor will confirm that you are not able to become pregnant.

If you are a male:

A small amount of lenalidomide is found in human semen. The risk to an unborn baby in females whose male partner is receiving lenalidomide is unknown at this time.

Male subjects (including those who have had a vasectomy) must practice complete abstinence or must use a condom during sexual contact with a pregnant female or a female that can become pregnant:

While you are taking lenalidomide

During breaks (dose interruptions) of lenalidomide

For at least 28 days after the last dose of lenalidomide

Male subjects should not donate sperm or semen while taking lenalidomide, during breaks (dose interruptions) and for at least 28 days after the last dose of lenalidomide.

If you suspect that your partner is pregnant any time during the study, you must immediately inform your study doctor. The study doctor will report all cases of pregnancy to Sponsor

Corporation. Your partner should call their healthcare provider immediately if they become pregnant.

2. **All subjects:**

Do not share lenalidomide with other people. It must be kept out of the reach of children and should never be given to any other person.

Do not donate blood while you take lenalidomide, during breaks (dose interruptions) and for at least 28 days after the last dose of lenalidomide.

Do not break, chew, or open lenalidomide capsules at any point.

You will get no more than a 28-day supply of lenalidomide at one time.

Return unused lenalidomide capsules to your study doctor.

Additional information is provided in the informed consent form and you can ask your study doctor for more information.

APPENDIX 13– INFORMED CONSENT FORM

Included as a separate document

APPENDIX 14– DOCUMENT OF REPORTING OF SERIOUS ADVERSE EVENTS

Protocol Code: GEM21menos65 Patient Number: _____ Site: _____

Investigator Information		Patient Information		
Date of Report: ____/____/____ Day Month Year	**Report type:** ☐ **Initial** ☐ **Follow-up** ☐ **Final**	**Date of birth:** ____/____/____ Day Month Year		**Race:** ☐ **Caucasian** ☐ **Black**
Date of Initial Report: ____/____/____ Day Month Year		**Gender:** ☐ **Male**	☐ **Female**	

Date the investigator staff became aware of: ____/____/____ Day Month Year		**Medication (Randomisation) No:**	☐ **Asian** ☐ **Other**
Principal Investigator´s Name:	**Tel.( ) _____**		
Principal Investigator´s Address:	**Fax ( ) _____** **Email: _____**	**Height: ____ cm / or ____ in** **Weight: ____ Kg / or ____ in**	

Study Drug Information (_____)

Indication for use:	**Indication diagnosis date:** ____/____/____ Day Month Year			**Regimen:**		
Study drug name 1: BORTEZOMIB	**First dose:** ____/____/____ Day Month Year	**Last dose prior to event:** ____/____/____ Day Month Year	**and Cycle ___, Day ___ (Oncology only)**	**Dose (in mg):**	**Route:**	**Lot:**
Study drug name 2: LENALIDOMIDE	**First dose:** ____/____/____ Day Month Year	**Last dose prior to event:** ____/____/____ Day Month Year	**and Cycle ___, Day ___ (Oncology only)**	**Dose (in mg):**	**Route:**	**Lot:**
Study drug name 3: DEXAMETHASONE	**First dose:** ____/____/____ Day Month Year	**Last dose prior to event:** ____/____/____ Day Month Year	**and Cycle ___, Day ___ (Oncology only)**	**Dose (in mg):**	**Route:**	**Lot:**
Study drug name 4: MELPHALAN	**First dose:** ____/____/____ Day Month Year	**Last dose prior to event:** ____/____/____ Day Month Year	**and Cycle ___, Day ___ (Oncology only)**	**Dose (in mg):**	**Route:**	**Lot:**
Study drug name 5: BUSULFAN	**First dose:** ____/____/____ Day Month Year	**Last dose prior to event:** ____/____/____ Day Month Year	**and Cycle ___, Day ___ (Oncology only)**	**Dose (in mg):**	**Route:**	**Lot:**

Please attach copies of the following completed CRF pages (if you consider it relevant)

- **Medical History**
- **History of Therapy and Events Related to Disease Under Study**
- **Concomitant Medication**

Relevant Laboratory and Diagnostic Tests **(Please provide relevant data below and attach copies of reports, if available)**

Test	**Date** Day Month Year	**Value/Units**	**Results**
	____/____/____		
	____/____/____		
	____/____/____		
	____/____/____		

Study contact completing form _____ _____ **Principal Investigator** _____ _____
Initials **Date** **Initials** **Date**

Serious Adverse Event Form

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PLEASE FAX TO +34 91 456 11 26 WITHIN 24 HOURS

Protocol Code: GEM2012MENOS65 Patient Number: _____ Site: _____

Action taken with study drug 1: BORTEZOMIB (check all that apply)

☐ **Dose continued unchanged**

☐ **Dose/regimen reduced** **Date decreased:** ____/____/____ Day Month Year **New Dose:** _____

☐ **Dose/regimen increased** **Date increased:** ____/____/____ Day Month Year **New Dose:** _____

☐ **Dose/regimen temporally held** **Date held:** ____/____/____ Day Month Year

☐ **Discontinued permanently due to this Adverse Event** **Date of discontinuation:** ____/____/____

☐ Patient no longer receive study drug (explain the reason in narrative) Date of last dose: ____/____/____ (Day Month Year)

☐ Not applicable

Relationship:	Assessment – The SAE is probably associated with:
☐ Non related ☐ Unlikely ☐ Possible ☐ Probable ☐ Definite	☐ Another drug/combination therapy drug. Please specify _____ ☐ Another condition (e.g. condition under study, intercurrent illness). Please spec_____ ☐ Protocol design or procedures (but not to study drug). Please specify _____

Action taken with study drug 2: LENALIDOMIDE (check all that apply)

☐ Dose continued unchanged

☐ Dose/regimen reduced Date decreased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen increased Date increased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen temporally held Date held: ____/____/____ (Day Month Year)

☐ Discontinued permanently due to this Adverse Event Date of discontinuation: ____/____/____ (Day Month Year)

☐ Patient no longer receive study drug (explain the reason in narrative) Date of last dose: ____/____/____ (Day Month Year)

☐ Not applicable

Relationship:	Assessment – The SAE is probably associated with:
☐ Non related ☐ Unlikely ☐ Possible ☐ Probable ☐ Definite	☐ Another drug/combination therapy drug. Please specify _____ ☐ Another condition (e.g. condition under study, intercurrent illness). Please spec_____ ☐ Protocol design or procedures (but not to study drug). Please specify _____

Action taken with study drug 3: DEXAMETHASONE (check all that apply)

☐ Dose continued unchanged

☐ Dose/regimen reduced Date decreased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen increased Date increased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen temporally held Date held: ____/____/____ (Day Month Year)

☐ Discontinued permanently due to this Adverse Event Date of discontinuation: ____/____/____ (Day Month Year)

☐ Patient no longer receive study drug (explain the reason in narrative) Date of last dose: ____/____/____ (Day Month Year)

☐ Not applicable

Relationship:	Assessment – The SAE is probably associated with:
☐ Non related ☐ Unlikely ☐ Possible ☐ Probable ☐ Definite	☐ Another drug/combination therapy drug. Please specify _____ ☐ Another condition (e.g. condition under study, intercurrent illness). Please spec_____ ☐ Protocol design or procedures (but not to study drug). Please specify _____

Study contact completing form _____ (Initials) _____ (Date) Principal Investigator _____ (Initials) _____ (Date)

Serious Adverse Event Form Page 3 of 4

PLEASE FAX TO +34 91 456 11 26 WITHIN 24 HOURS

Protocol Code: GEM2012MENOS65 Patient Number: _____ Site: _____

Action taken with study drug 4: MELPHALAN (check all that apply)

☐ Dose continued unchanged

☐ Dose/regimen reduced Date decreased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen increased Date increased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen temporally held Date held: ____/____/____ (Day Month Year)

☐ Discontinued permanently due to this Adverse Event Date of discontinuation: ____/____/____ (Day Month Year)

☐ Patient no longer receive study drug (explain the reason in narrative) Date of last dose: ____/____/____
Day Month Year

☐ Not applicable

Relationship:	Assessment – The SAE is probably associated with:
☐ Non related ☐ Unlikely ☐ Possible ☐ Probable ☐ Definite	☐ Another drug/combination therapy drug. Please specify _____ ☐ Another condition (e.g. condition under study, intercurrent illness). Please specify _____ ☐ Protocol design or procedures (but not to study drug). Please specify _____

Action taken with study drug 5: BUSULFAN (check all that apply)

☐ Dose continued unchanged

☐ Dose/regimen reduced Date decreased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen increased Date increased: ____/____/____ (Day Month Year) New Dose: _____

☐ Dose/regimen temporally held Date held: ____/____/____ (Day Month Year)

☐ Discontinued permanently due to this Adverse Event Date of discontinuation: ____/____/____ (Day Month Year)

☐ Patient no longer receive study drug (explain the reason in narrative) Date of last dose: ____/____/____ (Day Month Year)

☐ Not applicable

Relationship:	Assessment – The SAE is probably associated with:
☐ Non related ☐ Unlikely ☐ Possible ☐ Probable ☐ Definite	☐ Another drug/combination therapy drug. Please specify _____ ☐ Another condition (e.g. condition under study, intercurrent illness). Please specify _____ ☐ Protocol design or procedures (but not to study drug). Please specify _____

Adverse Event Information

ADVERSE EVENT (Diagnostic term): _____

AE onset date: ____/____/____ (Day Month Year)

AE became serious: ____/____/____ (Day Month Year)

Serious Criteria (check all that apply):	Maximum intensity	Did AE (s) abate after stopping study drug?		
☐ Death ☐ Life threatening ☐ Hospitalization/prolonged hospitalization ☐ Persistent or significant disability/incapacity ☐ Congenital anomaly/birth defect ☐ Important medical event	☐ Grade 1/Mild ☐ Grade 2/Moderate ☐ Grade 3/Severe ☐ Grade 4/Life-threatening	☐ NO	☐ YES	☐ NA
		Did AE (s) reappear after reintroducing study drug?		
		☐ NO	☐ YES	☐ NA

Serious Adverse Event Form Page 4 of 4

Description of Adverse Event(s)

Please provide a brief narrative description of the SAE (presenting symptoms, clinical course, treatment etc.)
Attach extra pages
e. g hospital discharge summary if available.

Status of SAE at time of this report	
☐ **Fatal**	☐ **Not completely resolved**
☐ **Completely resolved: ___/___/_____** DD MM YYYY	☐ **ongoing and unchanged**
intensity	☐ **ongoing with increased**
	☐ **ongoing with decreased**
intensity	

Death Information	
Date of death: ___/___/_____ DD MM YYYY	**Cause of death per (specify):**
Autopsy performed?	
☐ **Yes** ☐ **No**	☐ **Autopsy Report** _____
Autopsy report attached?	☐ **Death Certificate** _____
☐ **Yes** ☐ **No**	
Death certificate attached?	☐ **Other** _____
☐ **Yes** ☐ **No**	

_____	_____	_____
Study contact completing form printed name	Study contact completing form signature	Date
_____	_____	_____
Principal investigator printed name	Principal investigator signature	Date

APPENDIX 15- DOCUMENT OF PREGNANCY REPORTING

Study Code:	**GEM21menos65**		**Study Site:**	
Subject No:		**Subject Initials:**		
Country:				
Investigational Product(s)				
Therapy start date: (yy-mmm-dd)		Therapy stop date: (yy-mmm-dd)		
Date of last menstruation: (yy-mmm-dd)		Date of expected delivery: (yy-mmm-dd)		
Investigator's signature:		Date: (yy-mmm-dd)		
Keep the original of this form in the Investigator's File and a copy in the CRF. Please contact with the CRO by e-mail to ADKNOMA, farmacovigilancia@adknoma.com or by fax at: +34 93 206 66 67				
Infant Follow up				
Pregnancy result (mark with X)				
Live birth	Elective abortion	Spontaneous abortion	Stillbirth	Unknown
Please fill in a SAE form in case of any spontaneous abortion, stillbirth, birth defect/congenital anomaly, death, or other serious infant condition.				
Infant delivery date (yy-mmm-dd)		Condition of infant		
Normal		Abnormal*		
* If abnormal, please comment				
Comments:				
Investigator's signature:		Date: (yy-mmm-dd)		
Printed name				